1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J50S MLB_S

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
<REV>	<ECN>	<ECO_DESCRIPTION>	<ECODATE>

Wed Nov 16 13:50:00 2011

ALIASES RESOLVED

DOCUMENTS / BOARDS / ASSEMBLIES

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-9342	1	SCH,MLB-S,J50S	SCH		
820-3228	1	PCBF,MLB-S,J50S	MLB		
639-3211		PCBA,MLB-S,J50S			
085-4181	1	DEV LIST,MLB-S,J50S	DEV1		DEVELOPMENT_LIST

DRAWING
TITLE=SHORTSTACK
ABBREV=DRAWING

DRAWING TITLE
SCH,MLB_S,J50S

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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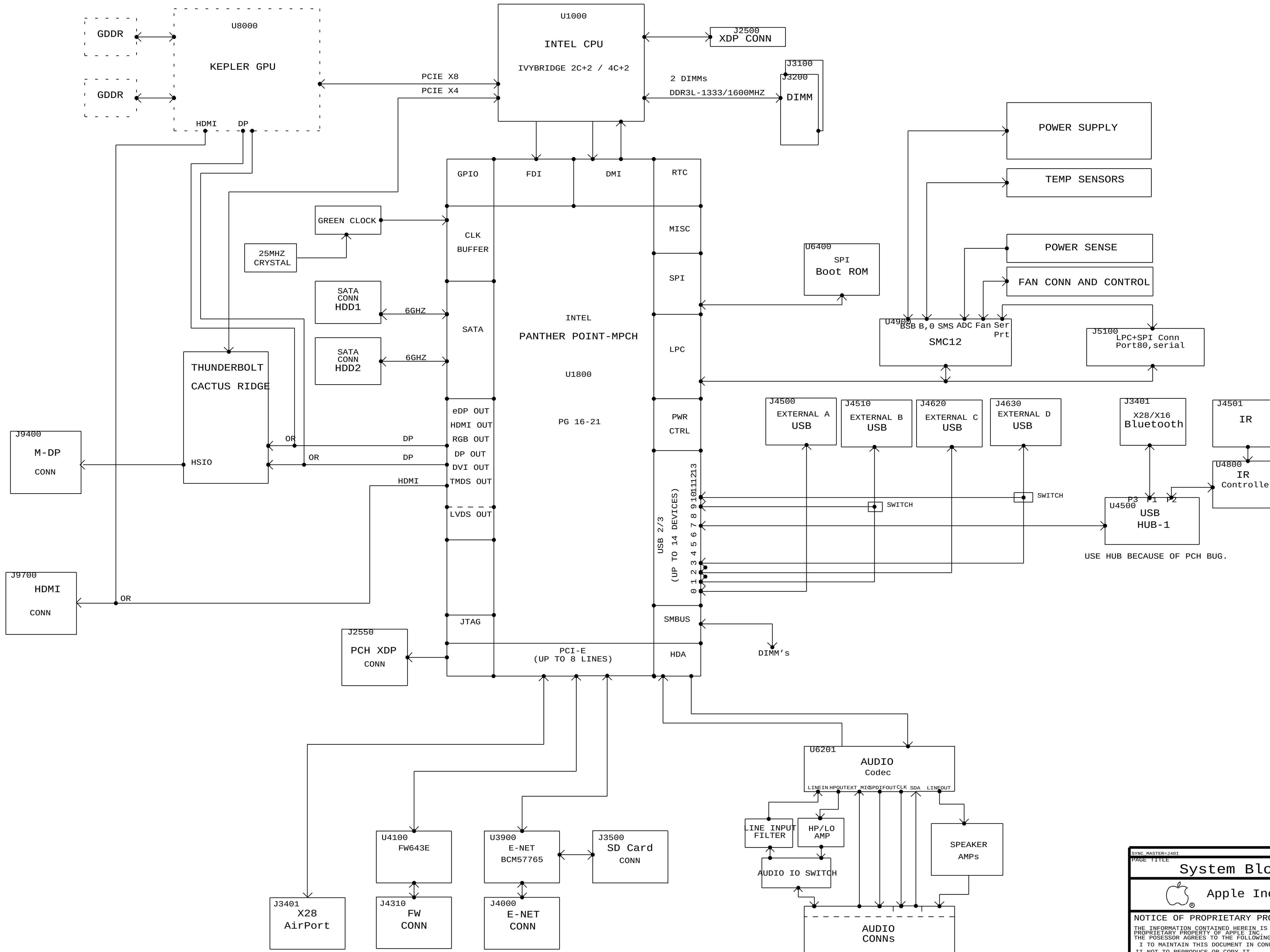

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

System Block Diagram

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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PAGE TITLE

Power Block Diagram

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Apple Inc.

PLATFORM CONTROL HUB (PCH)

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4143	1	IC,PCH,PANTHER_POINT,BGA,SSU,ES1	U1800	CRITICAL	PANTHER_POINT:ES1
337S4155	1	IC,PCH,PANTHER_POINT,BGA,SSU,ES2,QPEG	U1800	CRITICAL	PANTHER_POINT:ES2
337S4207	1	IC,PCH,PANTHER_POINT,BGA,SSU,PRE-QS,QPJ4	U1800	CRITICAL	PANTHER_POINT:PQS
337S4230	1	IC,PCH,PPPT-MB,HM77,C1,QS,QPRG	U1800	CRITICAL	PANTHER_POINT:QS
337S4231	1	IC,PCH,PPPT-MB TDB,HM77,C1,QS,QPRG	U1800	CRITICAL	PANTHER_POINT:QS-TDP

THUNDERBOLT

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
338S1047	1	IC,TBT,CR-4C,ES1,228 FCBGA,12X12MM	U9000	CRITICAL	

OTHER PARTS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION

PROGRAMMED PARTS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	
341S3360	1	IC,EFI BOOTROM,J50	U6400	CRITICAL		BASE: 335S0807,335S0812
338S1065	1	IC,SMC12,BLANK	U4900	CRITICAL	SMC:BLANK	BASE: 338S1065
341S3374	1	IC,SMC12,PROTO 0,J50I	U4900	CRITICAL	SMC:J50I	BASE: 338S1065
341S3372	1	IC,SMC12,PROTO 0,J50G	U4900	CRITICAL	SMC:J50G	BASE: 338S1065
341S3373	1	IC,SMC12,PROTO 0,J50S	U4900	CRITICAL	SMC:J50S	BASE: 338S1065
341S3366	1	IC,IR,PROTO 0,J50	U4700	CRITICAL		BASE: 338S0375
341S3200	1	IC,ENET ROM	U3990	CRITICAL		BASE: 335S0862 OR 335S0854

341S3198	341S3200		U3990	ALTERNATE ENET ROM

335S0865	1	IC,EEPROM,SERIAL,SPI,32KX8,1.8V,2X3QFN	U9090	CRITICAL	TBTROM:BLANK	
341S3362	1	IC,CONF ROM,TBT,J50I	U9090	CRITICAL	TBTROM:PROG_I	BASE: 335S0865
341S3362	1	IC,CONF ROM,TBT,J50G	U9090	CRITICAL	TBTROM:PROG_G	BASE: 335S0865
341S3429	1	IC,CONF ROM,TBT,J50S	U9090	CRITICAL	TBTROM:PROG_S	BASE: 335S0865

NON-SCHEMATIC PARTS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	
825-6838	1	LBL,SERIAL NUMBER	LBL			
946-3189	1	MLB UV EDGE BOND,J40G	UVEB	CRITICAL	J50X:J50G	0.78 GRAM
946-3379	1	MLB UV EDGE BOND,J40I	UVEB	CRITICAL	J50X:J50I	0.61 GRAM
946-3379	1	MLB UV EDGE BOND,J40S	UVEB	CRITICAL	J50X:J50S	0.61 GRAM
946-3190	1	MLB ADHESIVE,LOCTITE	ADHS	CRITICAL		0.048GRAM

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
128S0286	128S0248	?	?	39UF CAP FROM SANYO AND CHEMI-CON
132S0099	132S0203	?	?	0.1UF CAP,X7R
138S0676	138S0691		?	22UF CAP FROM MURATA
152S1447	152S1269	?	?	INDUCTOR FROM CYNTEC
152S1282	152S1155	?	?	INDUCTOR FROM CYNTEC
152S1366	152S1289	?	?	INDUCTOR FROM TOKO
155S0641	155S0397	?	?	FERRITE FROM LAIRD
157S0058	157S0055		ALL	DELTA ALT TO TDK MAGNETICS
371S0709	371S0652	?	?	PIN DIODE FROM NXP
376S0914	376S0881		?	SI7121DN AS ALTERNATE
376S0957	376S0999		?	DUAL FET FROM FAIRCHILD
376S0958	376S0953		?	DUAL FET FROM FAIRCHILD
376S1000	376S0960		?	DUAL FET FROM FAIRCHILD
376S0972	376S0612		?	FET FROM ROHM
376S0613	376S0859		?	OLDER DUAL FET FROM TOSHIBA
376S0855	376S0859		?	DUAL FET FROM DDS
376S0977	376S0859		?	DUAL FET FROM DDS
376S0926	376S0610		?	DUAL FET FROM FAIRCHILD
376S0572	376S0659		?	FET FROM ALPHA OMEGA
377S0124	377S0057		?	VARISTOR FROM AMOTECH
152S1493	152S1300	?	?	INDUCTOR FROM COILCRAFT

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

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COMMON PARTS TABLE

Apple Inc.

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Revision History

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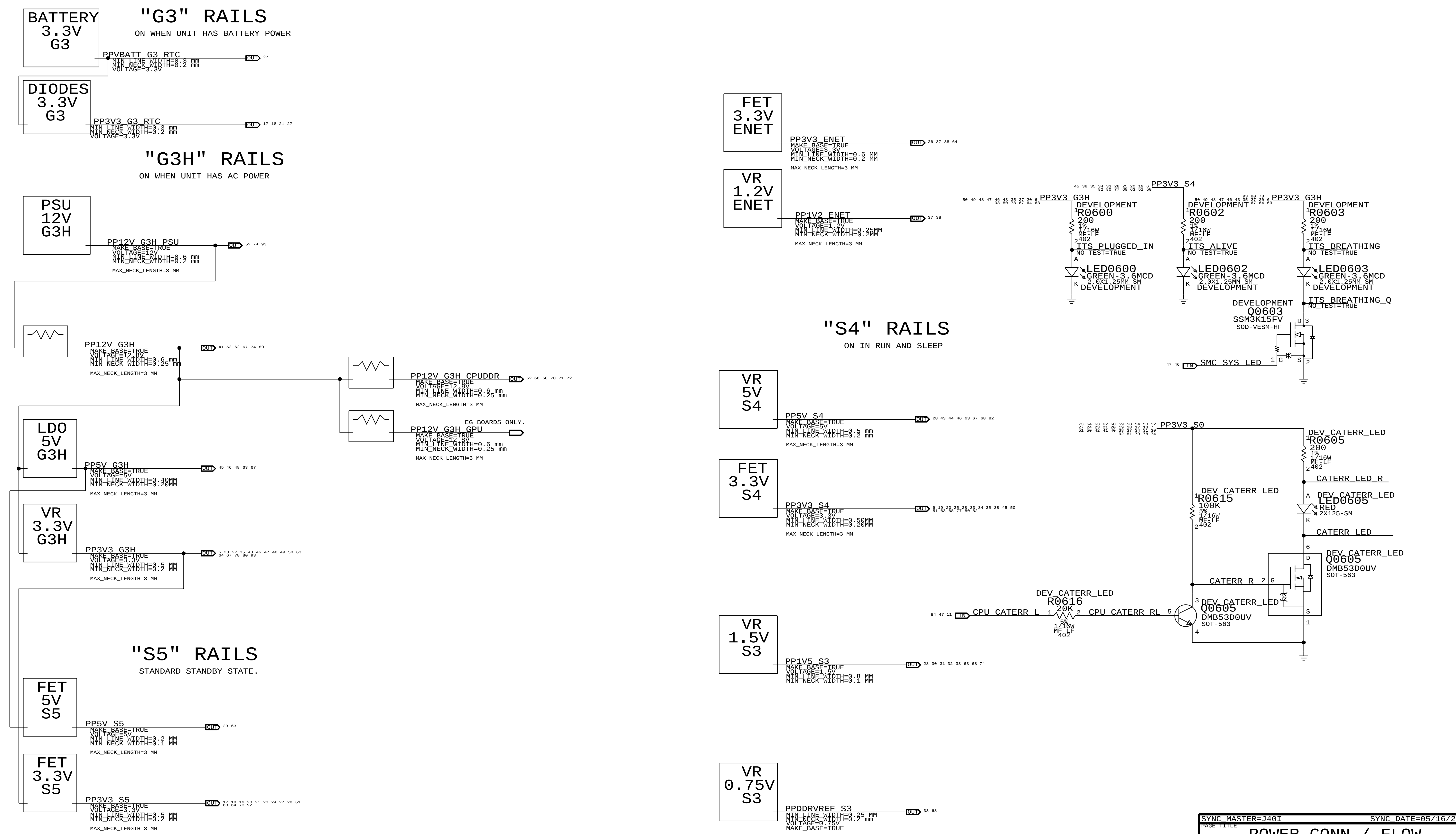

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POWER CONN / FLOW

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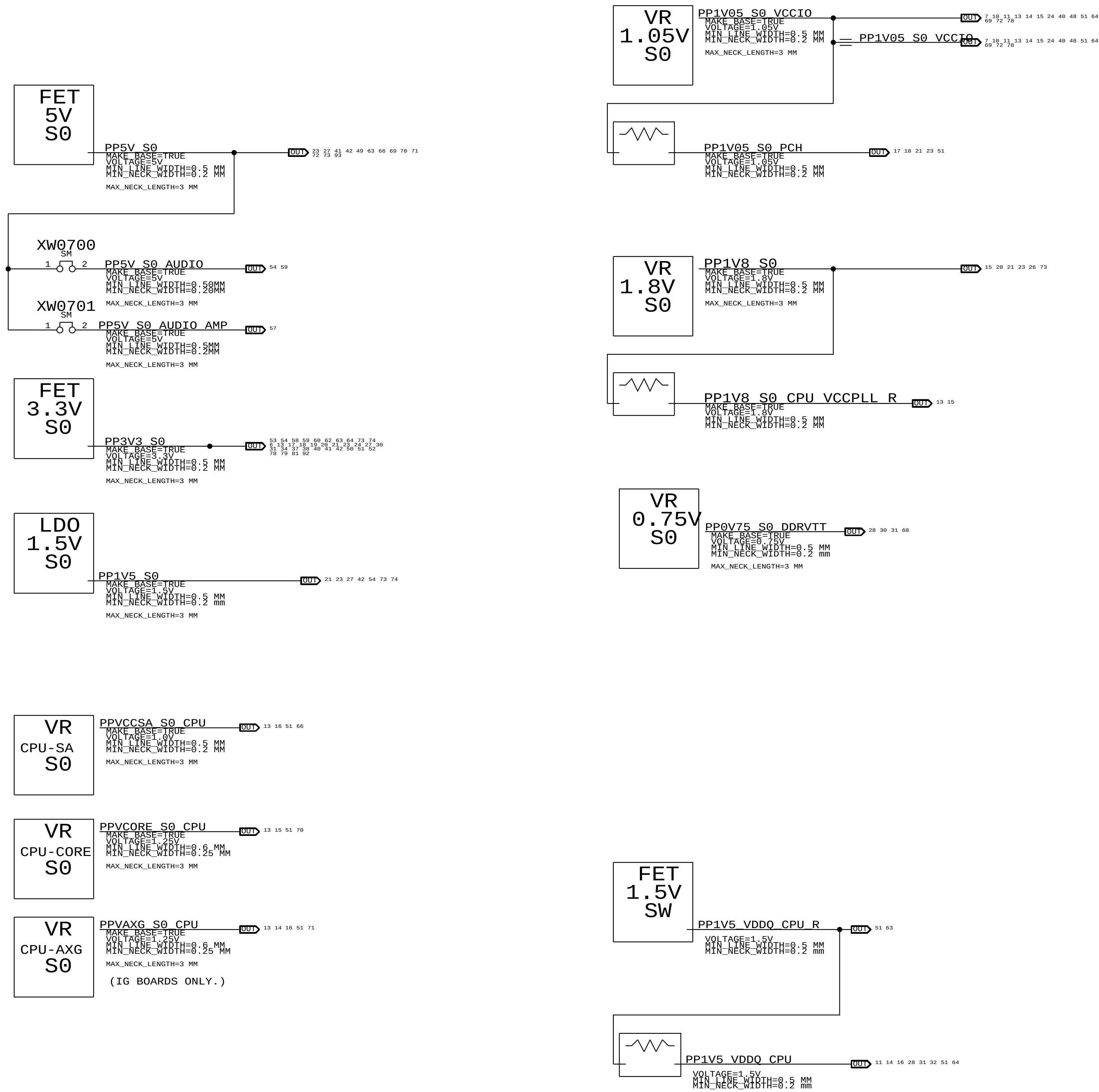
1.05V Rails



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POWER FLOW CONT.

Apple Inc.

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<SCH_NUM>	D

REVISION <E4LABEL>

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INTEL INTEGRATED GRAPHICS

FDI BUS.

10	IN	=FDI_DATA_P<7..0>	MAKE_BASE=TRUE	FDI_DATA_P<7..0>	OUT	18 84
10	IN	=FDI_DATA_N<7..0>	MAKE_BASE=TRUE	FDI_DATA_N<7..0>	OUT	18 84
84 18 10 8	OUT	FDI_INT	MAKE_BASE=TRUE	FDI_INT	IN	8 10 18 84
10	OUT	=FDI_FSYNC<1..0>	MAKE_BASE=TRUE	FDI_FSYNC<1..0>	IN	18 84
10	OUT	=FDI_LSYNC<1..0>	MAKE_BASE=TRUE	FDI_LSYNC<1..0>	IN	18 84

NO PCIE TO EXTERNAL GRAPHICS

	OUT	=PEG_D2R_N<7..0>	MAKE_BASE=TRUE	NC_PEG_D2RN<7..0>	10 93
	OUT	=PEG_D2R_P<7..0>	MAKE_BASE=TRUE	NC_PEG_D2RP<7..0>	10 93
	IN	=PEG_R2D_C_N<7..0>	MAKE_BASE=TRUE	NC_PEG_R2D_CN<7..0>	10 93
	IN	=PEG_R2D_C_P<7..0>	MAKE_BASE=TRUE	NC_PEG_R2D_CP<7..0>	10 93
93 87 17 8	IN	NC_PEG_CLK100MN	MAKE_BASE=TRUE	NC_PEG_CLK100MN	8 17 87 93
93 87 17 8	IN	NC_PEG_CLK100MP	MAKE_BASE=TRUE	NC_PEG_CLK100MP	8 17 87 93

HDMI

86 81 18 8	BI	HDMI_LS_SCL	MAKE_BASE=TRUE	HDMI_LS_SCL	BI	8 18 81 86
86 81 18 8	BI	HDMI_LS_SDA	MAKE_BASE=TRUE	HDMI_LS_SDA	BI	8 18 81 86
86 81 18 8	OUT	HDMI_LS_HPD	MAKE_BASE=TRUE	HDMI_LS_HPD	IN	8 18 81 86
18	IN	=IG_HDMI_DATA_C_N<2..0>	MAKE_BASE=TRUE	HDMI_DATA_C_N<2..0>	OUT	81 86
18	IN	=IG_HDMI_DATA_C_P<2..0>	MAKE_BASE=TRUE	HDMI_DATA_C_P<2..0>	OUT	81 86
86 81 18 8	IN	HDMI_CLK_C_N	MAKE_BASE=TRUE	HDMI_CLK_C_N	OUT	8 18 81 86
86 81 18 8	IN	HDMI_CLK_C_P	MAKE_BASE=TRUE	HDMI_CLK_C_P	OUT	8 18 81 86

THUNDERBOLT (TBT)

PCIE

FUNCTION 0, REVERSED BIT ORDER

OUT	=PE_0R_TBT_D2R_N<3..0>	MAKE_BASE=TRUE	PCIE_TBT_D2R_N<3..0>	IN	10 76 87
OUT	=PE_0R_TBT_D2R_P<3..0>	MAKE_BASE=TRUE	PCIE_TBT_D2R_P<3..0>	IN	10 76 87
IN	=PE_0R_TBT_R2D_C_N<3..0>	MAKE_BASE=TRUE	PCIE_TBT_R2D_C_N<3..0>	OUT	10 76 87
IN	=PE_0R_TBT_R2D_C_P<3..0>	MAKE_BASE=TRUE	PCIE_TBT_R2D_C_P<3..0>	OUT	10 76 87

USE WITH BOM OPTION PEG:1X16R

FUNCTION 1, FORWARD BIT ORDER

OUT	=PE_1F_TBT_D2R_N<3..0>	MAKE_BASE=TRUE	NC_PE_1F_D2RN<3..0>	10 93
OUT	=PE_1F_TBT_D2R_P<3..0>	MAKE_BASE=TRUE	NC_PE_1F_D2RP<3..0>	10 93
IN	=PE_1F_TBT_R2D_C_N<3..0>	MAKE_BASE=TRUE	NC_PE_1F_R2D_CN<3..0>	10 93
IN	=PE_1F_TBT_R2D_C_P<3..0>	MAKE_BASE=TRUE	NC_PE_1F_R2D_CP<3..0>	10 93

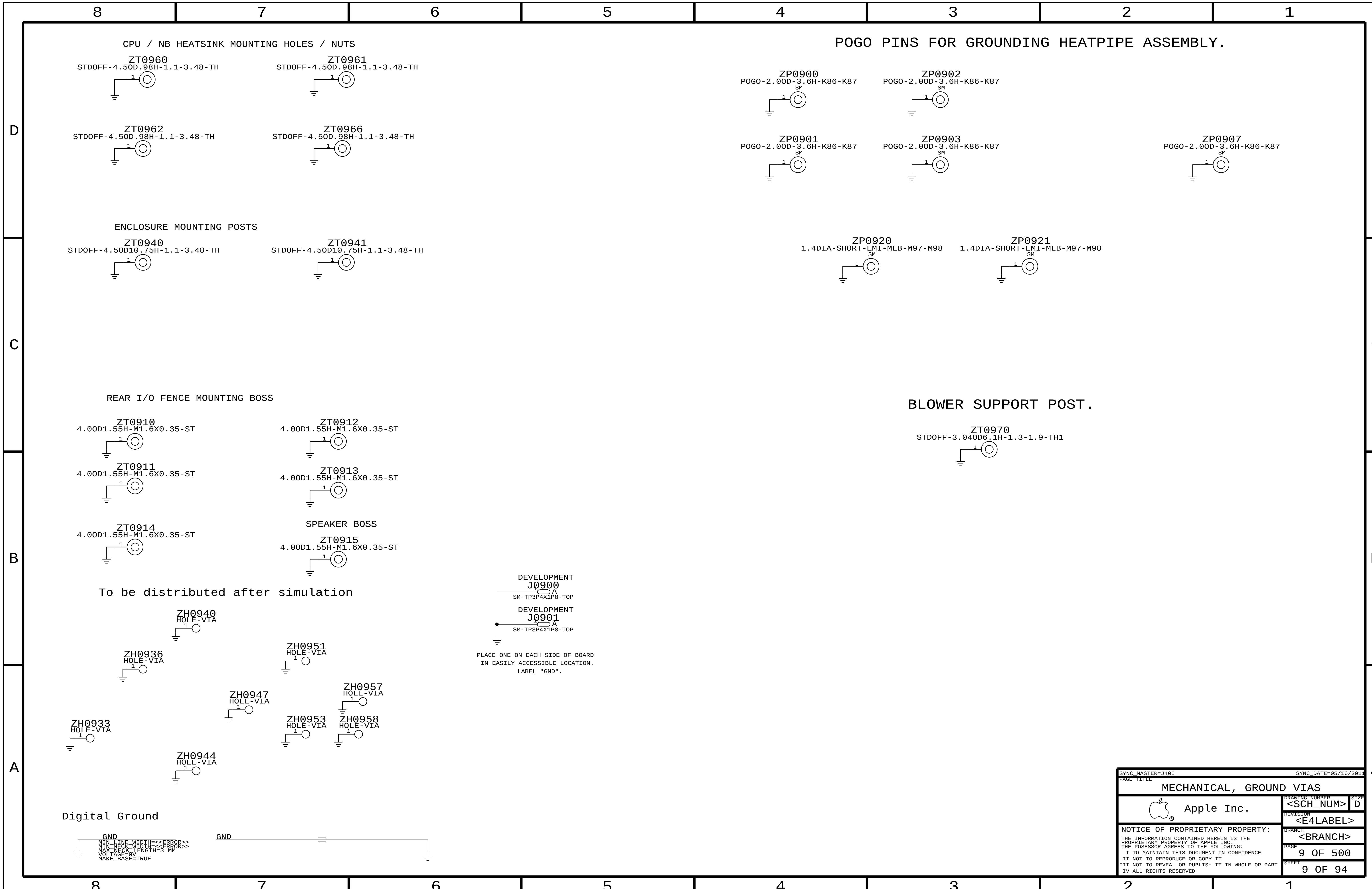
USE WITH BOM OPTION PEG:2X8F.

TBT DISPLAY PORT INPUT

86 80 79 18 8	BI	DP_TBTSNK0_CTRL_CLK	MAKE_BASE=TRUE	DP_TBTSNK0_CTRL_CLK	BI	8 18 79 80 86
86 80 79 18 8	BI	DP_TBTSNK0_CTRL_DATA	MAKE_BASE=TRUE	DP_TBTSNK0_CTRL_DATA	BI	8 18 79 80 86
86 76 18 8	BI	DP_TBTSNK0_AUXCH_C_N	MAKE_BASE=TRUE	DP_TBTSNK0_AUXCH_C_N	BI	8 18 76 86
86 76 18 8	BI	DP_TBTSNK0_AUXCH_C_P	MAKE_BASE=TRUE	DP_TBTSNK0_AUXCH_C_P	BI	8 18 76 86
86 76 18 8	OUT	DP_TBTSNK0_HPD	MAKE_BASE=TRUE	DP_TBTSNK0_HPD	IN	8 18 76 86
	IN	=IG_DP_TBTSNK0_ML_C_N<3..0>	MAKE_BASE=TRUE	DP_TBTSNK0_ML_C_N<3..0>	OUT	18 76 86
	IN	=IG_DP_TBTSNK0_ML_C_P<3..0>	MAKE_BASE=TRUE	DP_TBTSNK0_ML_C_P<3..0>	OUT	18 76 86

TBT DISPLAY PORT INPUT

Pages	Port	Net	Property	Alias	Port	Pages
86 79 18 8	BI	DP_TBTSNK1_CTRL_CLK	MAKE_BASE=TRUE	DP_TBTSNK1_CTRL_CLK	BI	8 18 79 86
86 79 18 8	BI	DP_TBTSNK1_CTRL_DATA	MAKE_BASE=TRUE	DP_TBTSNK1_CTRL_DATA	BI	8 18 79 86
86 76 18 8	BI	DP_TBTSNK1_AUXCH_C_N	MAKE_BASE=TRUE	DP_TBTSNK1_AUXCH_C_N	BI	8 18 76 86
86 76 18 8	BI	DP_TBTSNK1_AUXCH_C_P	MAKE_BASE=TRUE	DP_TBTSNK1_AUXCH_C_P	BI	8 18 76 86
86 76 18 8	OUT	DP_TBTSNK1_HPD	MAKE_BASE=TRUE	DP_TBTSNK1_HPD	IN	8 18 76 86
	IN	=IG_DP_TBTSNK1_ML_C_N<3..0>	MAKE_BASE=TRUE	DP_TBTSNK1_ML_C_N<3..0>	OUT	18 76 86
	IN	=IG_DP_TBTSNK1_ML_C_P<3..0>	MAKE_BASE=TRUE	DP_TBTSNK1_ML_C_P<3..0>	OUT	18 76 86


CPU / NB HEATSINK MOUNTING HOLES / NUTS
ZT0960
STDOFF-4.5OD.98H-1.1-3.48-TH
ZT0961
STDOFF-4.5OD.98H-1.1-3.48-TH
ZT0962
STDOFF-4.5OD.98H-1.1-3.48-TH
ZT0966
STDOFF-4.5OD.98H-1.1-3.48-TH
ENCLOSURE MOUNTING POSTS
ZT0940
STDOFF-4.5OD10.75H-1.1-3.48-TH
ZT0941
STDOFF-4.5OD10.75H-1.1-3.48-TH
REAR I/O FENCE MOUNTING BOSS
ZT0910
4.0OD1.55H-M1.6X0.35-ST
ZT0912
4.0OD1.55H-M1.6X0.35-ST
ZT0911
4.0OD1.55H-M1.6X0.35-ST
ZT0913
4.0OD1.55H-M1.6X0.35-ST
ZT0914
4.0OD1.55H-M1.6X0.35-ST
SPEAKER BOSS
ZT0915
4.0OD1.55H-M1.6X0.35-ST
To be distributed after simulation
ZH0940
HOLE-VIA
ZH0936
HOLE-VIA
ZH0951
HOLE-VIA
ZH0947
HOLE-VIA
ZH0957
HOLE-VIA
ZH0933
HOLE-VIA
ZH0953
HOLE-VIA
ZH0958
HOLE-VIA
ZH0944
HOLE-VIA
Digital Ground
GND
MIN_LINE_WIDTH=<<ERROR>>
MIN_NECK_WIDTH=<<ERROR>>
MAX_NECK_LENGTH=3 MM
VOLTAGE=0V
MAKE_BASE=TRUE
GND
DEVELOPMENT
J0900
SM-TP3P4X1P8-TOP
DEVELOPMENT
J0901
SM-TP3P4X1P8-TOP
PLACE ONE ON EACH SIDE OF BOARD
IN EASILY ACCESSIBLE LOCATION.
LABEL "GND".
POGO PINS FOR GROUNDING HEATPIPE ASSEMBLY.
ZP0900
POGO-2.0OD-3.6H-K86-K87
SM
ZP0902
POGO-2.0OD-3.6H-K86-K87
SM
ZP0901
POGO-2.0OD-3.6H-K86-K87
SM
ZP0903
POGO-2.0OD-3.6H-K86-K87
SM
ZP0907
POGO-2.0OD-3.6H-K86-K87
SM
ZP0920
1.4DIA-SHORT-EMI-MLB-M97-M98
SM
ZP0921
1.4DIA-SHORT-EMI-MLB-M97-M98
SM
BLOWER SUPPORT POST.
ZT0970
STDOFF-3.04OD6.1H-1.3-1.9-TH1
SYNC_MASTER=J40I
SYNC_DATE=05/16/2011
PAGE TITLE
MECHANICAL, GROUND VIAS
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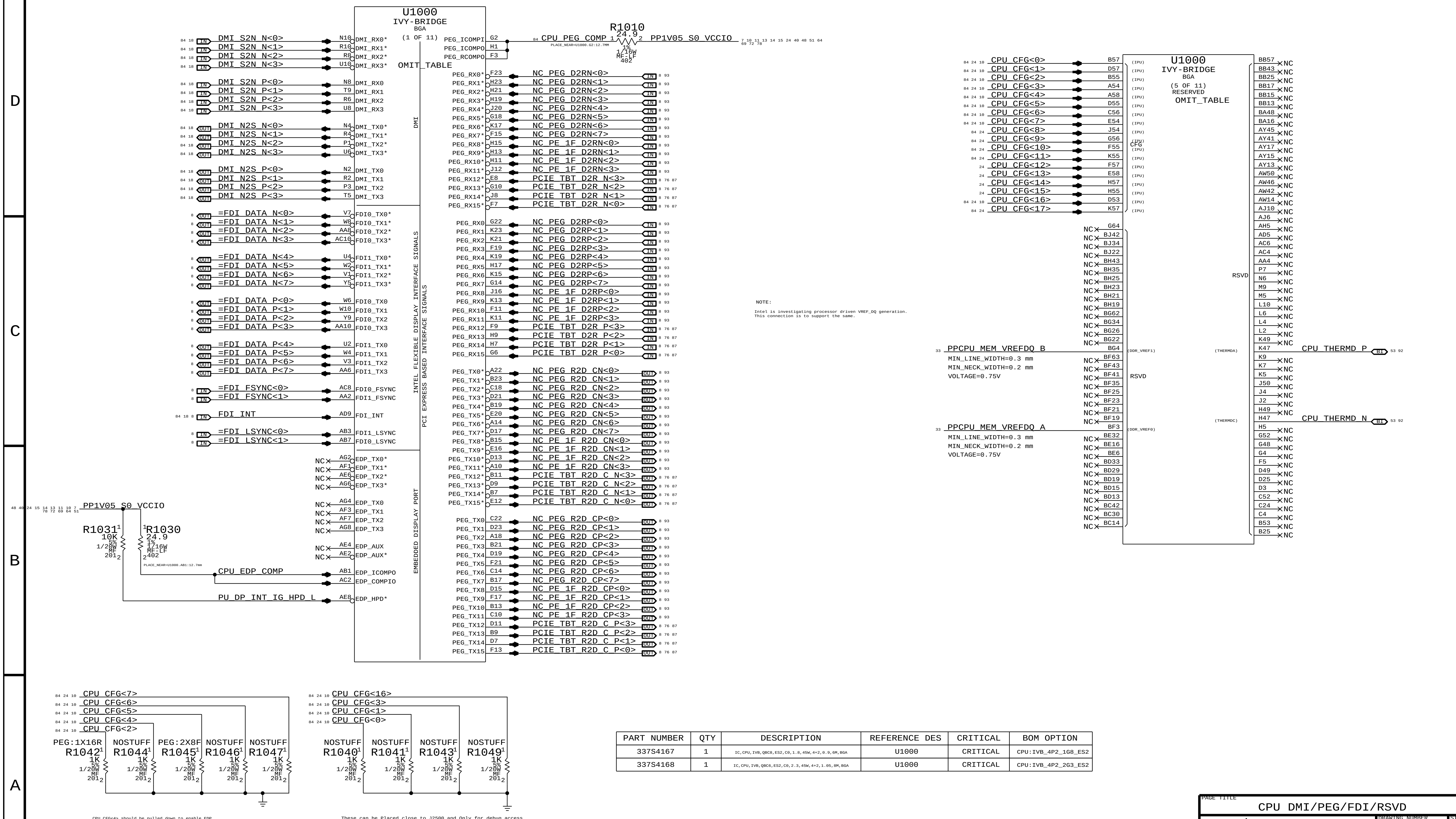

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4167	1	IC,CPU,IVB,QBC8,ES2,C0,1.8,45W,4+2,0.9,6M,BGA	U1000	CRITICAL	CPU:IVB_4P2_1G8_ES2
337S4168	1	IC,CPU,IVB,QBC6,ES2,C0,2.3,45W,4+2,1.05,8M,BGA	U1000	CRITICAL	CPU:IVB_4P2_2G3_ES2

CFG [7] :PEG DEFER TRAINING 1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB 0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION 11 = 1 X16 (DEFAULT) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [4] :eDP ENABLE/DISABLE 1 = DISABLED 0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED

PAGE TITLE
CPU DMI/PEG/FDI/RSVD

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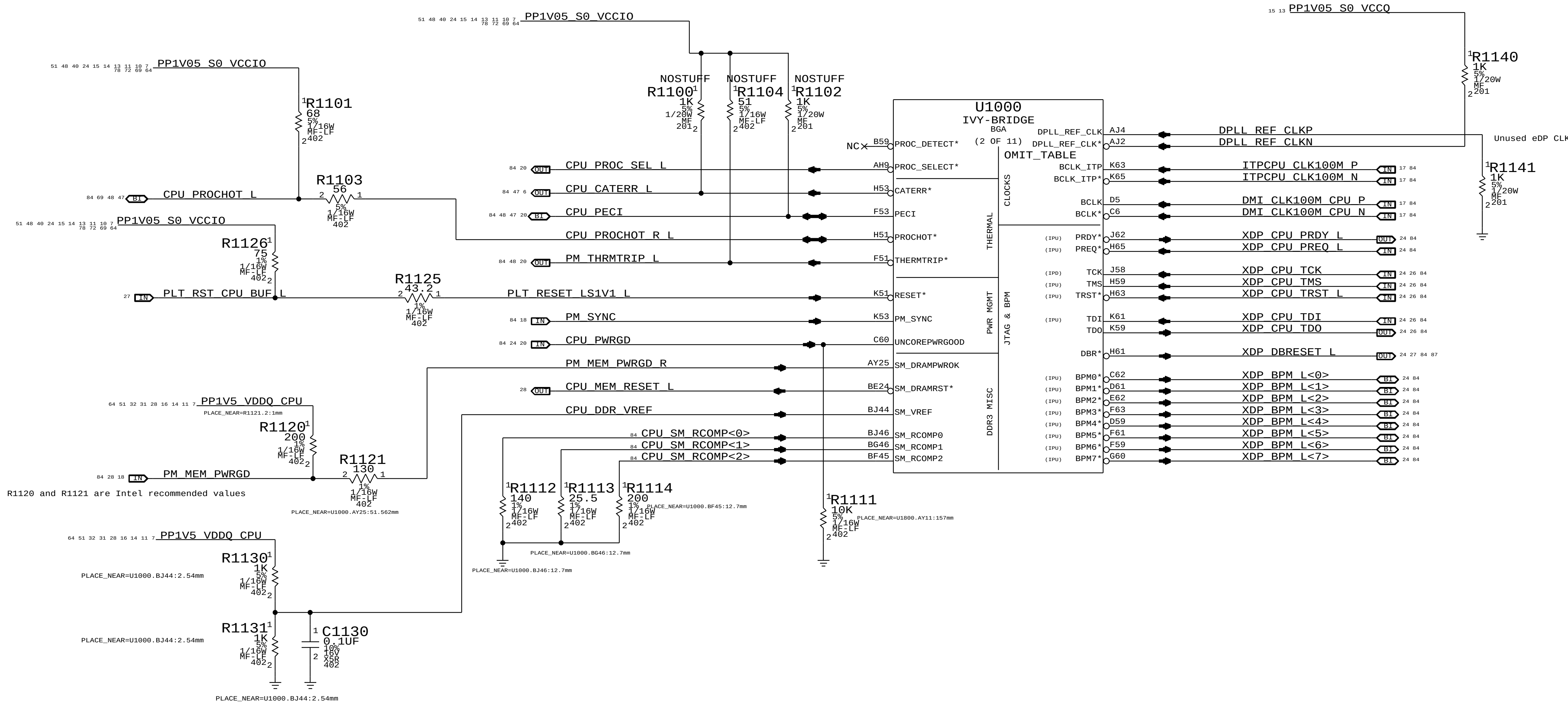

PAGE TITLE

CPU CLOCK/MISC/JTAG

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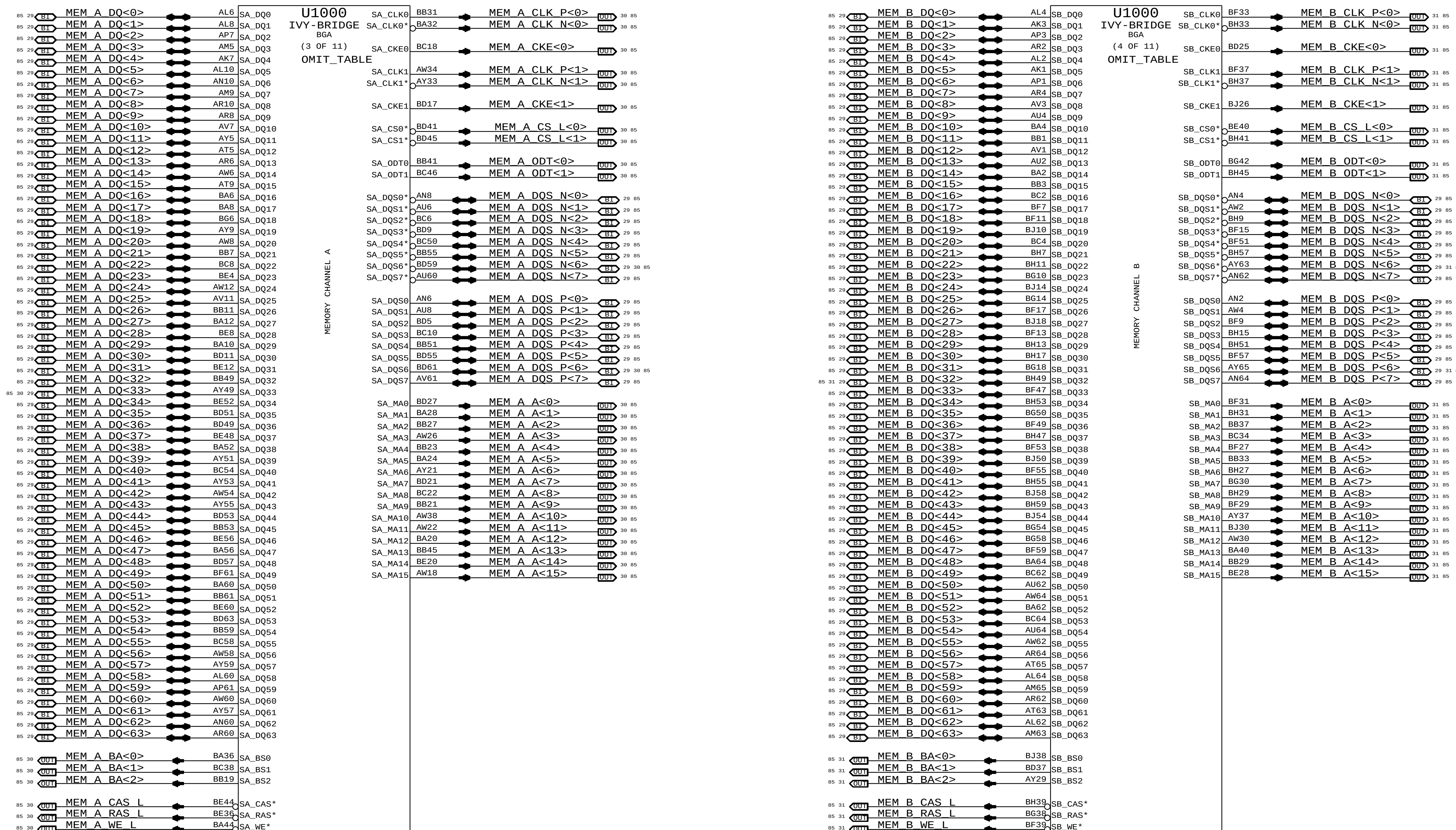

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PAGE TITLE
CPU DDR3 INTERFACES

Apple Inc.

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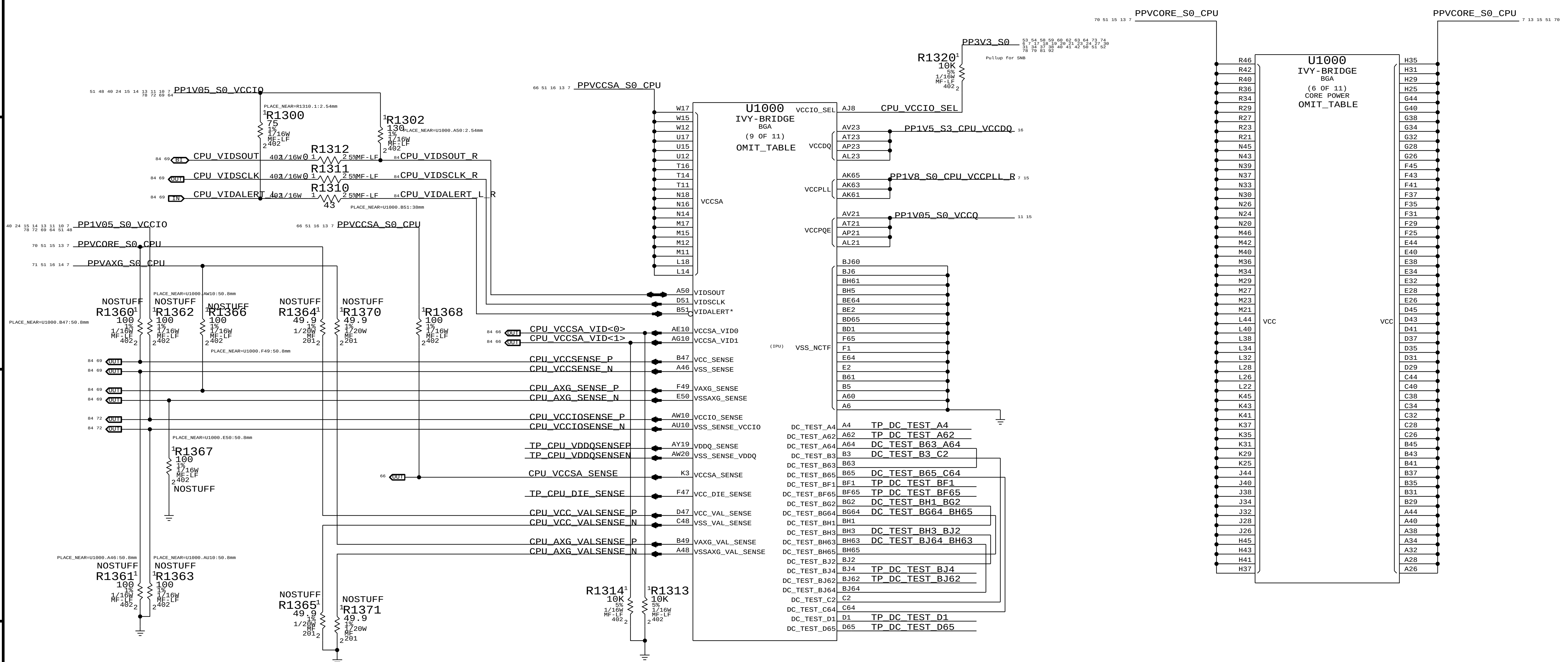

NOTE: Intel validation sense lines per doc 439028 rev1.0
HR_PPDG sections 6.2.1 and 6.3.1.

SYNC_MASTER=K92_MLB
SYNC_DATE=08/03/2010

PAGE TITLE
CPU POWER

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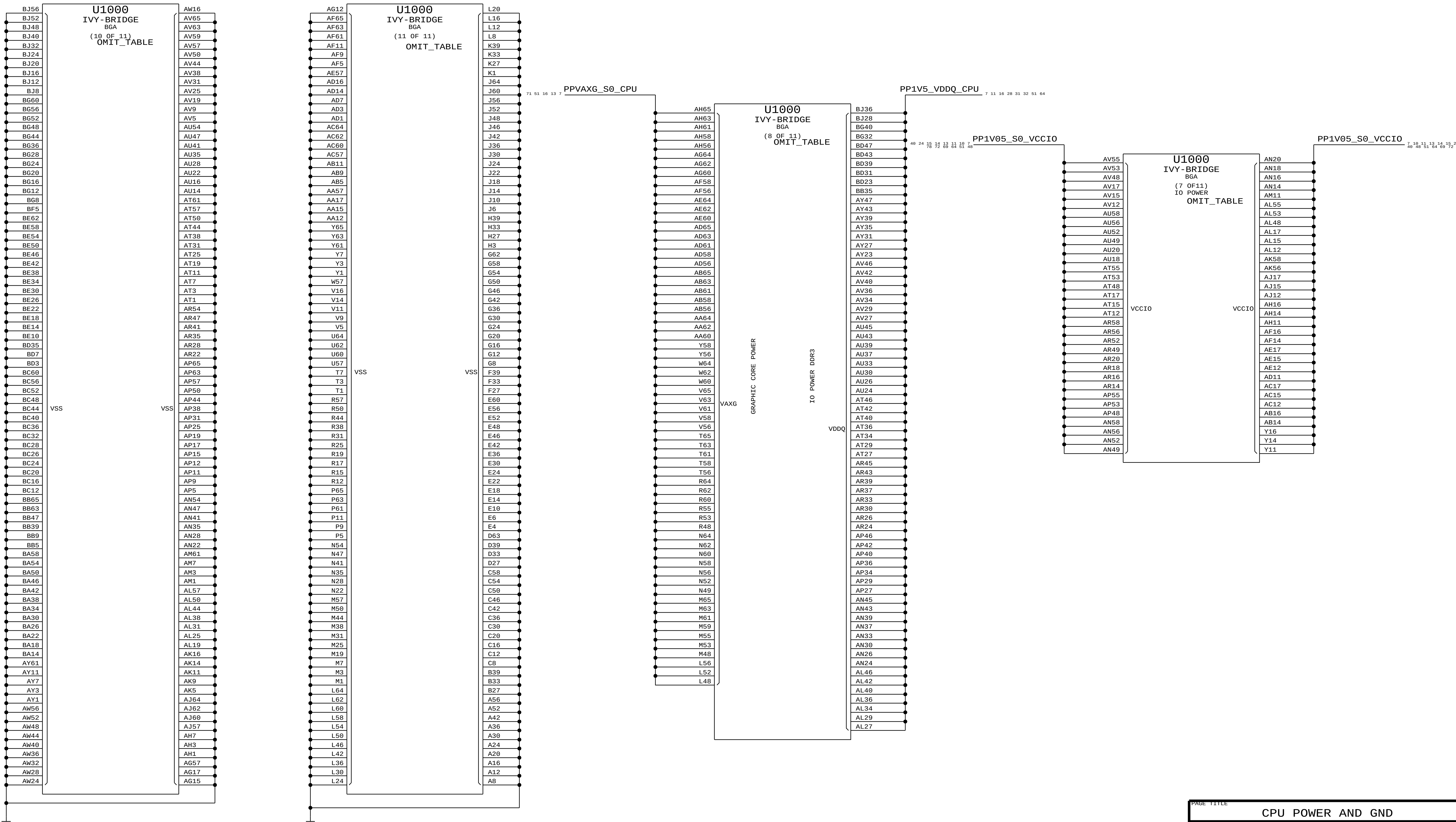

PAGE TITLE

CPU POWER AND GND

Apple Inc.

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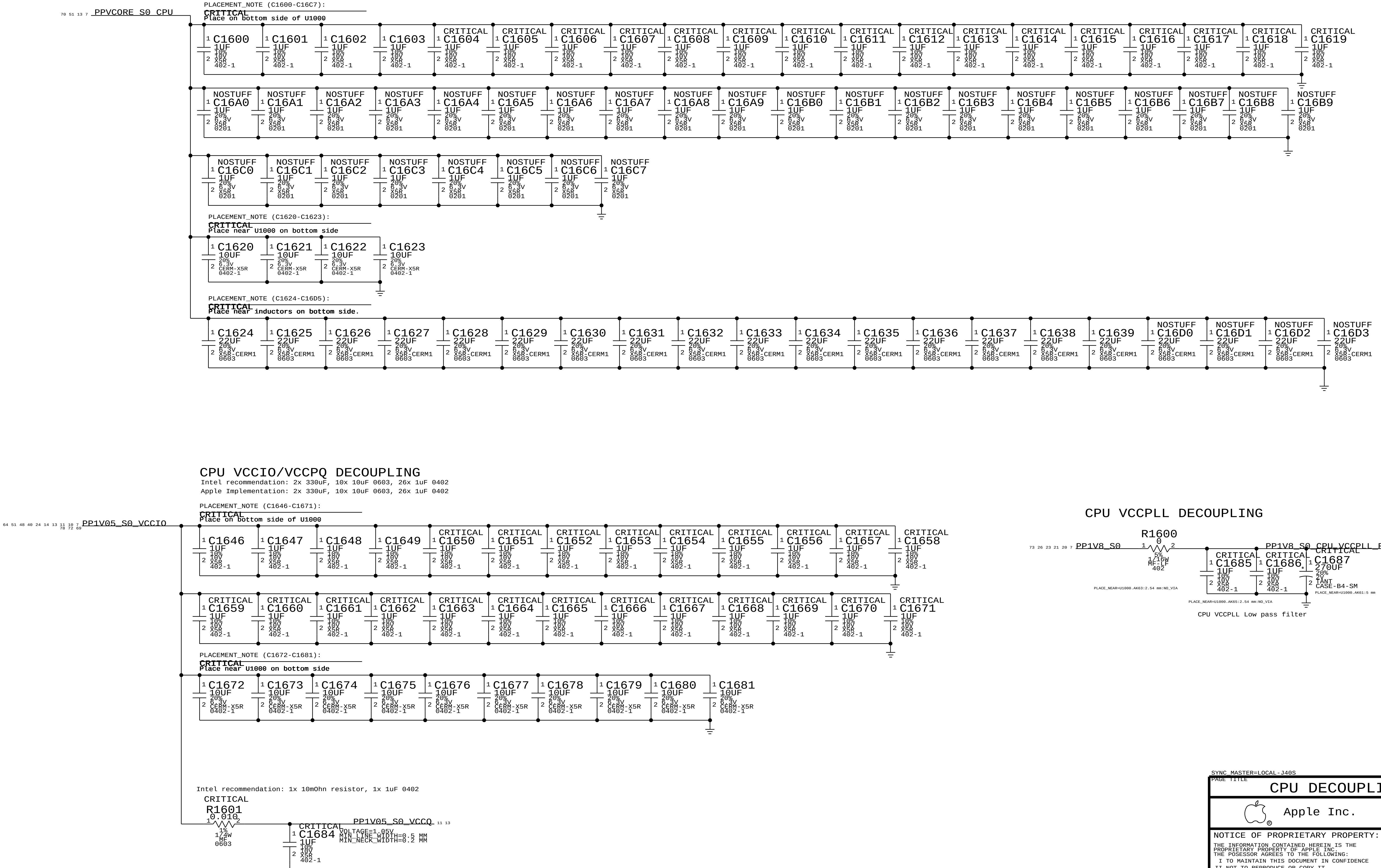
CPU VCORE DECOUPLING

Intel recommendation: 4x 470uF 4mOhm, 2x 470uF 4mOhm (NOSTUFF), 16x 22uF 0805, 4x 10uF 0603, 20x 1uF 0402, 28x 1uF 0402 (NOSTUFF)
Apple Implementation: 4x 470uF 4mOhm, 1x 470uF 4mOhm (NOSTUFF), 16x 22uF 0603, 4x 10uF 0402, 20x 1uF 0402, 28x 1uF 0201 (NOSTUFF), 4x 22uF 0603 (NOSTUFF)



CPU VCCIO/VCCPQ DECOUPLING

Intel recommendation: 2x 330uF, 10x 10uF 0603, 26x 1uF 0402
Apple Implementation: 2x 330uF, 10x 10uF 0603, 26x 1uF 0402

CPU VCCPLL DECOUPLING

SYNC_MASTER=LOCAL-J40S SYNC_DATE=05/16/2011

PAGE TITLE
CPU DECOUPLING-I

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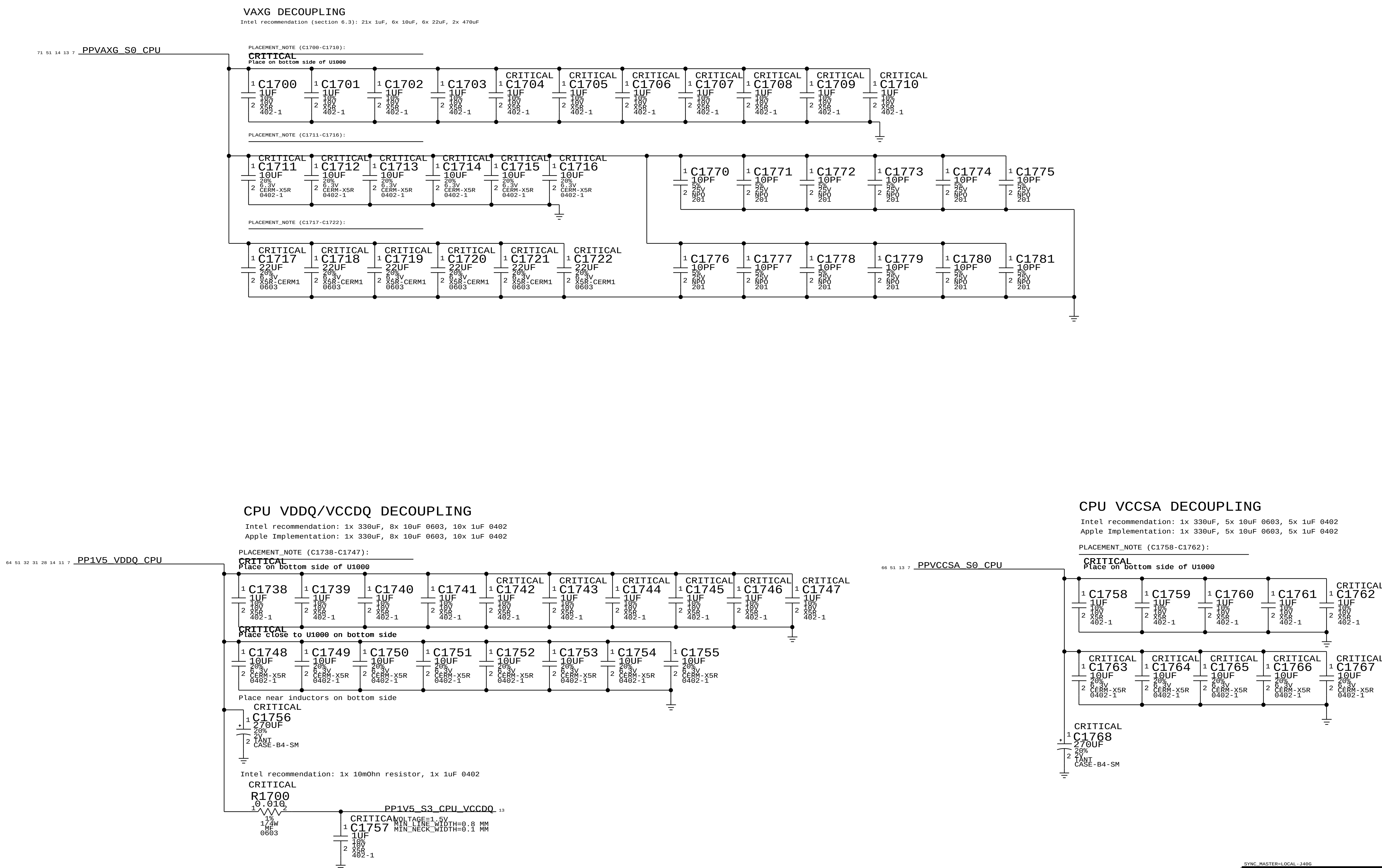

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CPU DECOUPLING-II

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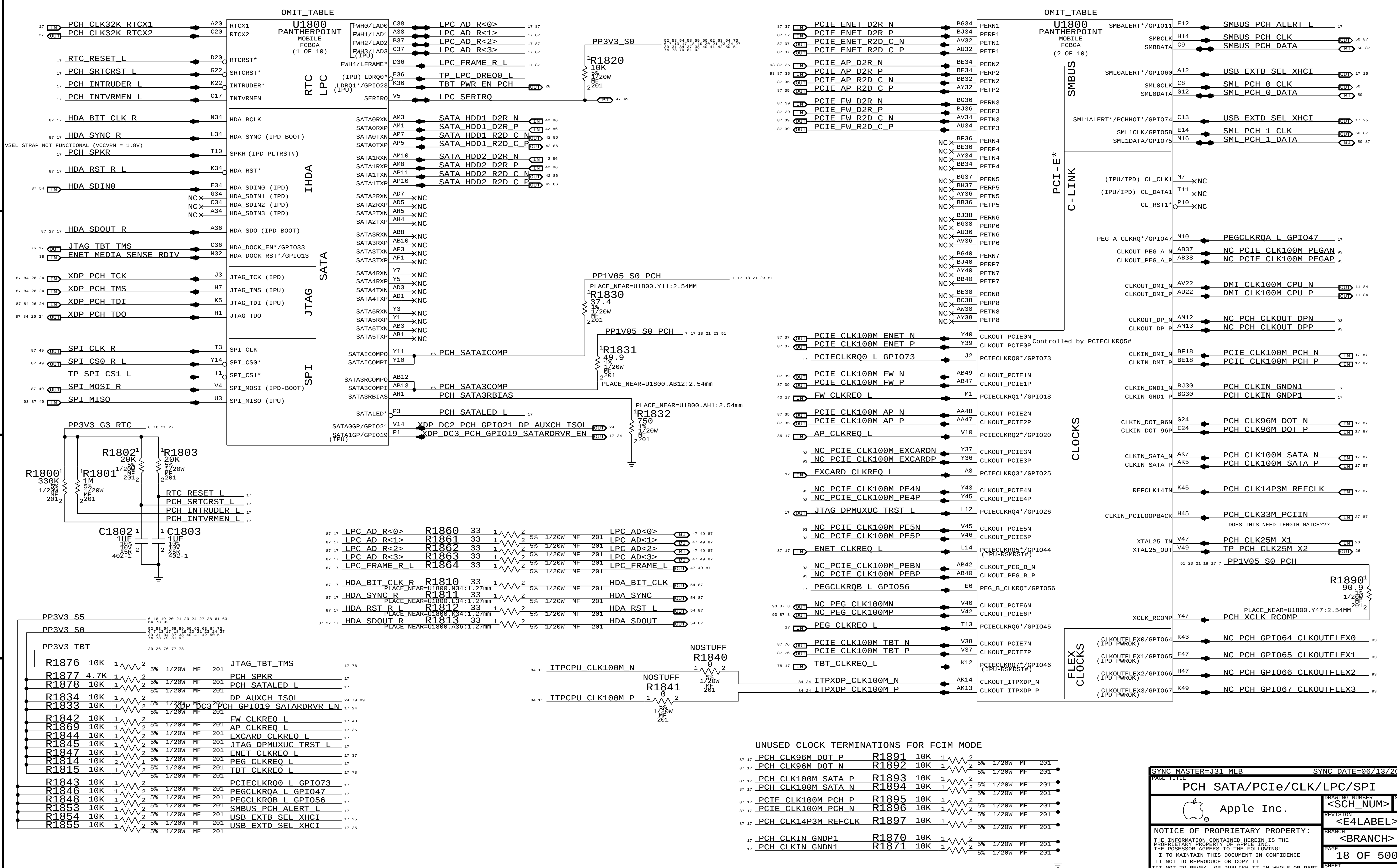

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PAGE TITLE

PCH SATA/PCIe/CLK/LPC/SPI

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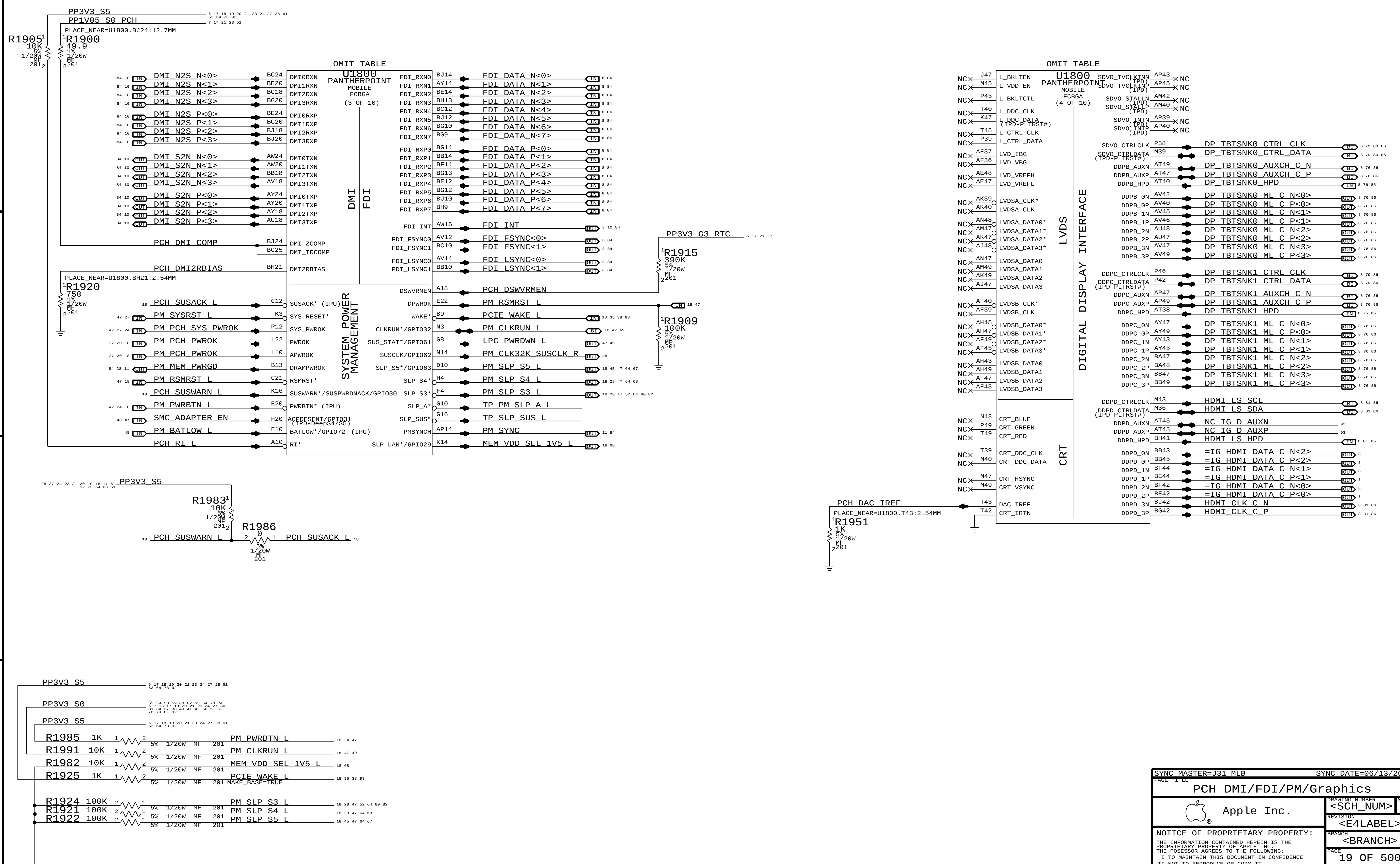

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PAGE TITLE
PCH DMI/FDI/PM/Graphics

Apple Inc.

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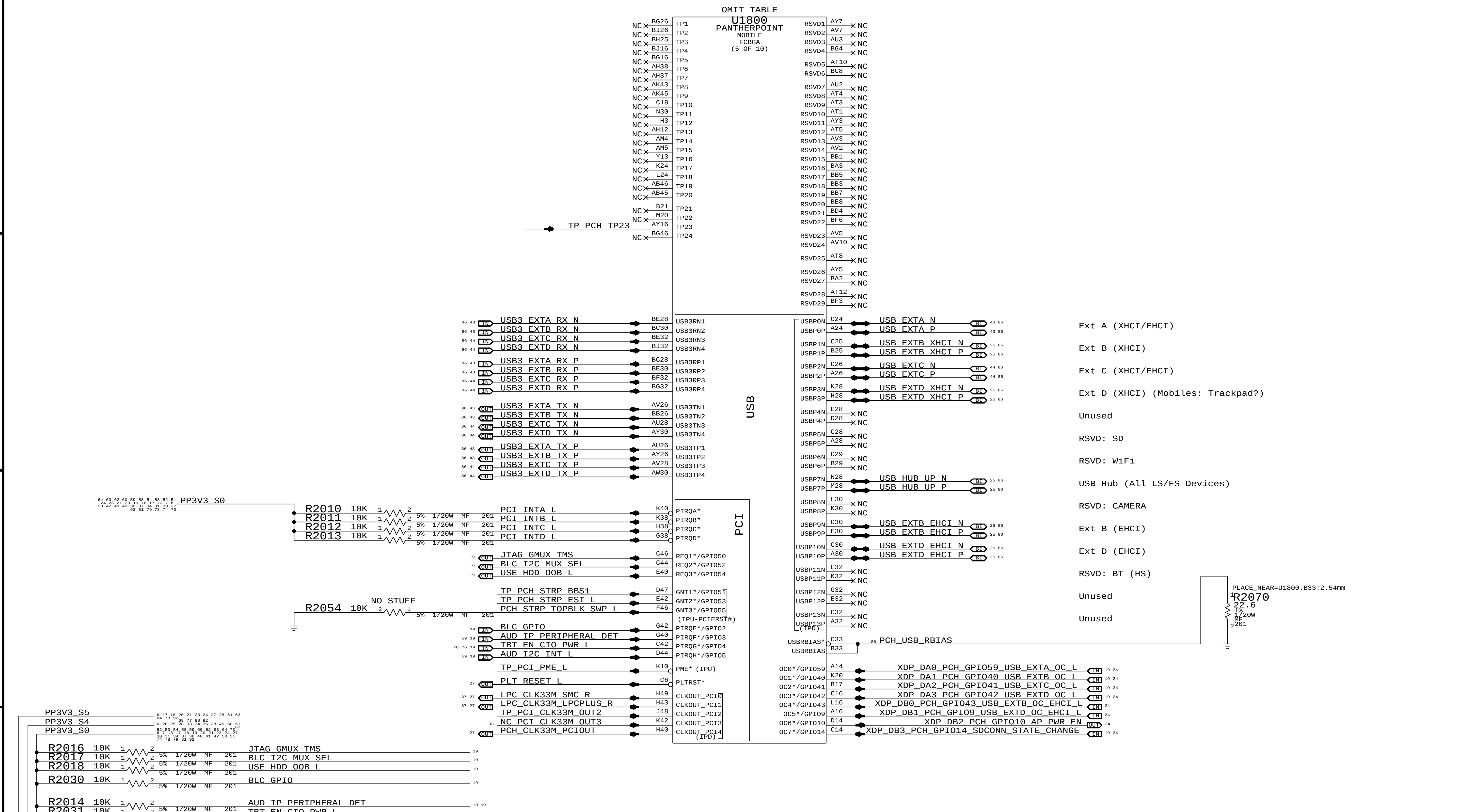

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PAGE TITLE: PCH PCI/USB/TP/RSVD

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

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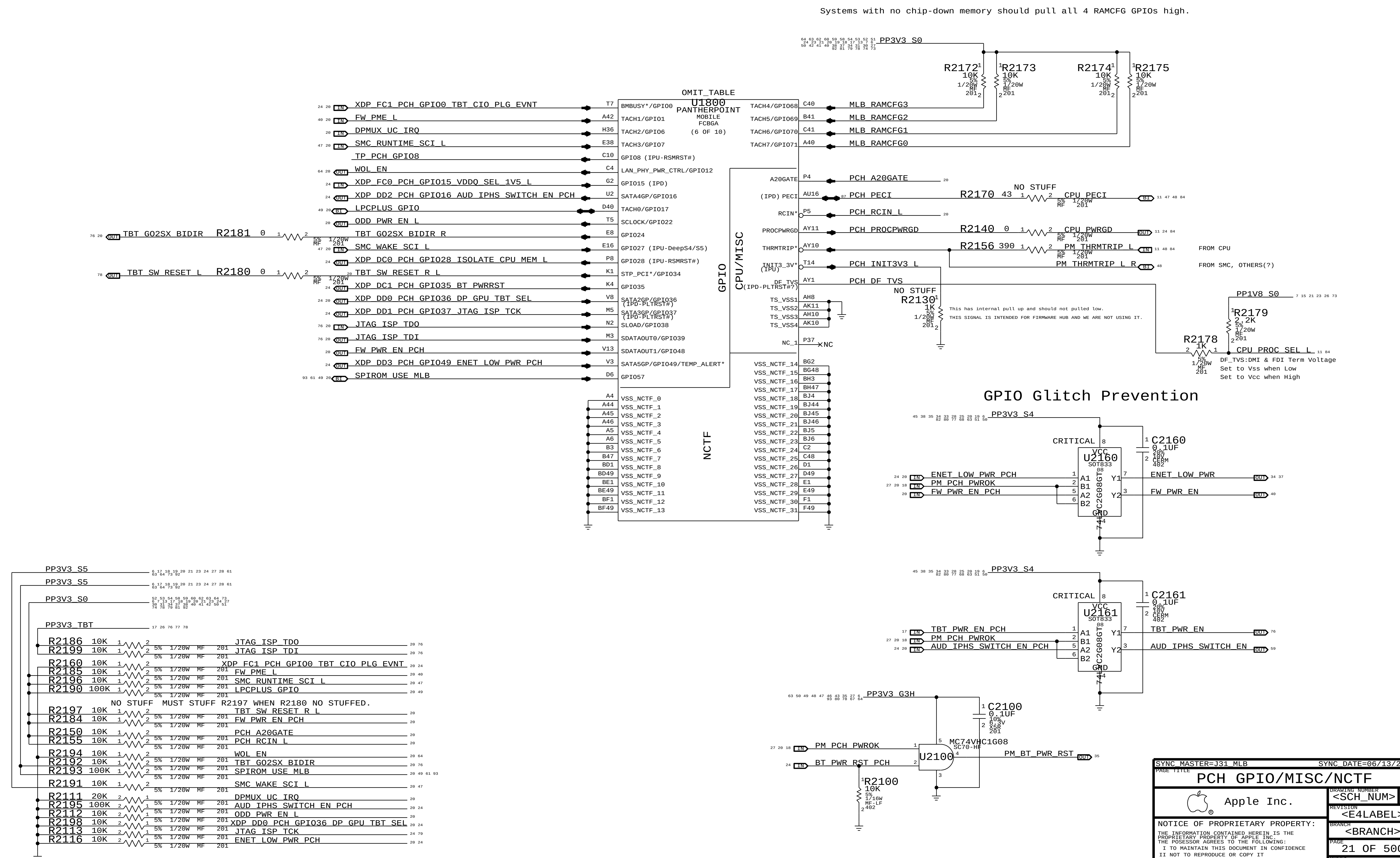
SHEET: 19 OF 94



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PCH GPIO/MISC/NCTF

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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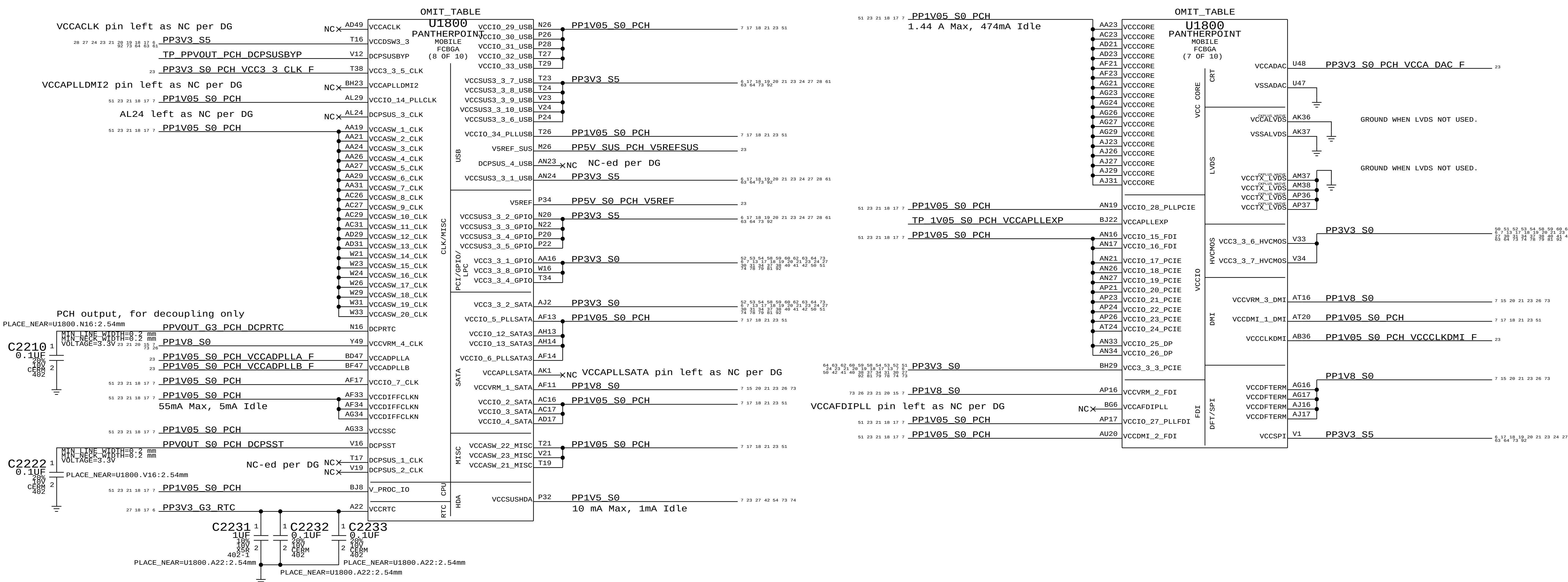
SHEET 20 OF 94



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PAGE TITLE

PCH POWER

Apple Inc.

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REVISION <E4LABEL>

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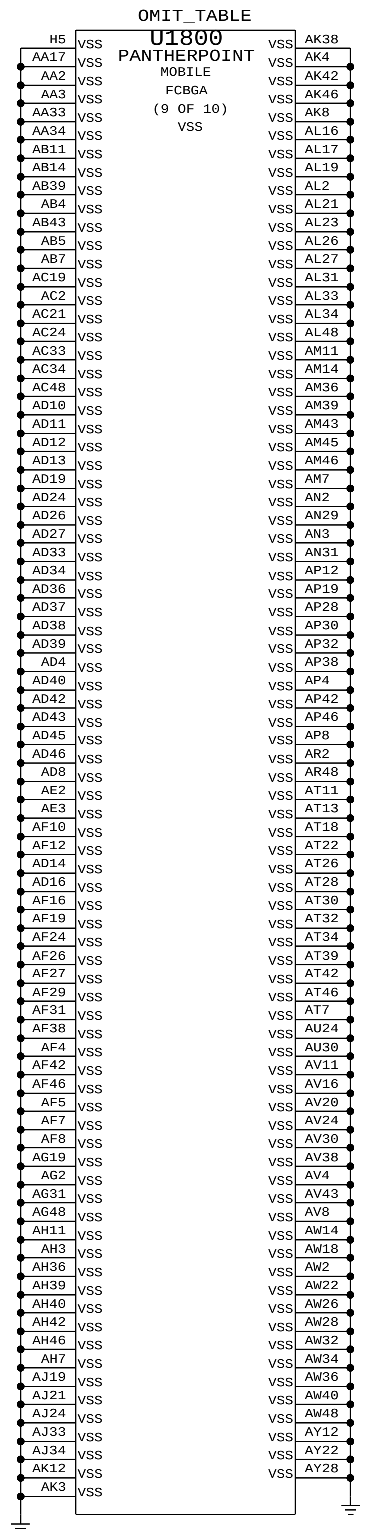
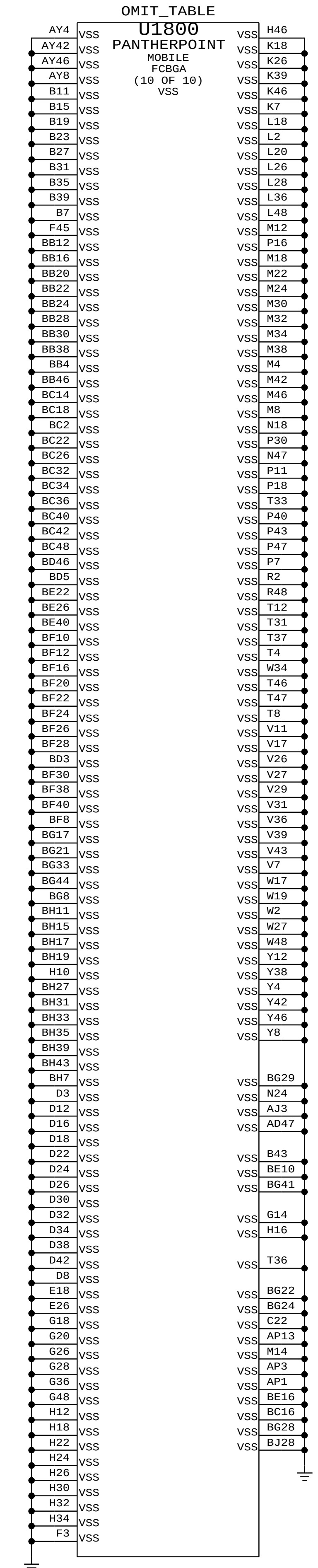
SHEET 21 OF 94





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PCH GROUNDS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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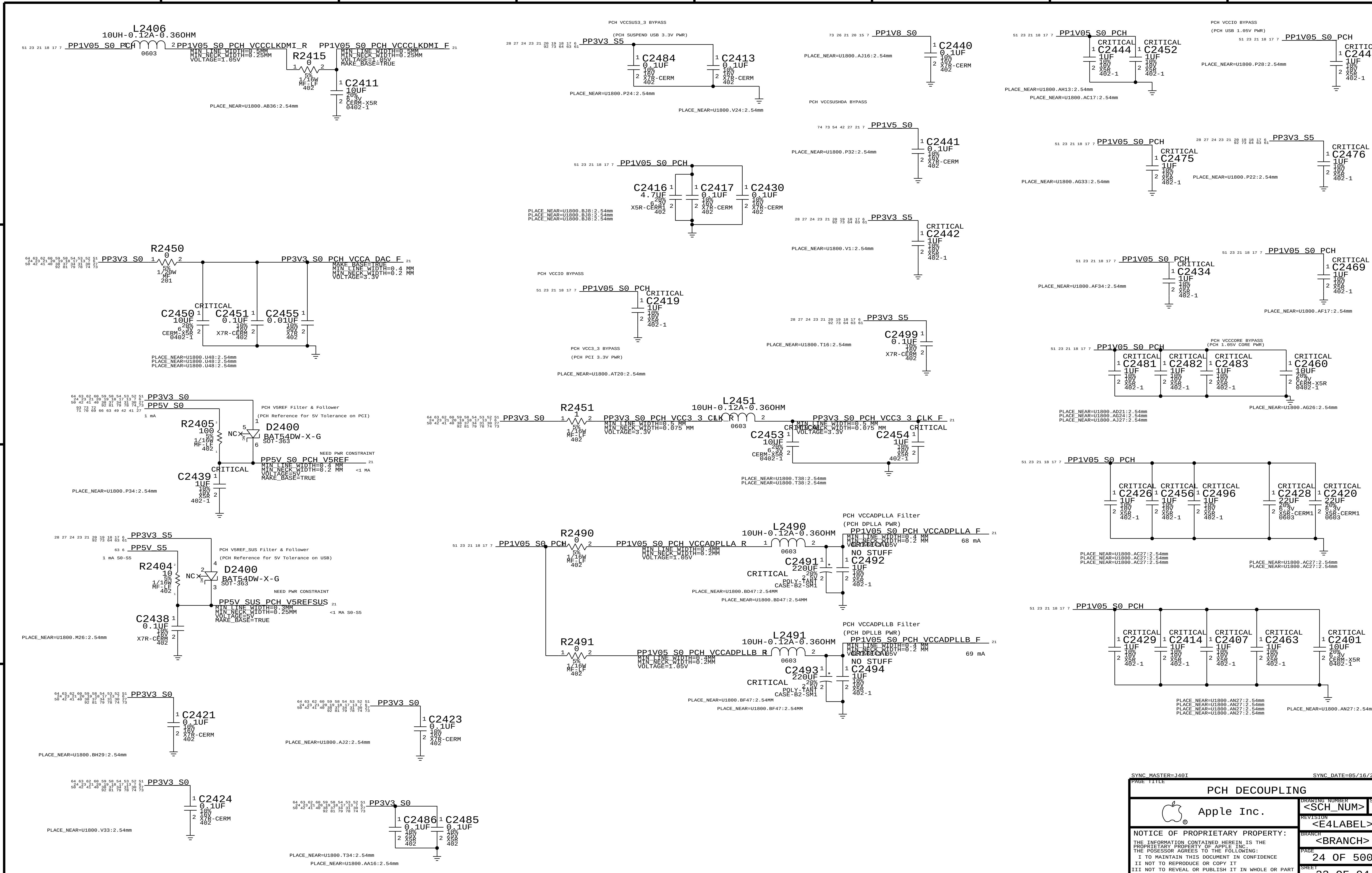
SHEET 22 OF 94



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

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PCH DECOUPLING

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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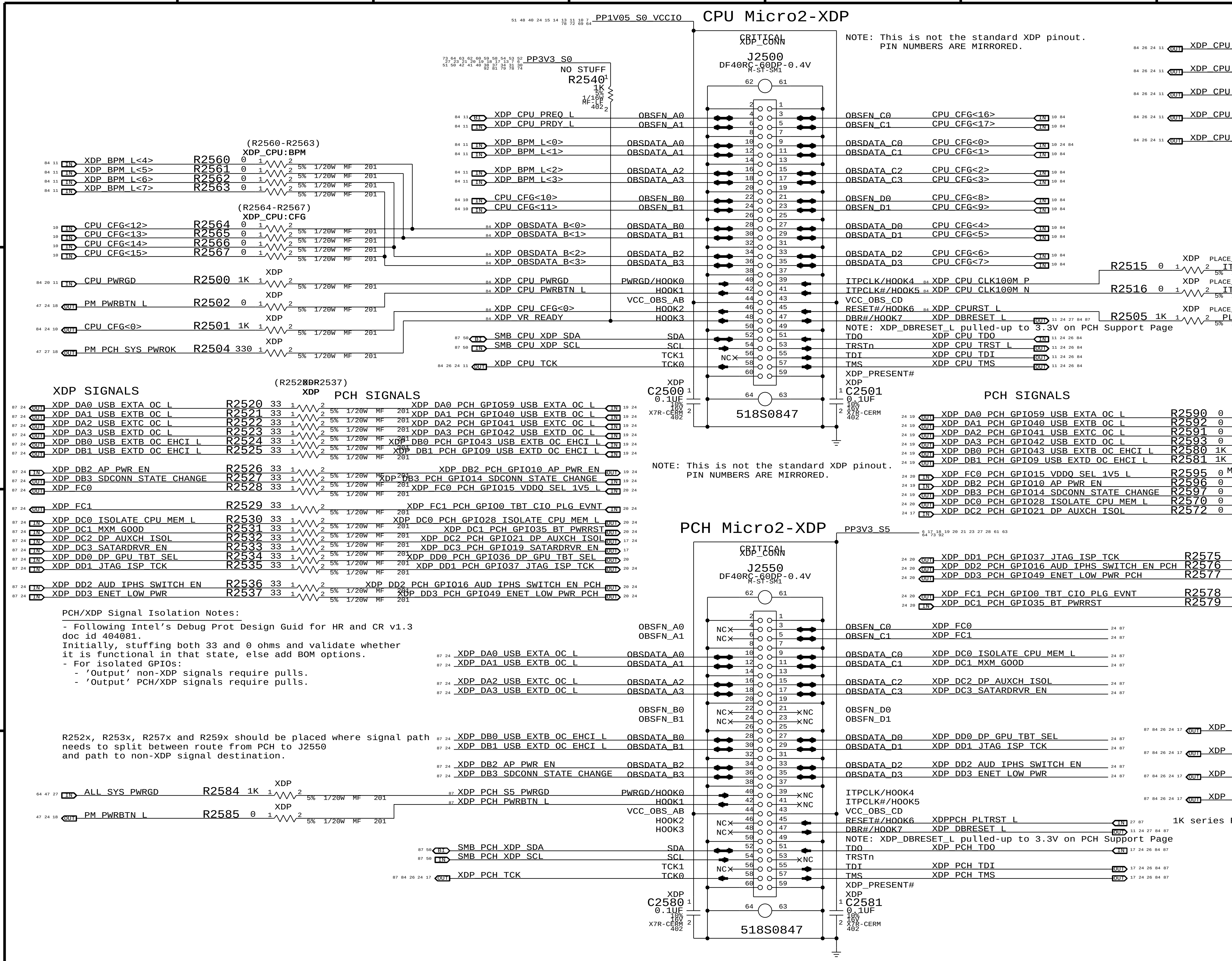

SYNC_MASTER=J30_MLB SYNC_DATE=07/27/2011

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XDP:CPU & PCH

Apple Inc.

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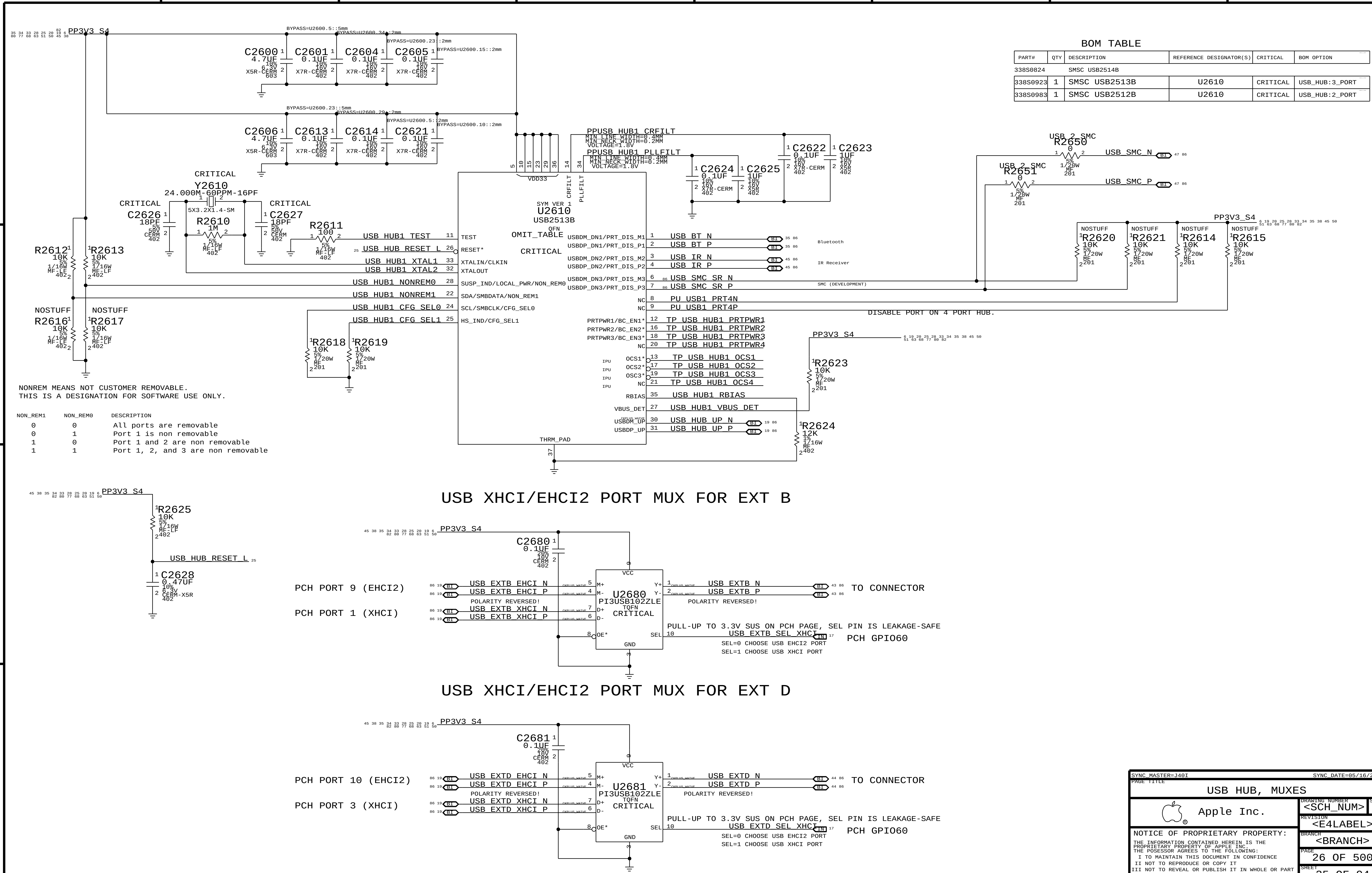

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0824		SMSC USB2514B			
338S0923	1	SMSC USB2513B	U2610	CRITICAL	USB_HUB:3_PORT
338S0983	1	SMSC USB2512B	U2610	CRITICAL	USB_HUB:2_PORT

NONREM MEANS NOT CUSTOMER REMOVABLE.
THIS IS A DESIGNATION FOR SOFTWARE USE ONLY.

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: USB HUB, MUXES

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

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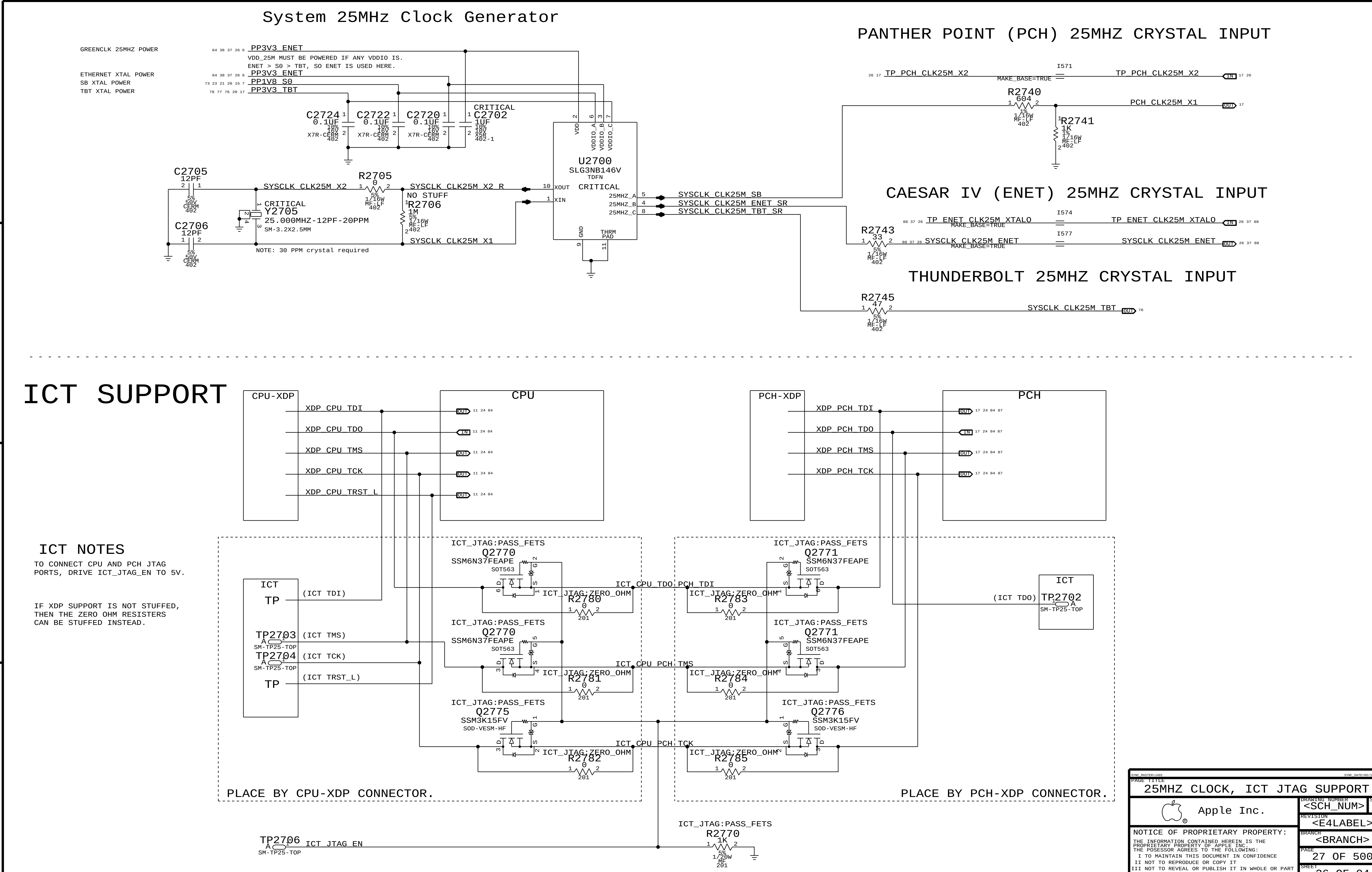

PAGE TITLE
25MHZ CLOCK, ICT JTAG SUPPORT

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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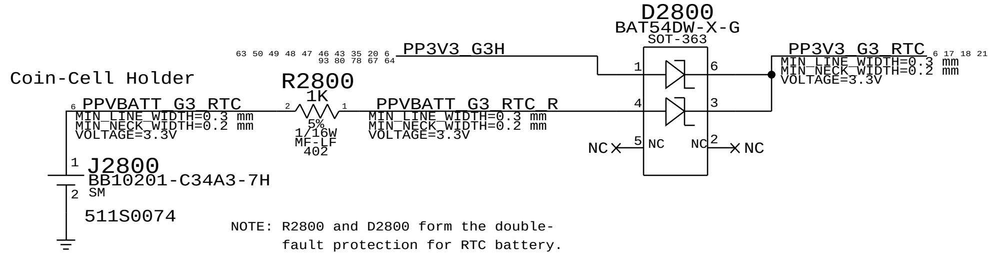
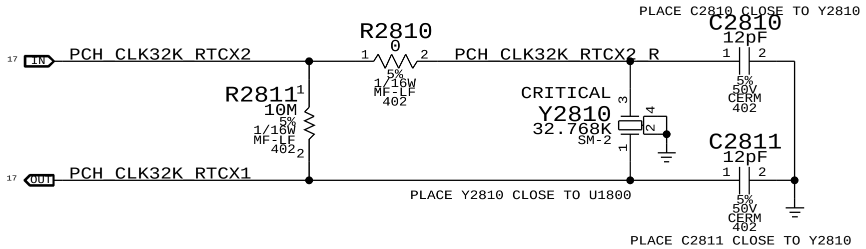
RTC Power Sources



NOTE: R2800 and D2800 form the double-fault protection for RTC battery.

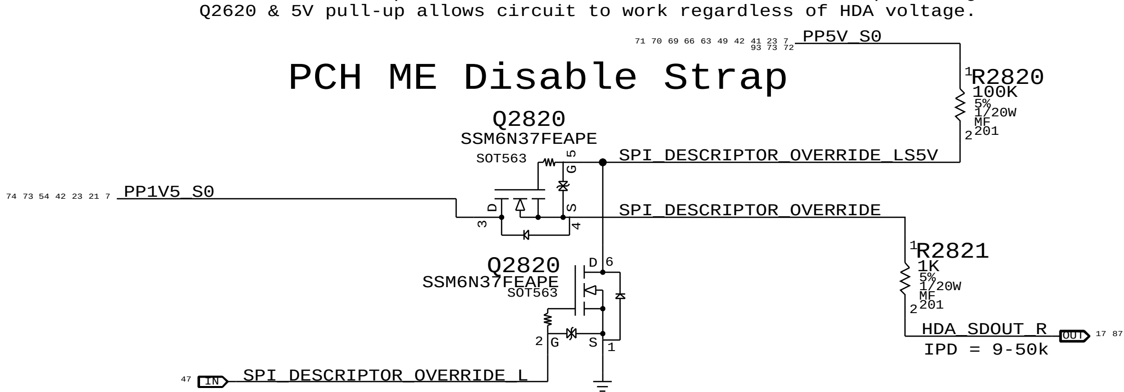
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
742-0062	1	BAT,COIN,LI/MAG,BR2032,3V,190MAH	BAT	CRITICAL	

PCH RTC Crystal

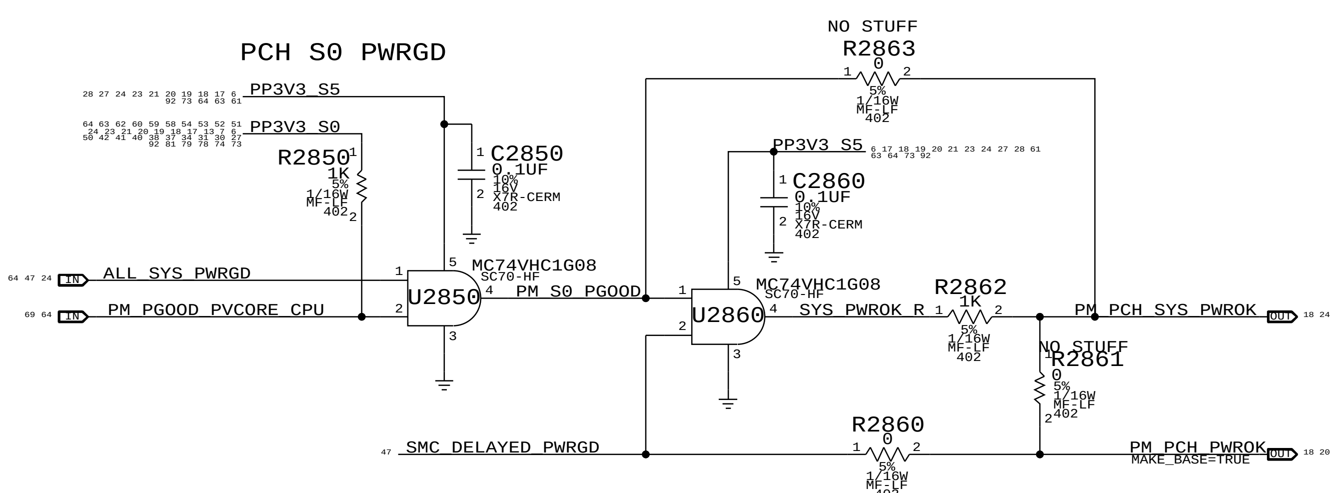


PCH uses HDA_SDO as a power-up strap. If low, ME functions normally.
If high, ME is disabled. This allows for full re-flashing of SPI ROM.
SMC controls strap enable to allow in-field control of strap setting.
Q2620 & 5V pull-up allows circuit to work regardless of HDA voltage.

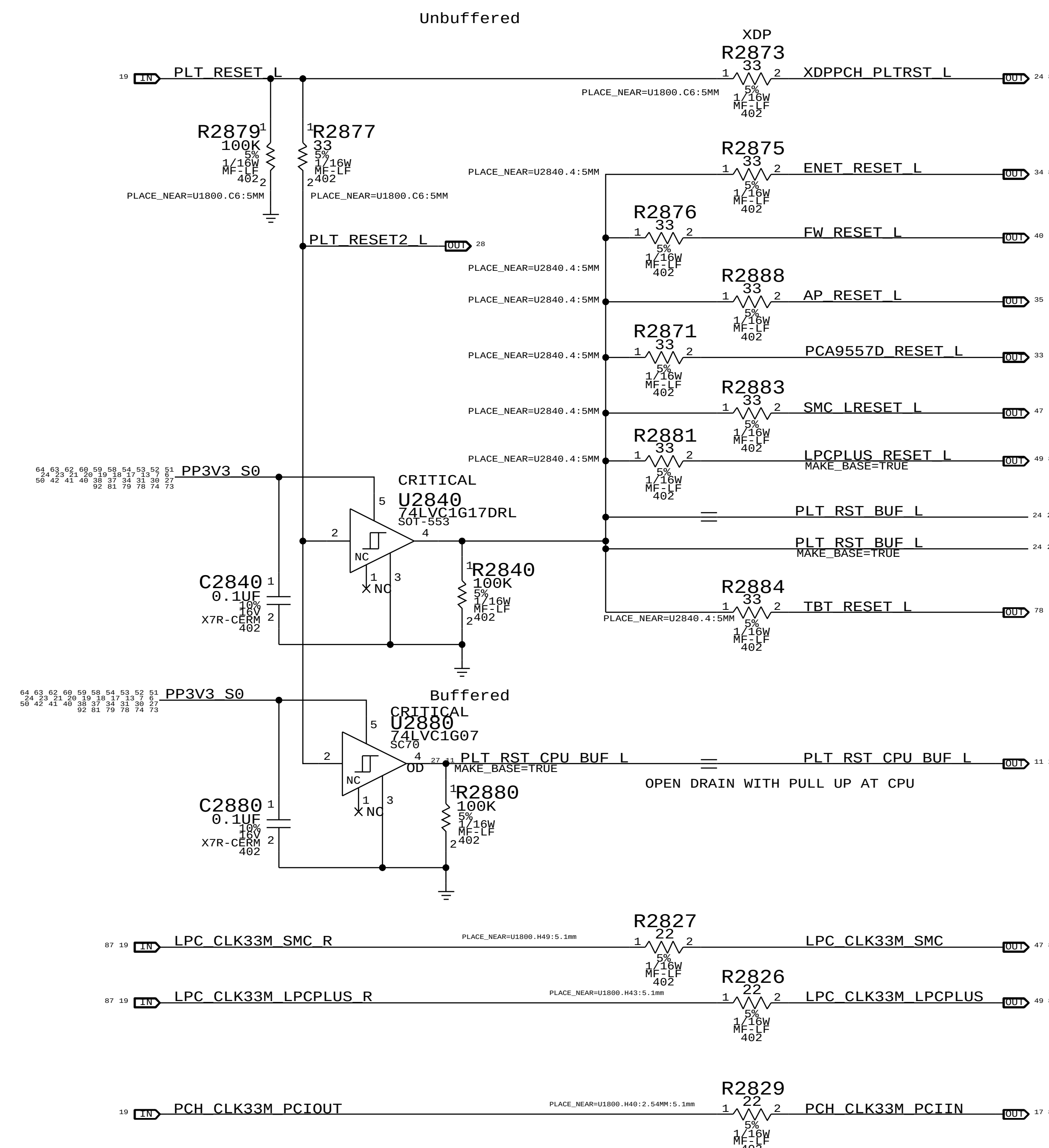
PCH ME Disable Strap



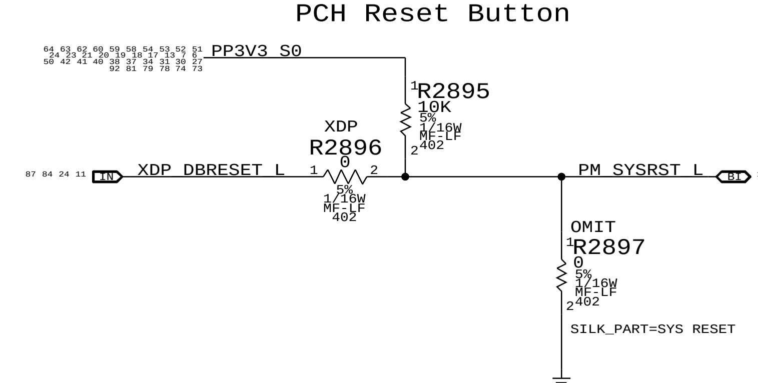
PCH S0 PWRGD



Platform Reset Connections



PCH Reset Button



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

RTC SUPPORT, RESETS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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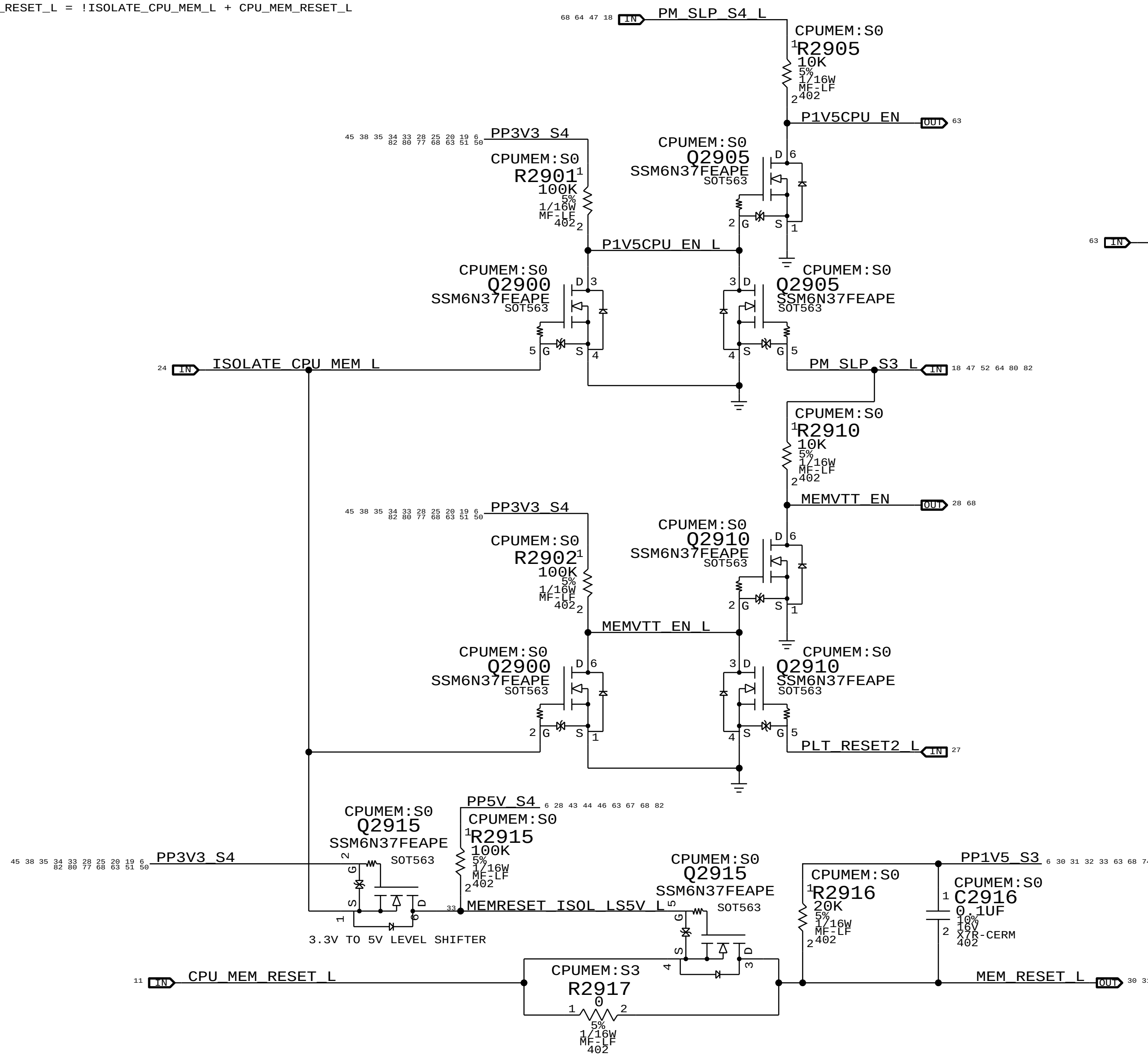
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well
as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

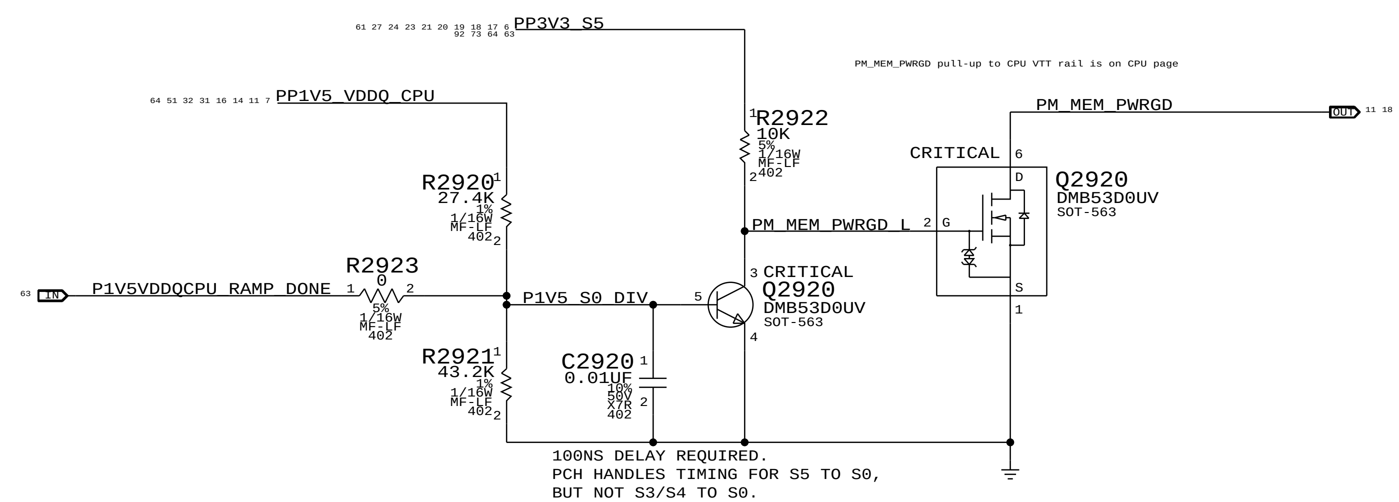


Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0 0	1	1	1	1	1	CPU_MEM_RESET_L	1	1
to 1	0	1	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
S3 3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
to 5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0 7	1	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

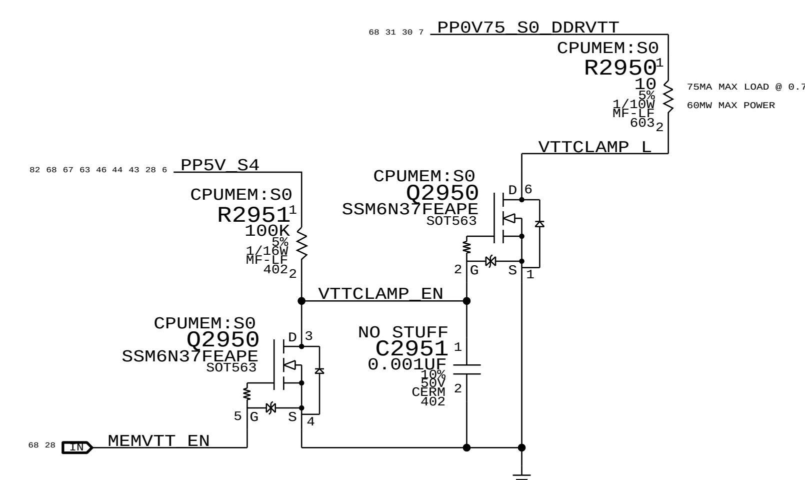
NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

CPU Memory S3 Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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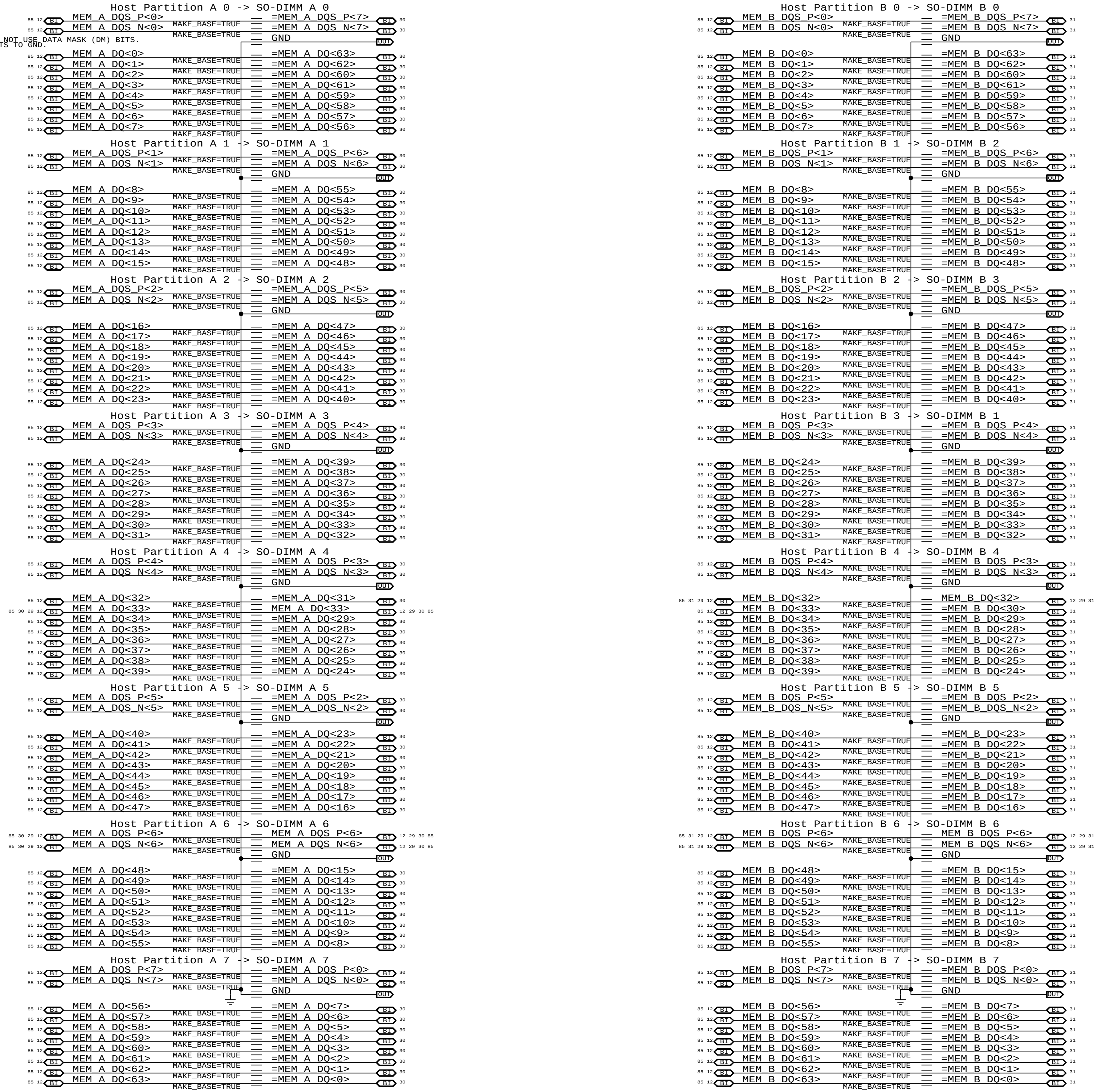
IVYBRIDGE CPU DOES NOT USE DATA MASK (DM) BITS.
PER PDG, TIE DM BITS TO GND.



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SO-DIMM Pinswaps

Apple Inc.

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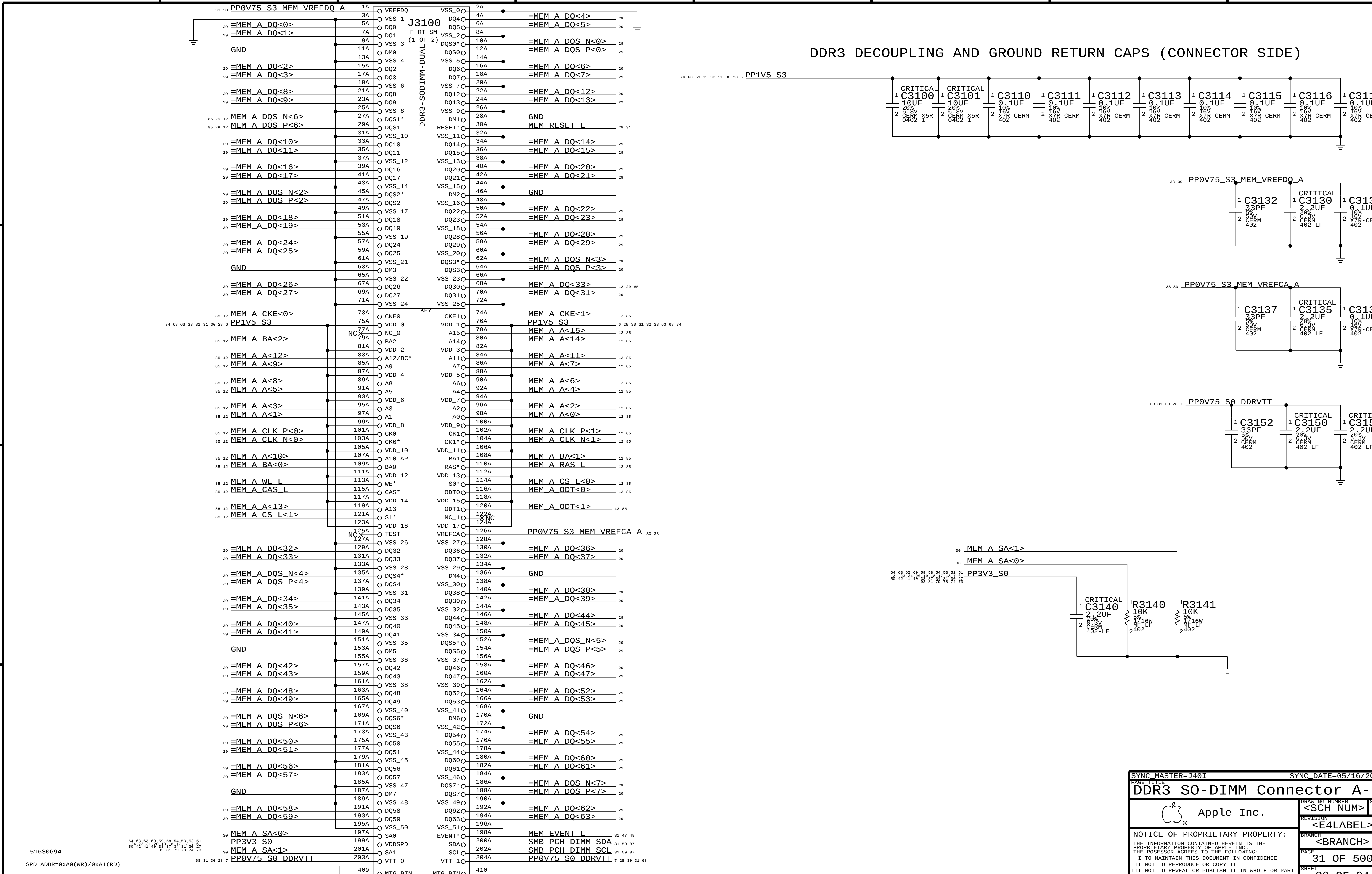

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

DDR3 SO-DIMM Connector A-1

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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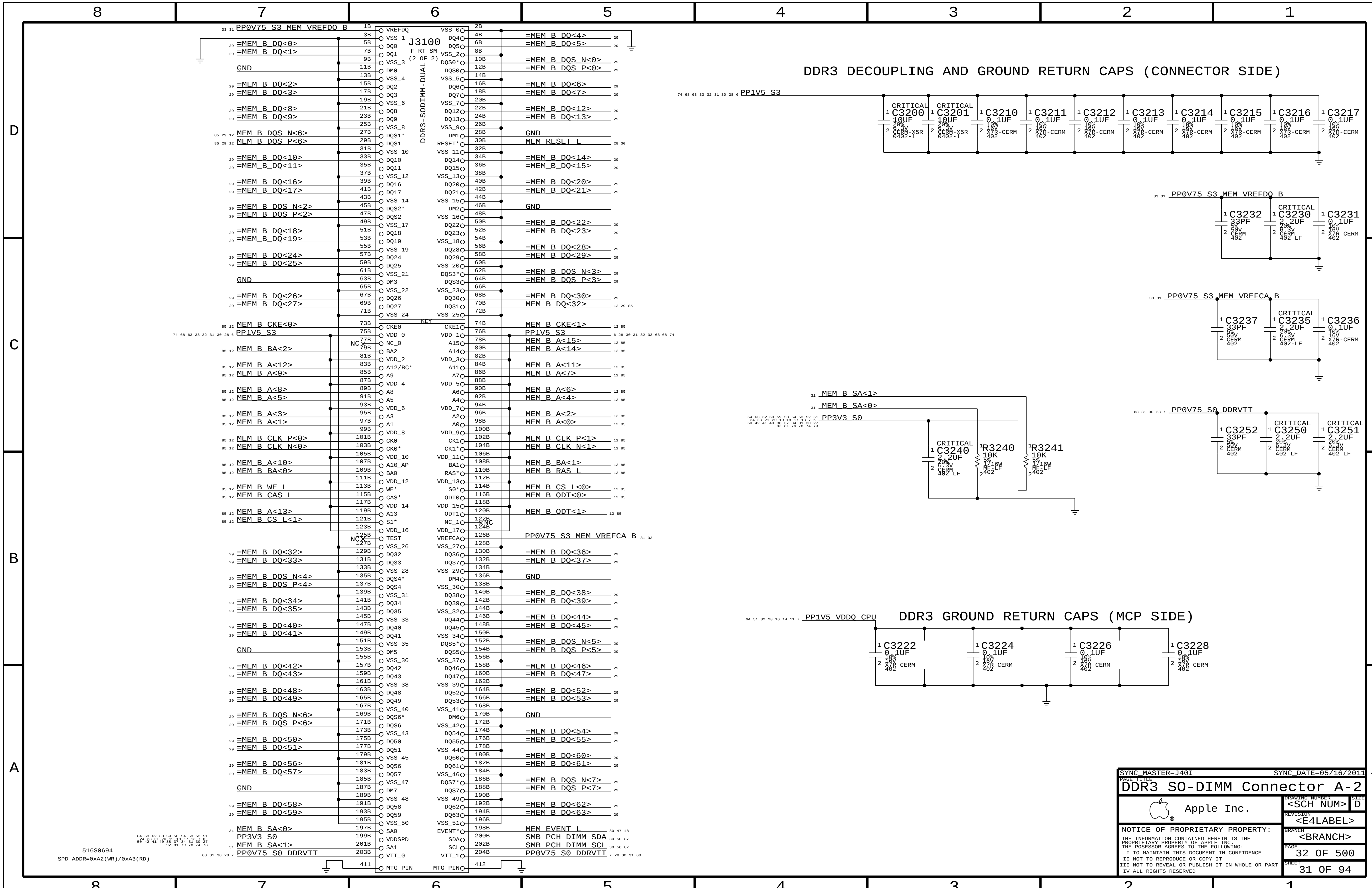
SHEET 30 OF 94



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

DDR3 SO-DIMM Connector A-2

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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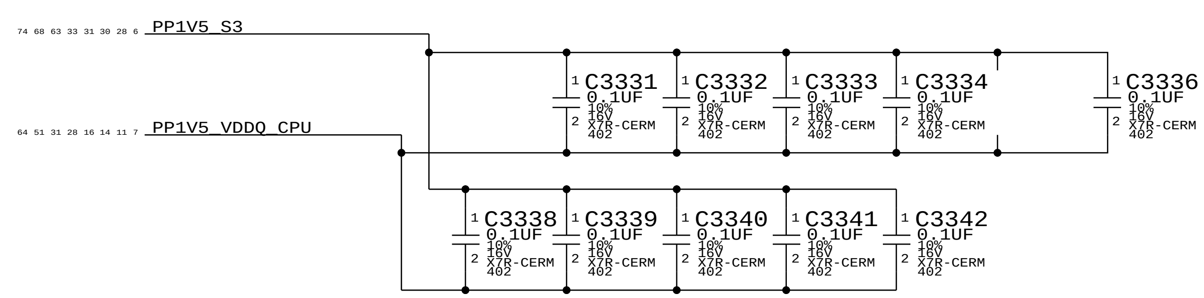


STICHING CAPS.

PLACE ACROSS PLANE SPLIT BETWEEN THESE RAIL
WHERE THE MEMORY ROUTES ARE.
EXTRAS CAN BE DELETED.

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

DDR3 Support

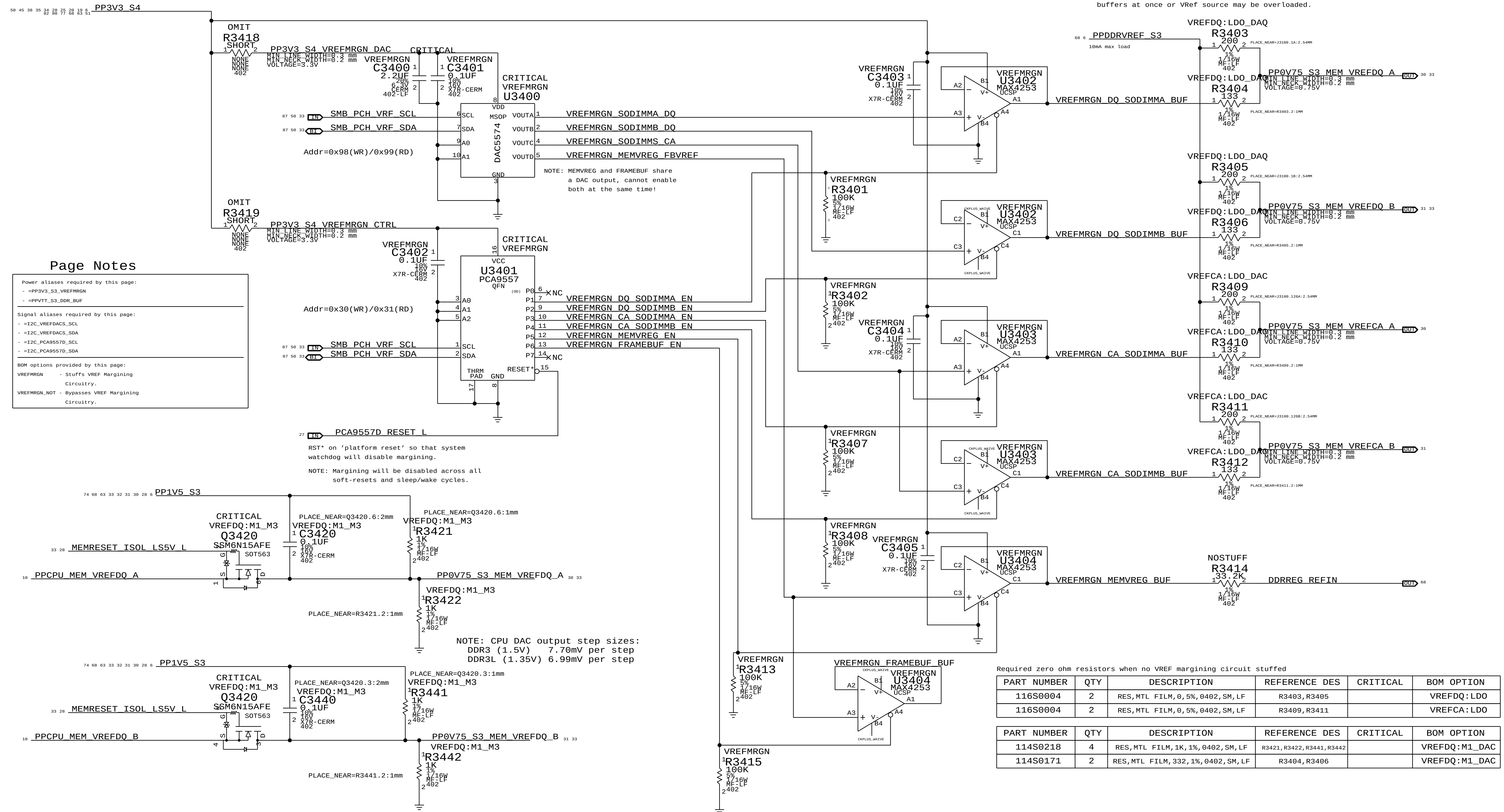
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

NOTE: Must not enable more than two SO-DIMM margining buffers at once or VRef source may be overloaded.



Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

VREFMRGN - Stuffs VREF Margining Circuitry.

VREFMRGN_NOT - Bypasses VREF Margining Circuitry.

Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES,MTL FILM,0,5%,0402,SM,LF	R3403,R3405		VREFDQ:LDO
116S0004	2	RES,MTL FILM,0,5%,0402,SM,LF	R3409,R3411		VREFCA:LDO

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0218	4	RES,MTL FILM,1K,1%,0402,SM,LF	R3421,R3422,R3441,R3442		VREFDQ:M1_DAC
114S0171	2	RES,MTL FILM,332,1%,0402,SM,LF	R3404,R3406		VREFDQ:M1_DAC

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value	0.75V (DAC: 0x3A)				1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)				1.998V - 1.002V (+/- 498mV)	1.056V - 1.442V (+/- 180mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)				0.000V - 1.501V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:	+3.4mA - -3.4mA (- = sourced)				+33uA - -33uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:	7.69mV / step @ output				8.59mV / step @ output	1.51mV / step @ output

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

FSB/DDR3/FRAMEBUF Vref Margining

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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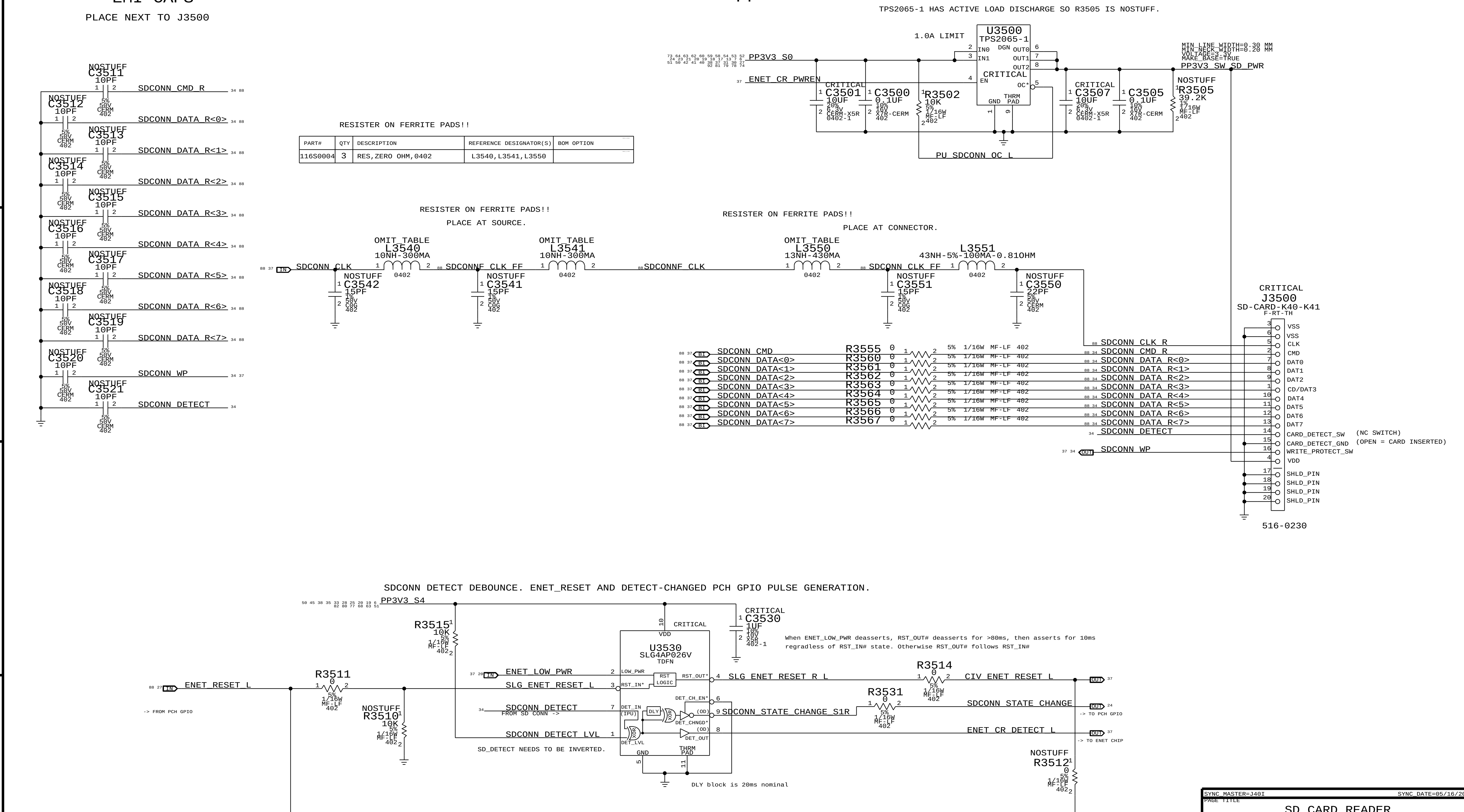
Caesar IV Support

EMI CAPS

PLACE NEXT TO J3500

RESISTER ON FERRITE PADS!!

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0004	3	RES,ZERO OHM,0402	L3540,L3541,L3550	



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: SD CARD READER

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

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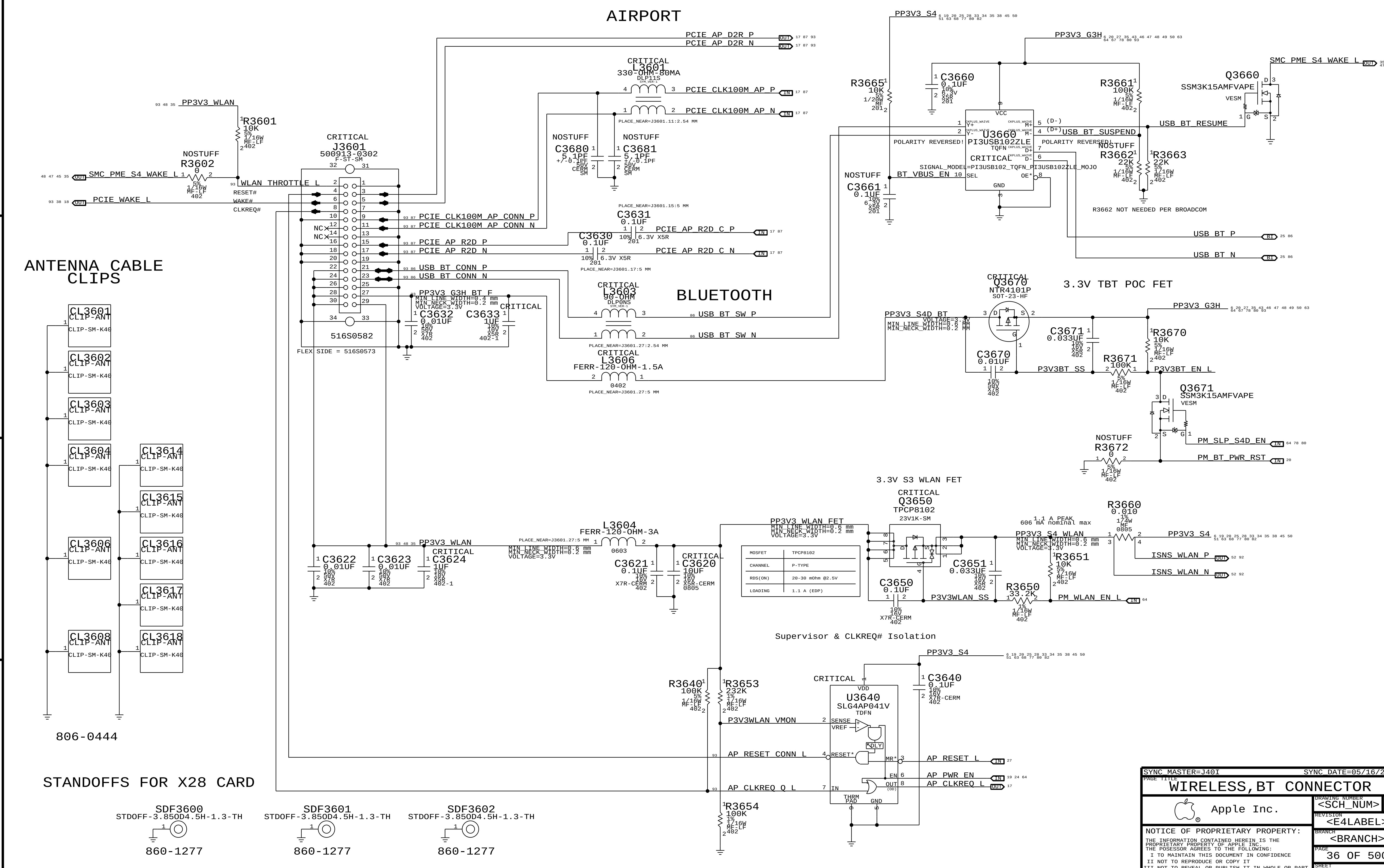

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: WIRELESS,BT CONNECTOR

DRAWING NUMBER: <SCH_NUM> SIZE: D

Apple Inc.

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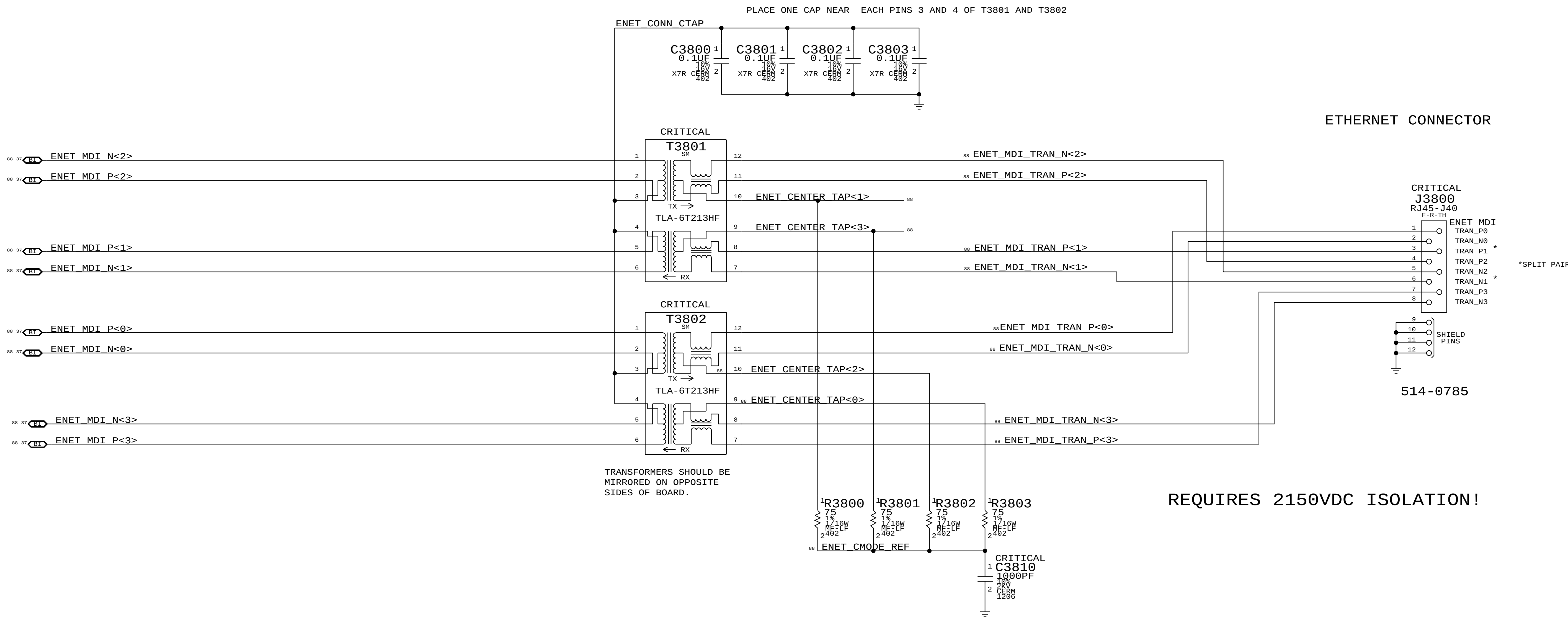

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PAGE TITLE

ETHERNET CONNECTOR

Apple Inc.

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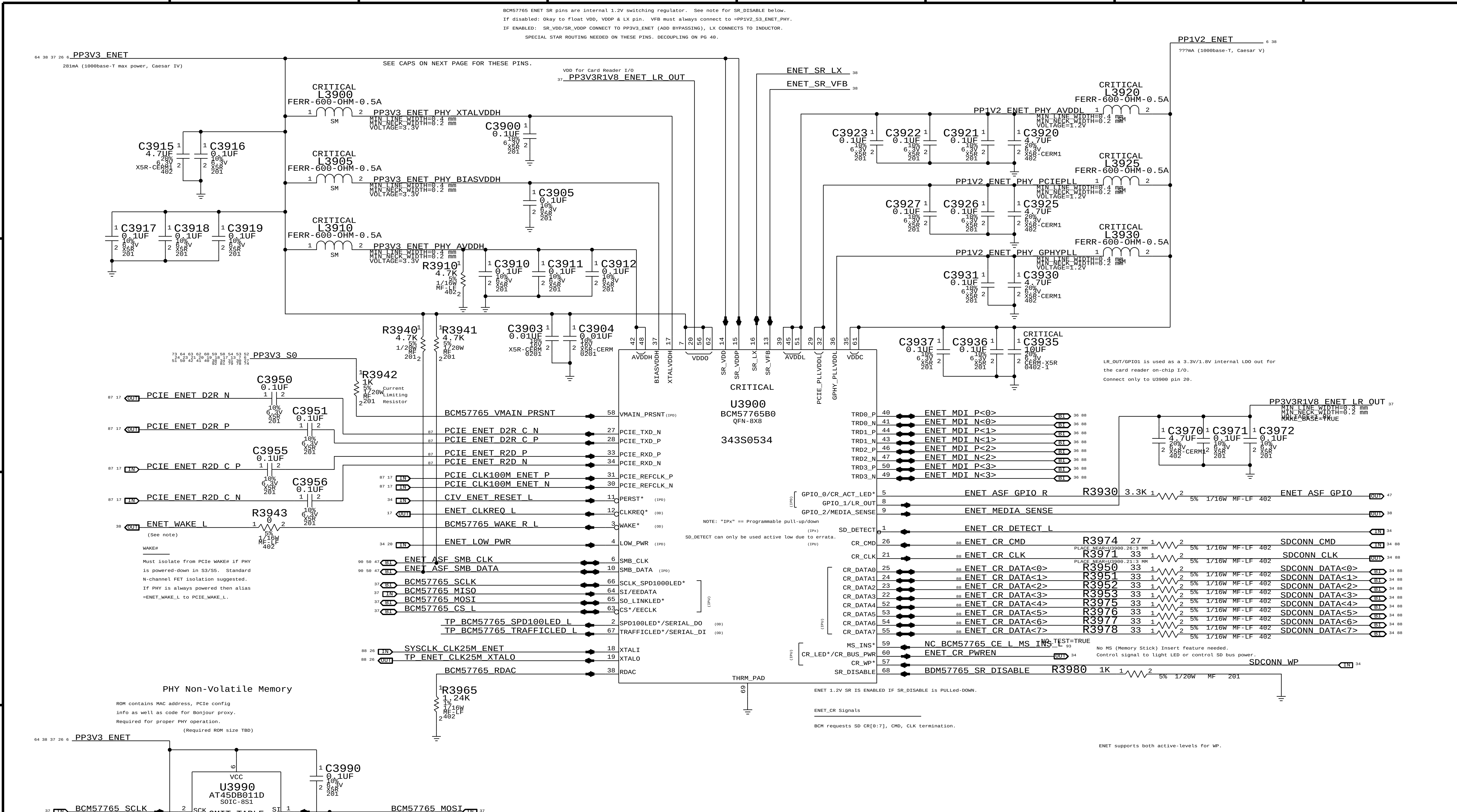

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PAGE TITLE: ETHERNET PHY (CAESAR IV)

Apple Inc.

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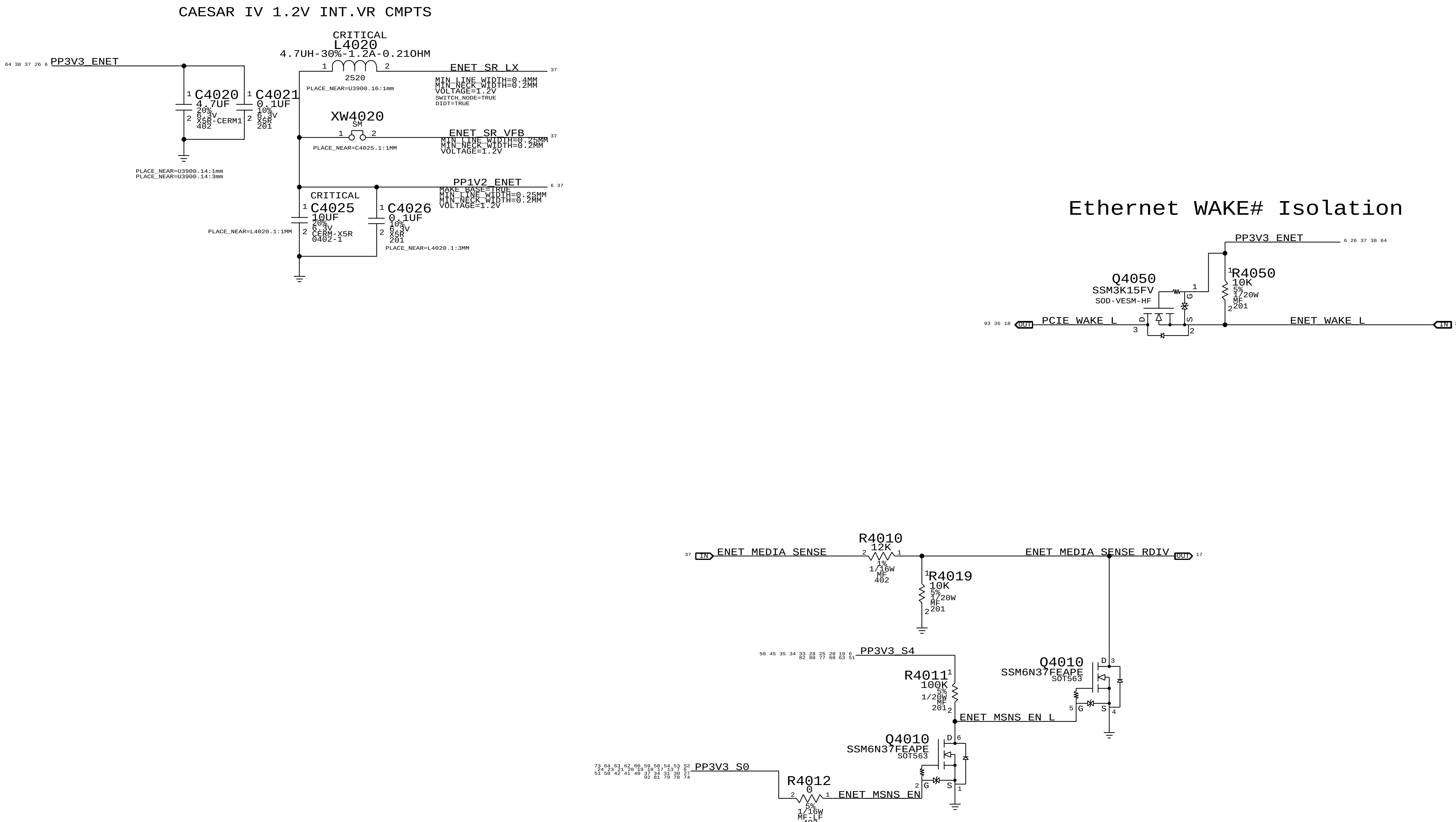

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

ETHERNET SUPPORT

Apple Inc.

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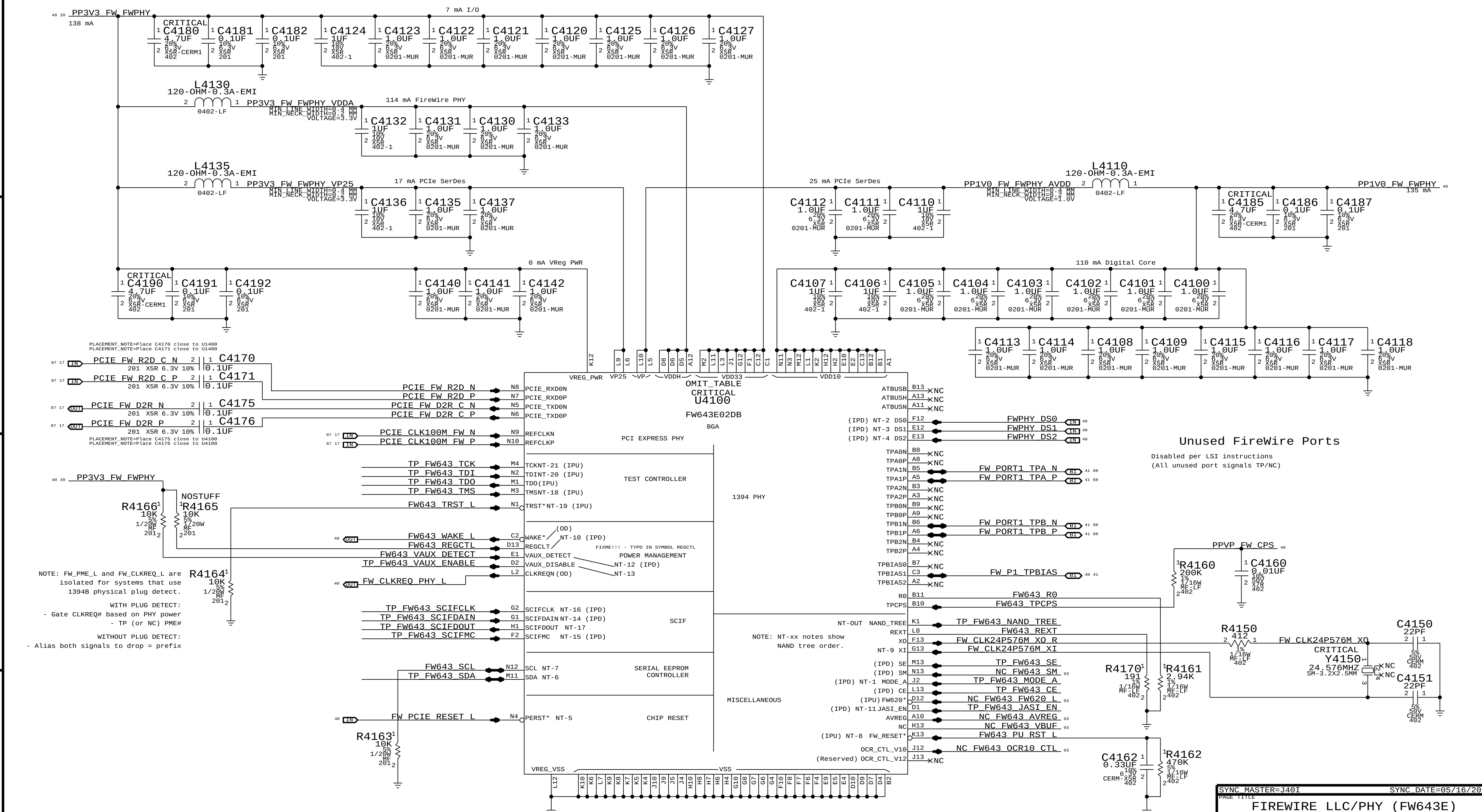
SHEET 38 OF 94



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
338S1078	1	IC,FW643E,1394B,PHY/OHCI/PCIE	U4100	CRITICAL	

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

FIREWIRE LLC/PHY (FW643E)

Apple Inc.

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REVISION <E4LABEL>

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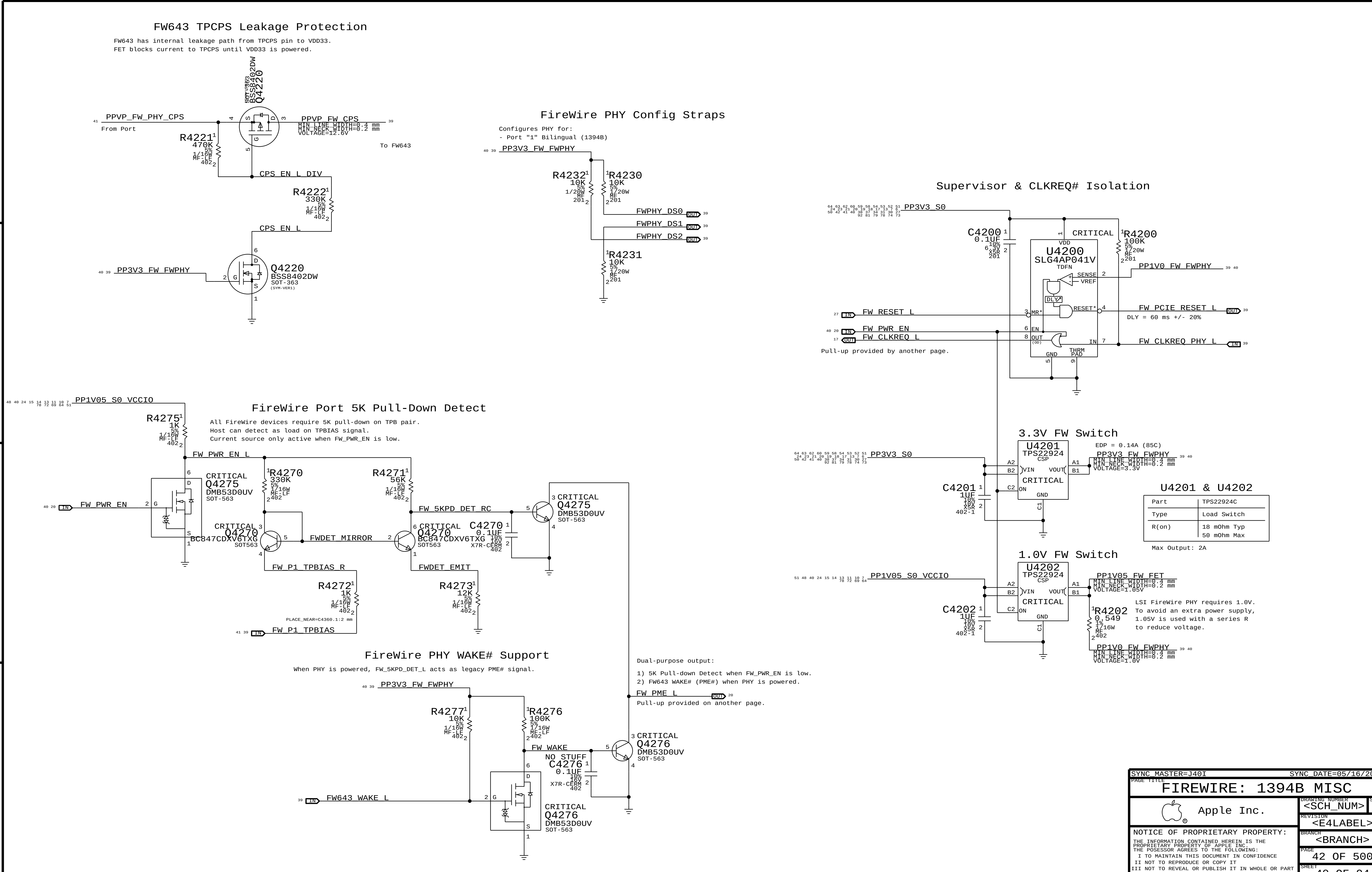

Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

Max Output: 2A

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

FIREWIRE: 1394B MISC

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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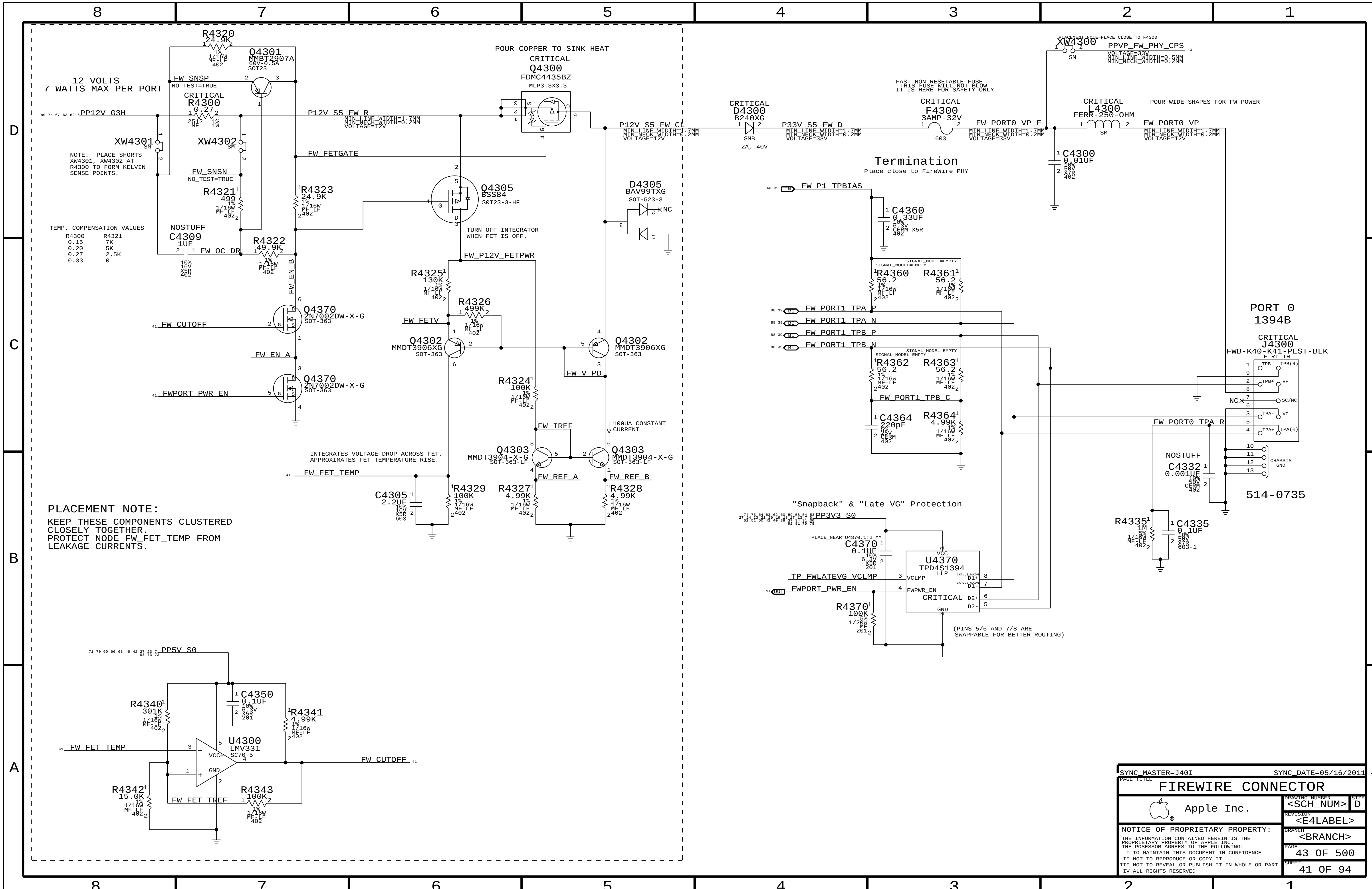

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PAGE TITLE
FIREWIRE CONNECTOR

Apple Inc.

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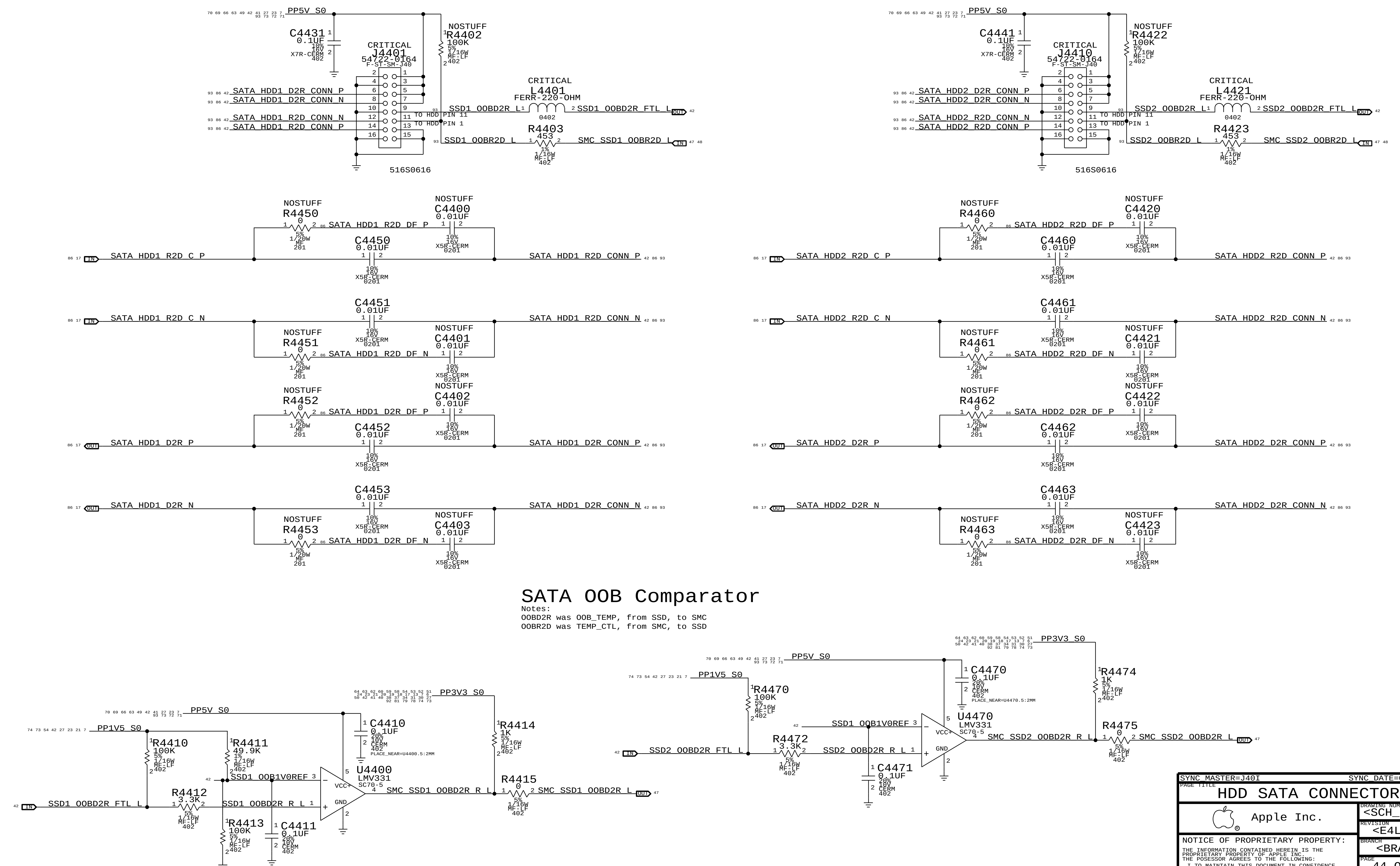
SATA CONNECTOR (HDD1)

SATA CONNECTOR (HDD2)



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HDD SATA CONNECTORS

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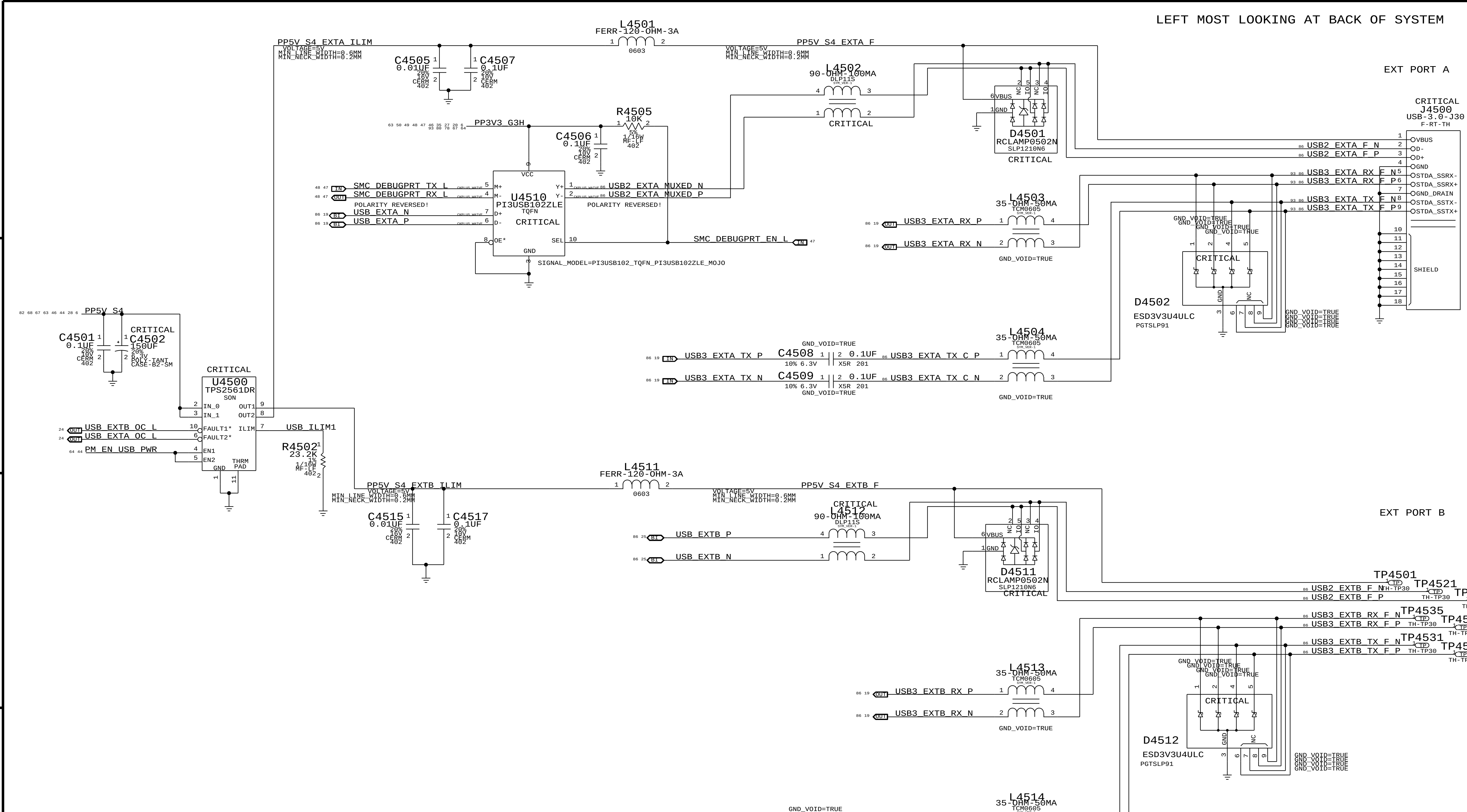

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USB EXTERNAL PORTS A & B

Apple Inc.

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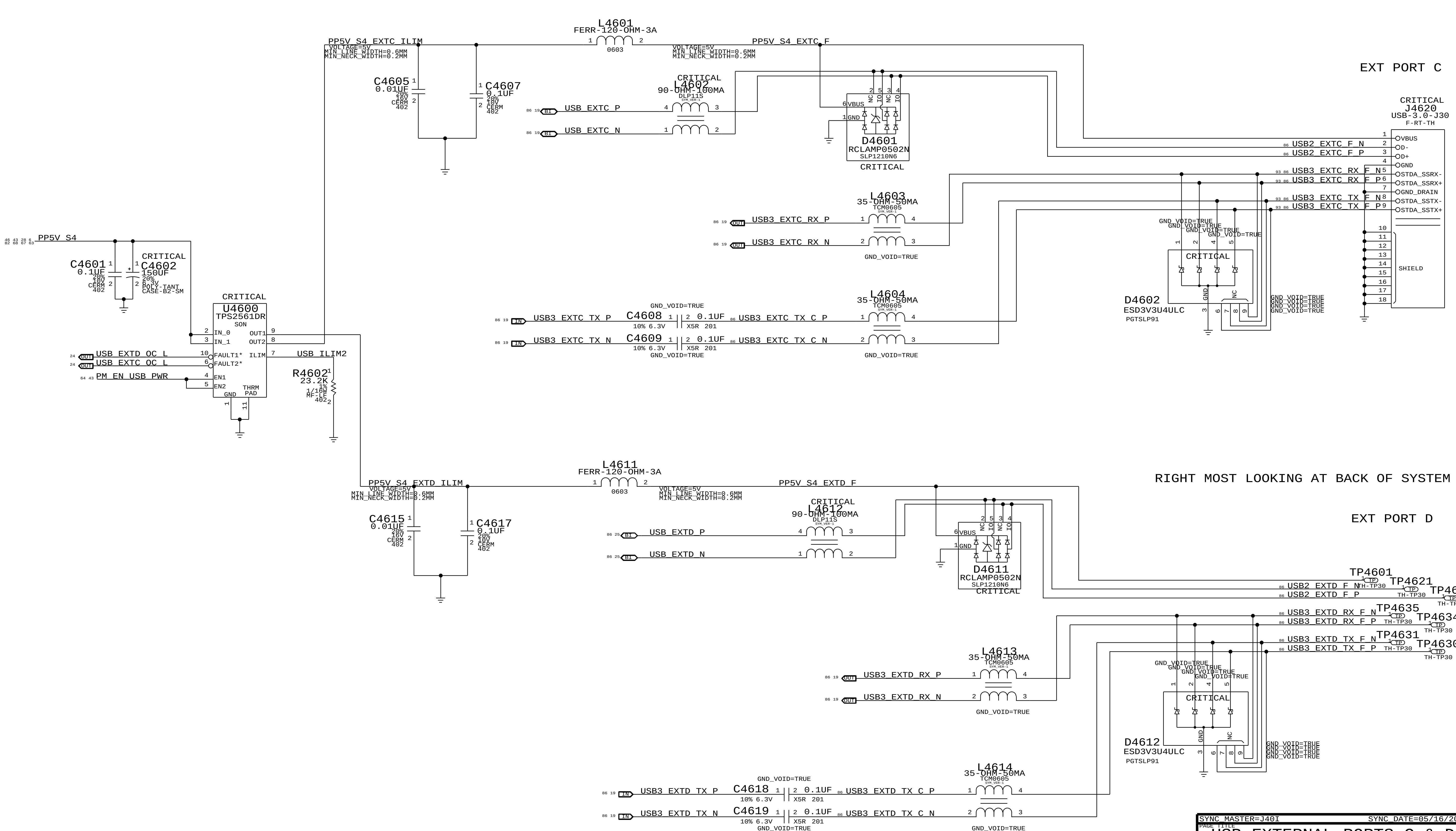

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PAGE TITLE

USB EXTERNAL PORTS C & D

Apple Inc.

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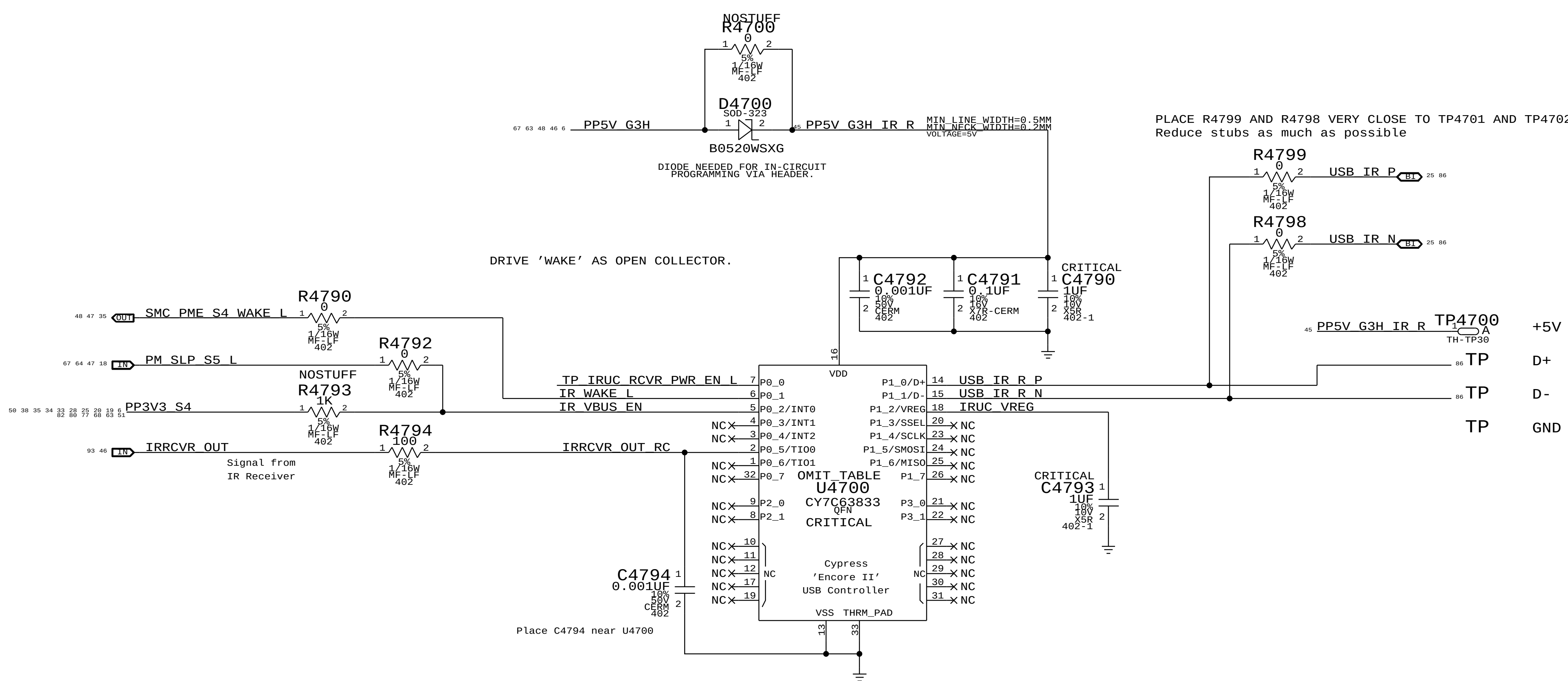
IR Controller



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PAGE TITLE: IR CONTROLLER		
Apple Inc.	DRAWING NUMBER: <SCH_NUM>	SIZE: D
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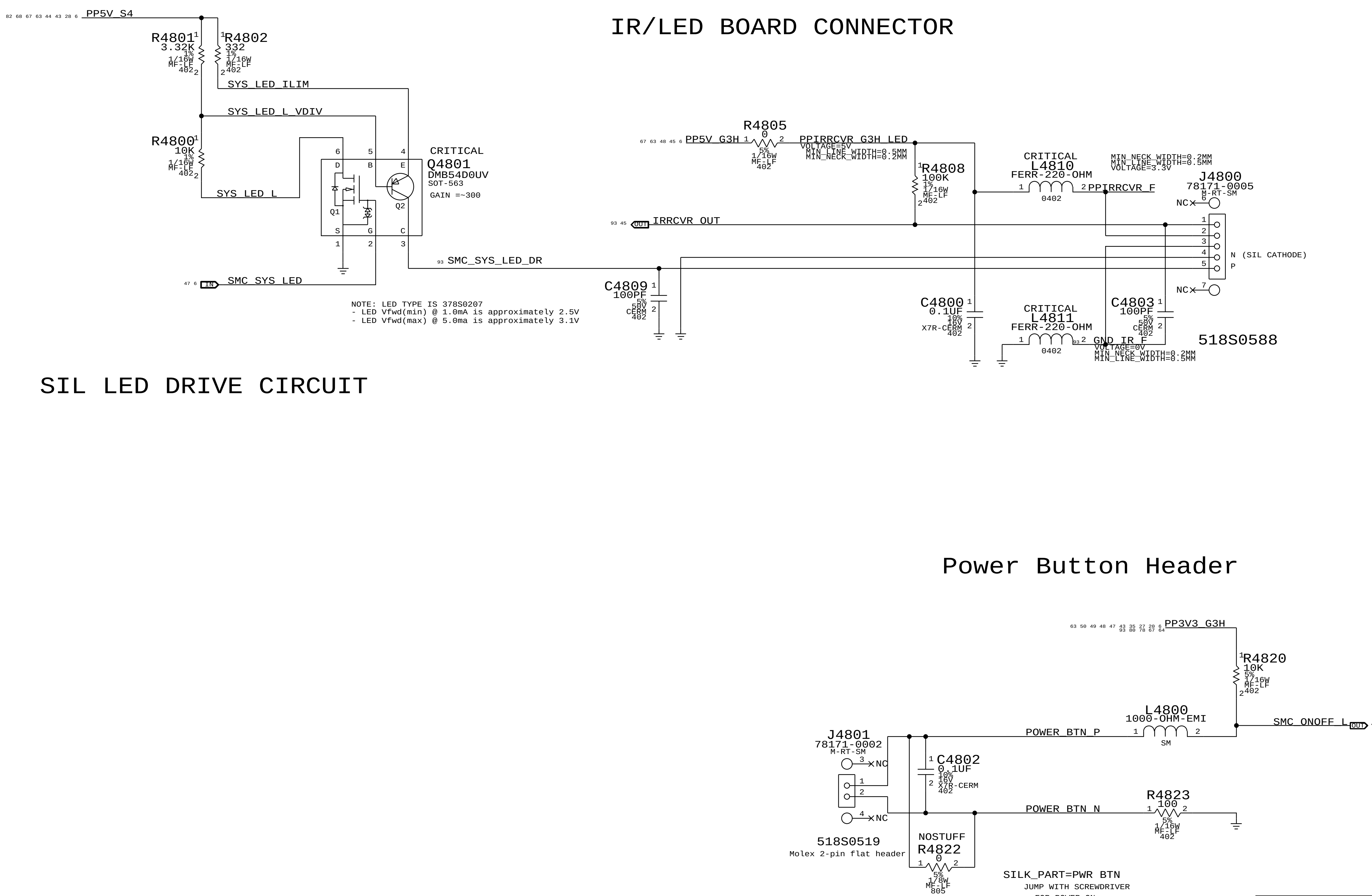
NOTICE OF PROPRIETARY PROPERTY:	BRANCH: <BRANCH>	
	PAGE: 47 OF 500	
	SHEET: 45 OF 94	



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

IR/LED, PWR BTTN

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

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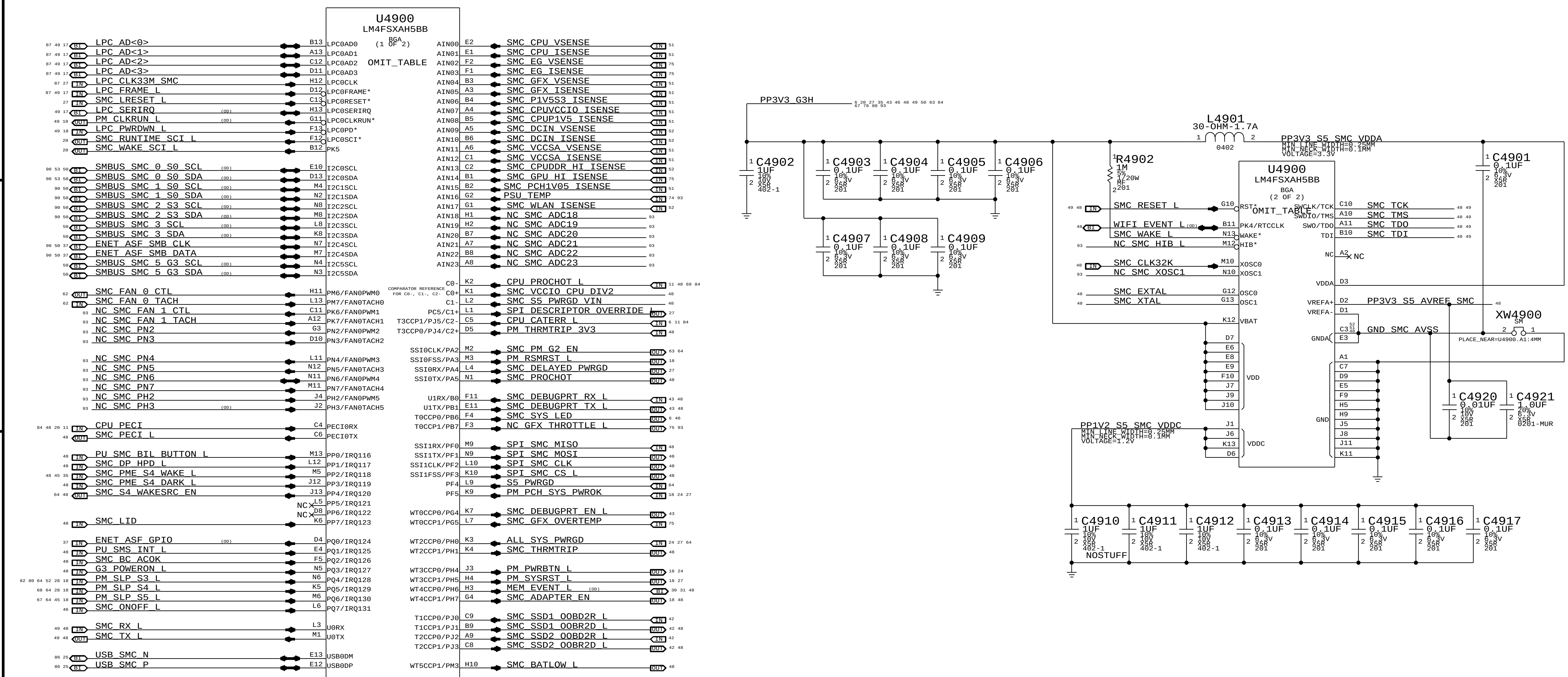
SHEET 46 OF 94



NOTE:
SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

NOTE:
Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

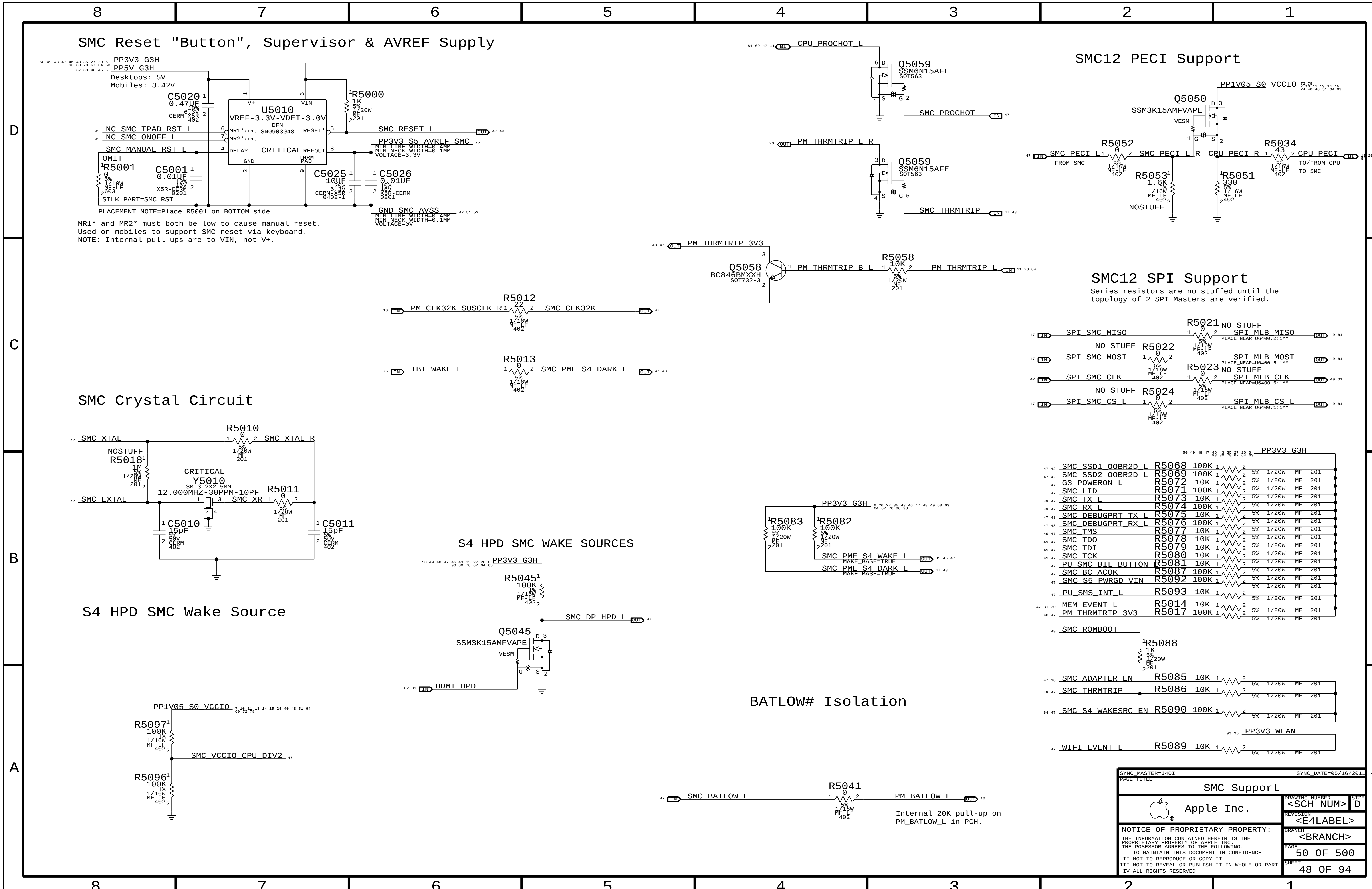
SMC

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>


SMC Reset "Button", Supervisor & AVREF Supply
PP3V3_G3H
PP5V_G3H
Desktops: 5V
Mobiles: 3.42V
C5020 0.47UF 10% 6.3V CERM-X5R 402
U5010 VREF-3.3V-VDET-3.0V DFN SN0903048
V+ VIN
MR1* (IPU)
MR2* (IPU)
RESET*
DELAY
CRITICAL
REFOUT
GND
THRM PAD
NC_SMC_TPAD_RST_L
NC_SMC_ONOFF_L
SMC_MANUAL_RST_L
R5000 1K 5% 1/20W MF 201
SMC_RESET_L
PP3V3_S5_AVREF_SMC
MIN_LINE_WIDTH=0.4MM
MIN_NECK_WIDTH=0.1MM
VOLTAGE=3.3V
OMIT
R5001 0 5% 1/10W MF-LF 603
SILK_PART=SMC_RST
C5001 0.01UF 10% 16V X5R-CERM 0201
C5025 10UF 20% 6.3V CERM-X5R 0402-1
C5026 0.01UF 10% 16V X5R-CERM 0201
GND_SMC_AVSS
MIN_LINE_WIDTH=0.4MM
MIN_NECK_WIDTH=0.1MM
VOLTAGE=0V
PLACEMENT_NOTE=Place R5001 on BOTTOM side
MR1* and MR2* must both be low to cause manual reset.
Used on mobiles to support SMC reset via keyboard.
NOTE: Internal pull-ups are to VIN, not V+.
CPU_PROCHOT_L
Q5059 SSM6N15AFE SOT563
SMC_PROCHOT
PM_THRMTRIP_L_R
Q5059 SSM6N15AFE SOT563
SMC_THRMTRIP
PM_THRMTRIP_3V3
Q5058 BC846BMXXH SOT732-3
PM_THRMTRIP_B_L
R5058 10K 5% 1/20W MF 201
PM_THRMTRIP_L
SMC12 PECI Support
PP1V05_S0_VCCIO
Q5050 SSM3K15AMFVAPE
VESM
SMC_PECI_L
FROM SMC
R5052 0 1/16W MF-LF 402
SMC_PECI_L_R
CPU_PECI_R
R5034 43 5% 1/16W MF-LF 402
CPU_PECI
TO/FROM CPU
TO SMC
R5053 1.6K 1/16W MF-LF 402
NOSTUFF
R5051 330 5% 1/16W MF-LF 402
SMC12 SPI Support
Series resistors are no stuffed until the
topology of 2 SPI Masters are verified.
SPI_SMC_MISO
R5021 0 NO STUFF
SPI_MLB_MISO
PLACE_NEAR=U6400.2:1MM
NO STUFF R5022 0
SPI_SMC_MOSI
SPI_MLB_MOSI
PLACE_NEAR=U6400.5:1MM
R5023 0 NO STUFF
SPI_SMC_CLK
SPI_MLB_CLK
PLACE_NEAR=U6400.6:1MM
NO STUFF R5024 0
SPI_SMC_CS_L
SPI_MLB_CS_L
PLACE_NEAR=U6400.1:1MM
R5012 22 5% 1/16W MF-LF 402
PM_CLK32K_SUSCLK_R
SMC_CLK32K
R5013 0 5% 1/16W MF-LF 402
TBT_WAKE_L
SMC_PME_S4_DARK_L
SMC Crystal Circuit
SMC_XTAL
R5010 0 5% 1/20W MF 201
SMC_XTAL_R
NOSTUFF
R5018 1M 5% 1/20W MF 201
CRITICAL
Y5010 SM-3.2X2.5MM 12.000MHZ-30PPM-10PF
R5011 0 5% 1/20W MF 201
SMC_EXTAL
SMC_XR
C5010 15pF 5% 50V CERM 402
C5011 15pF 5% 50V CERM 402
S4 HPD SMC Wake Source
S4 HPD SMC WAKE SOURCES
PP3V3_G3H
R5045 100K 1% 1/16W MF-LF 402
SMC_DP_HPD_L
Q5045 SSM3K15AMFVAPE
VESM
HDMI_HPD
PP3V3_G3H
R5083 100K 5% 1/20W MF 201
R5082 100K 5% 1/20W MF 201
SMC_PME_S4_WAKE_L
MAKE_BASE=TRUE
SMC_PME_S4_DARK_L
MAKE_BASE=TRUE
PP3V3_G3H
SMC_SSD1_OOBR2D_L R5068 100K
SMC_SSD2_OOBR2D_L R5069 100K
G3_POWERON_L R5072 10K
SMC_LID R5071 100K
SMC_TX_L R5073 10K
SMC_RX_L R5074 10K
SMC_DEBUGPRT_TX_L R5075 10K
SMC_DEBUGPRT_RX_L R5076 100K
SMC_TMS R5077 10K
SMC_TDO R5078 10K
SMC_TDI R5079 10K
SMC_TCK R5080 10K
PU_SMC_BIL_BUTTON_L R5081 10K
SMC_BC_ACOK R5087 100K
SMC_S5_PWRGD_VIN R5092 100K
PU_SMS_INT_L R5093 10K
MEM_EVENT_L R5014 10K
PM_THRMTRIP_3V3 R5017 100K
5% 1/20W MF 201
SMC_ROMBOOT
R5088 1K 5% 1/20W MF 201
SMC_ADAPTER_EN R5085 10K
SMC_THRMTRIP R5086 10K
SMC_S4_WAKESRC_EN R5090 100K
PP3V3_WLAN
WIFI_EVENT_L R5089 10K
PP1V05_S0_VCCIO
R5097 100K 1% 1/16W MF-LF 402
SMC_VCCIO_CPU_DIV2
R5096 100K 1% 1/16W MF-LF 402
BATLOW# Isolation
SMC_BATLOW_L
R5041 0 5% 1/16W MF-LF 402
PM_BATLOW_L
Internal 20K pull-up on
PM_BATLOW_L in PCH.
SYNC_MASTER=J40I
SYNC_DATE=05/16/2011
PAGE TITLE
SMC Support
Apple Inc.
DRAWING NUMBER <SCH_NUM>
SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>
PAGE 50 OF 500
SHEET 48 OF 94
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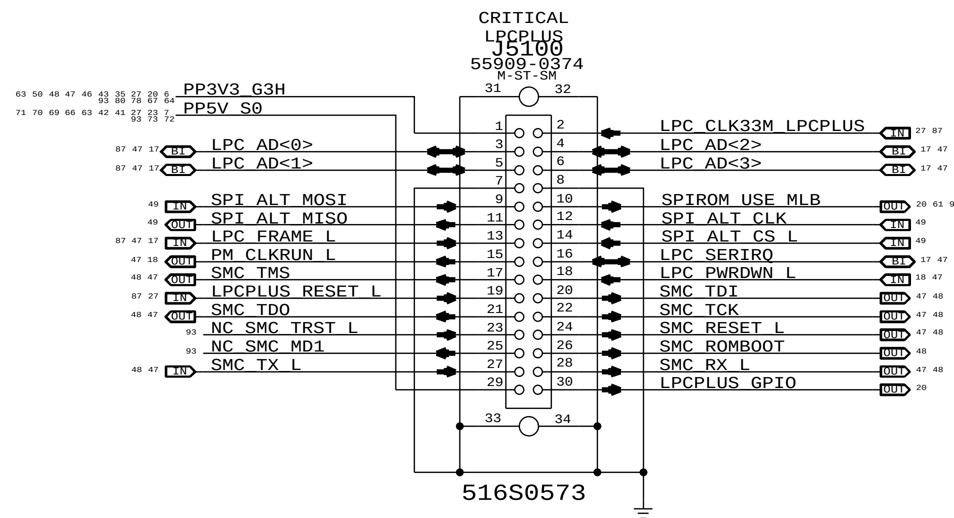
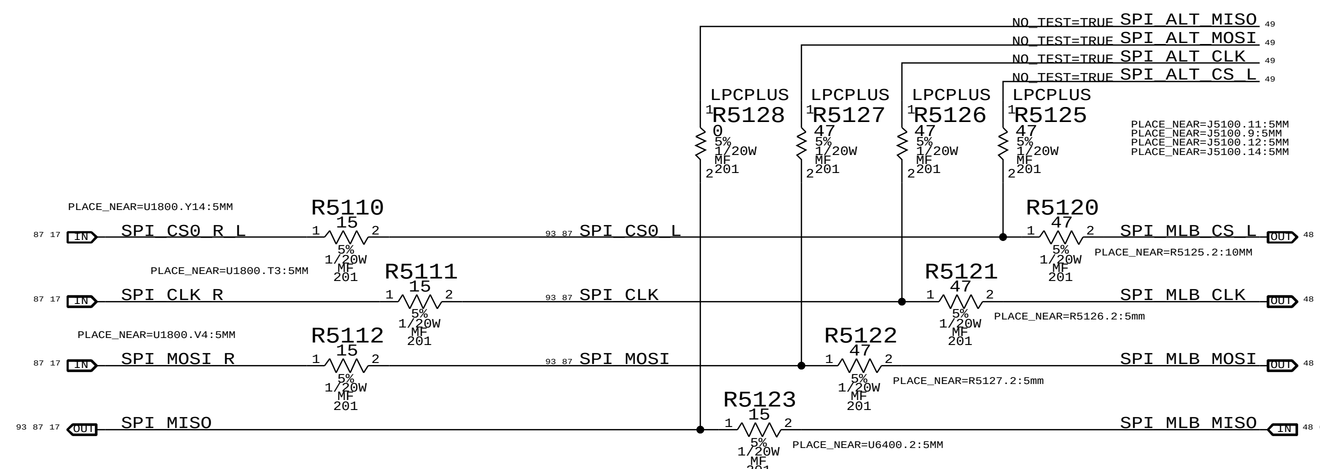
MATT CARD
LPC+SPI Connector



SPI Bus Series Termination



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

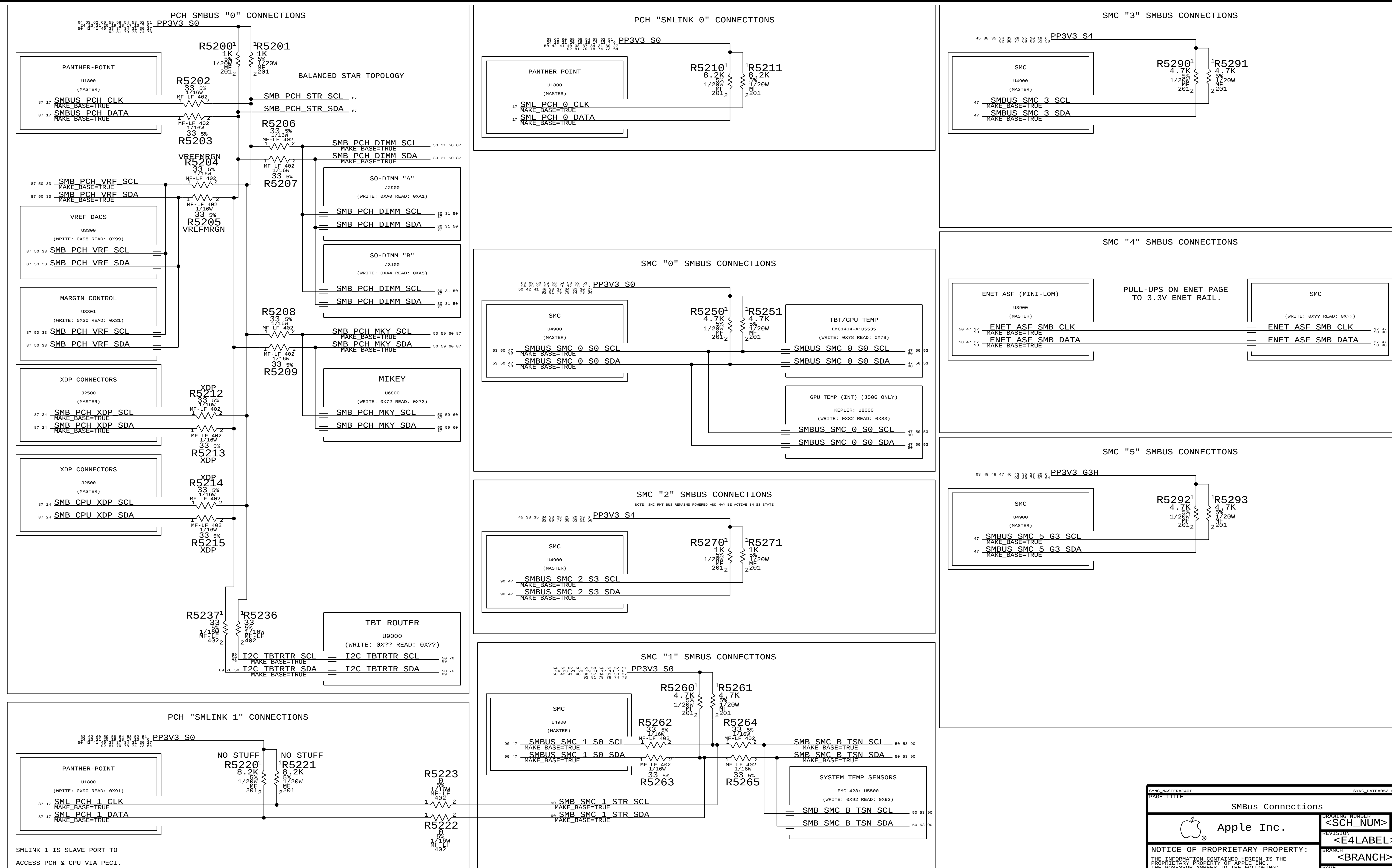
PAGE TITLE: LPC+ Debug Connector

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

SMBus Connections

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

REVISION <E4LABEL>

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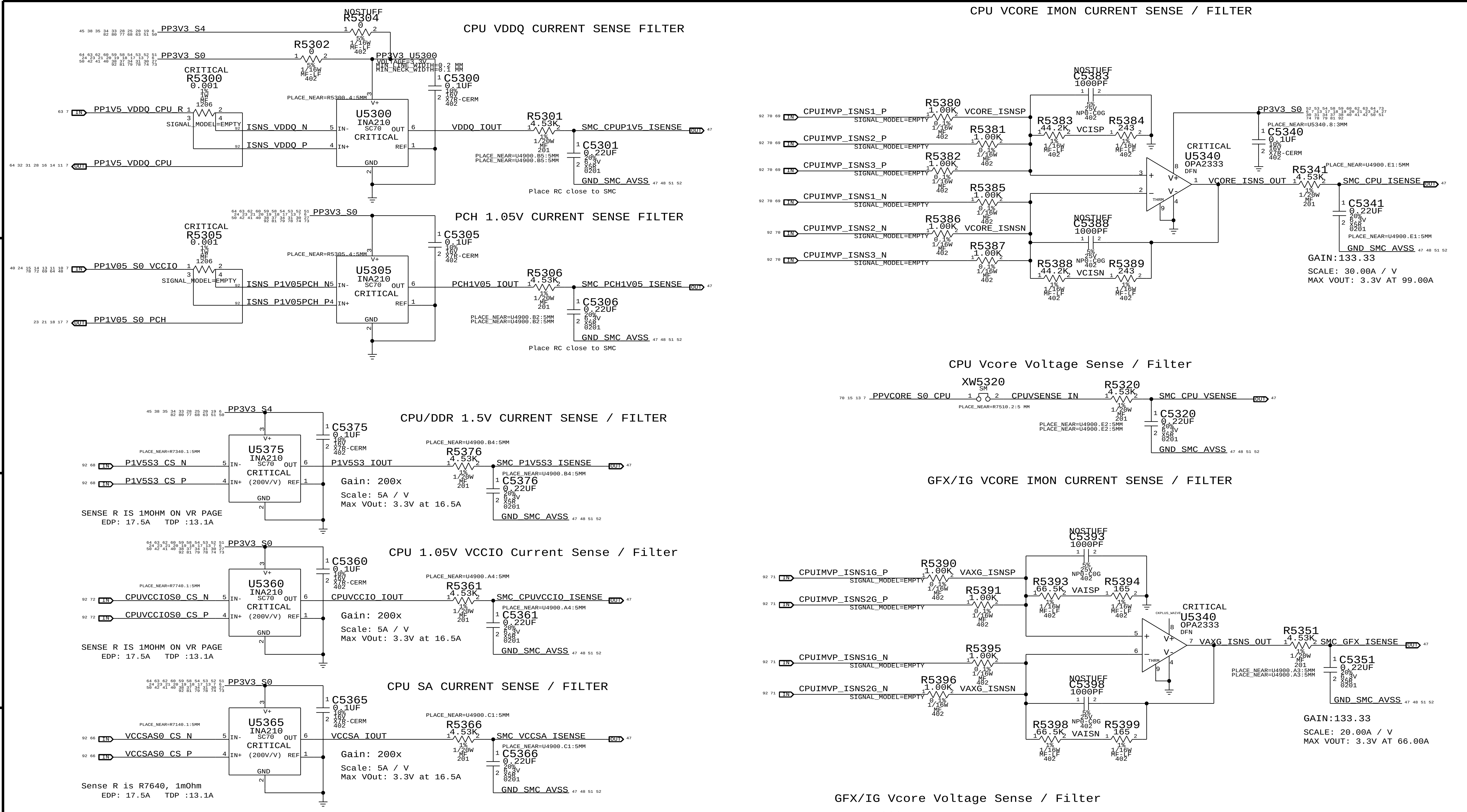

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

Voltage & Load Side Current Sensing

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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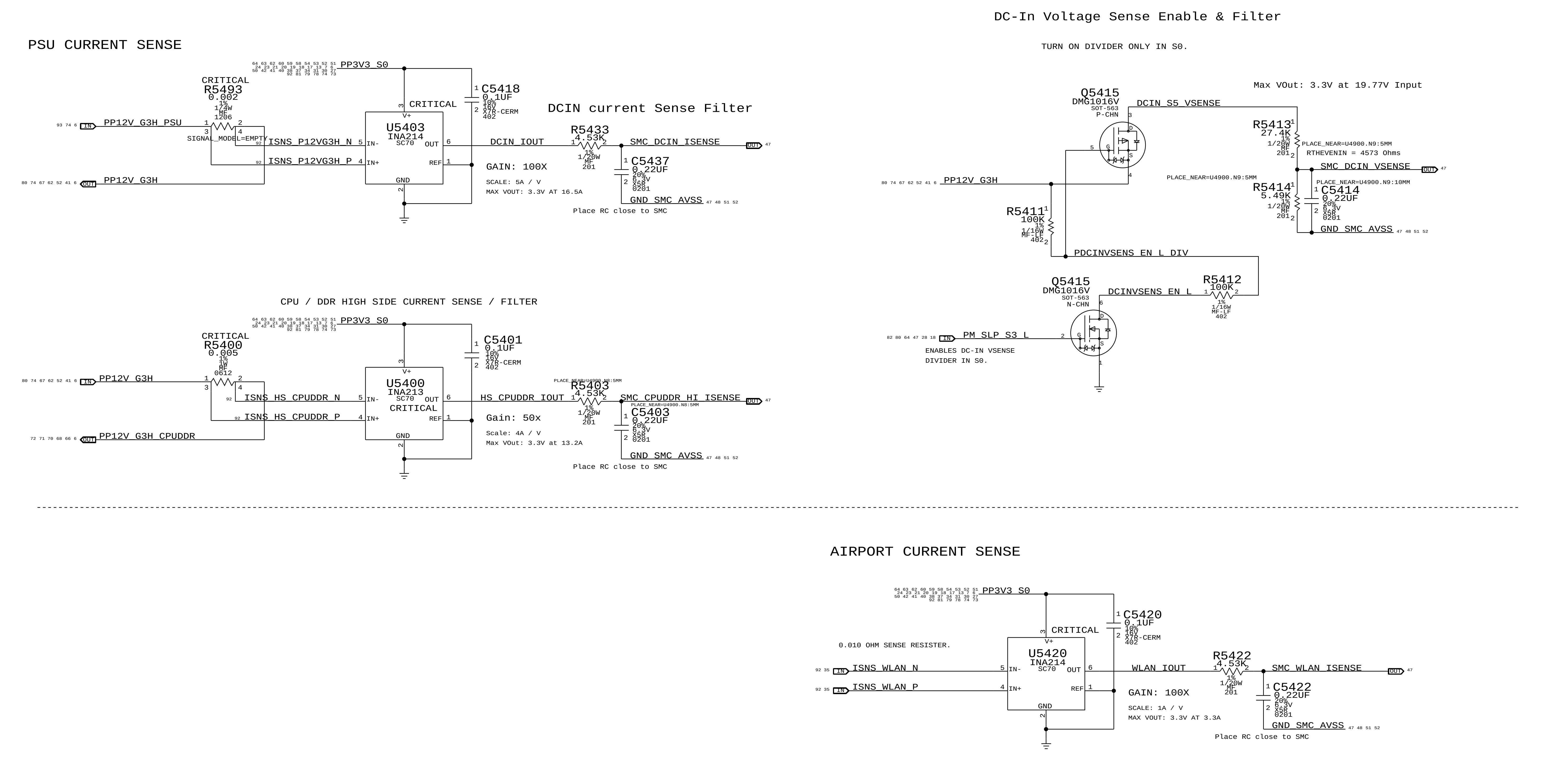
J40I, J40S, AND J40



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: High Side Current Sensing

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

PAGE: 54 OF 500

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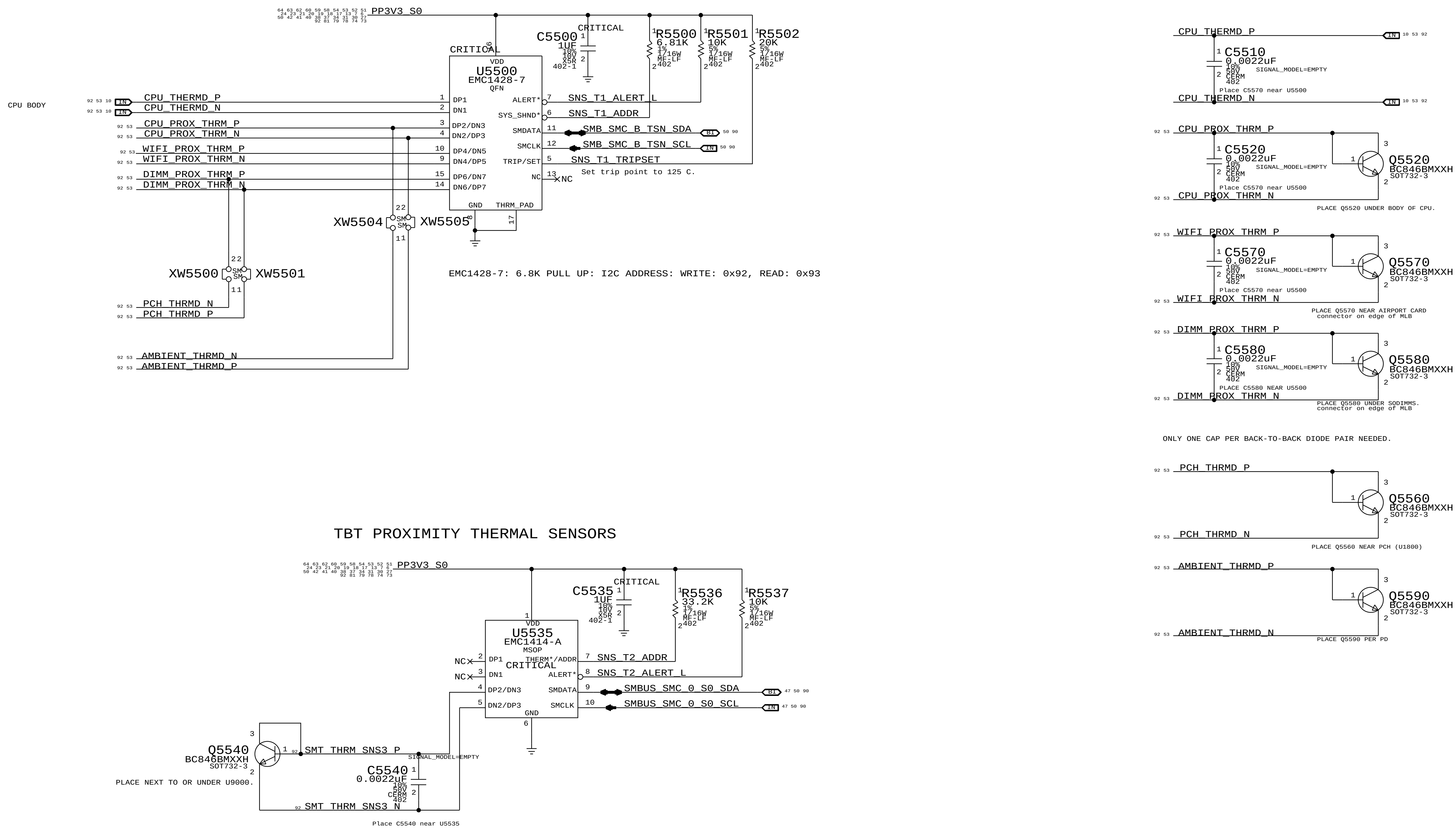


SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

THERMAL SENSORS

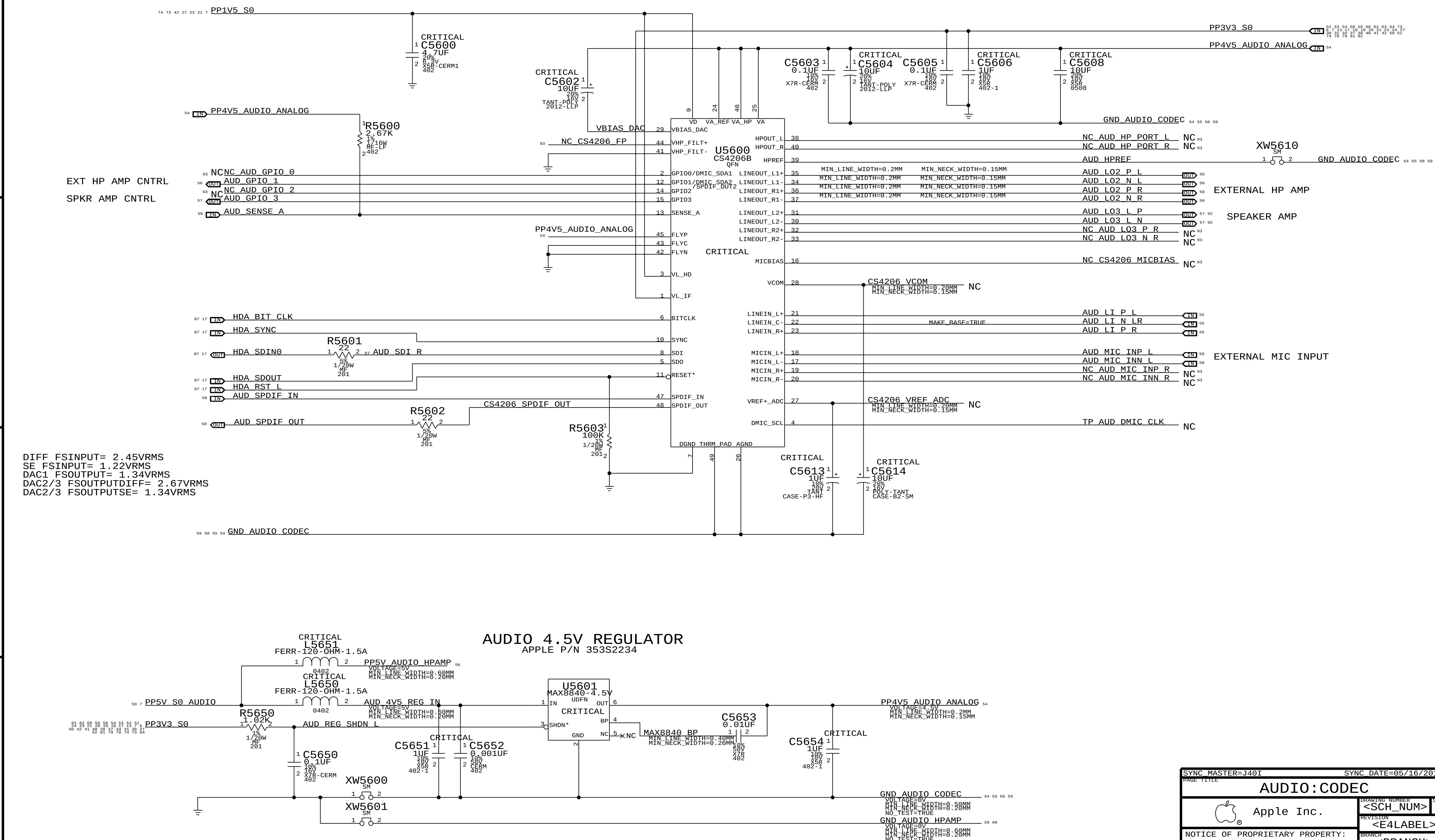
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

AUDIO CODEC
APPLE P/N 353S3199



DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

AUDIO 4.5V REGULATOR
APPLE P/N 353S2234

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: AUDIO:CODEC

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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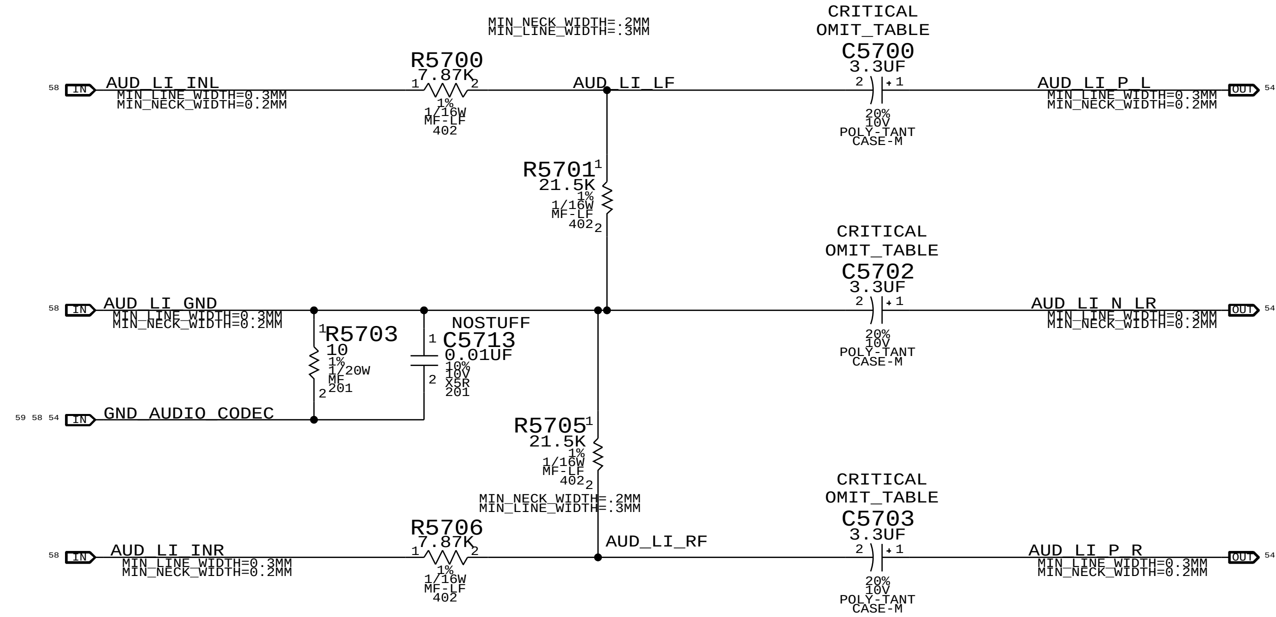
CODEC Nom SE RIN = 20K OHMS
FC = 5 HZ Max
VIN = 2VRMS, CODEC VIN = 1.14 VRMS
NET RIN = 18K OHMS



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
128S0309	3	CAP,TANT,POLY,4.7UF,20%,10V,SMD	C5700,C5702,C5703	CRITICAL	?

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

AUDIO:LINE-IN

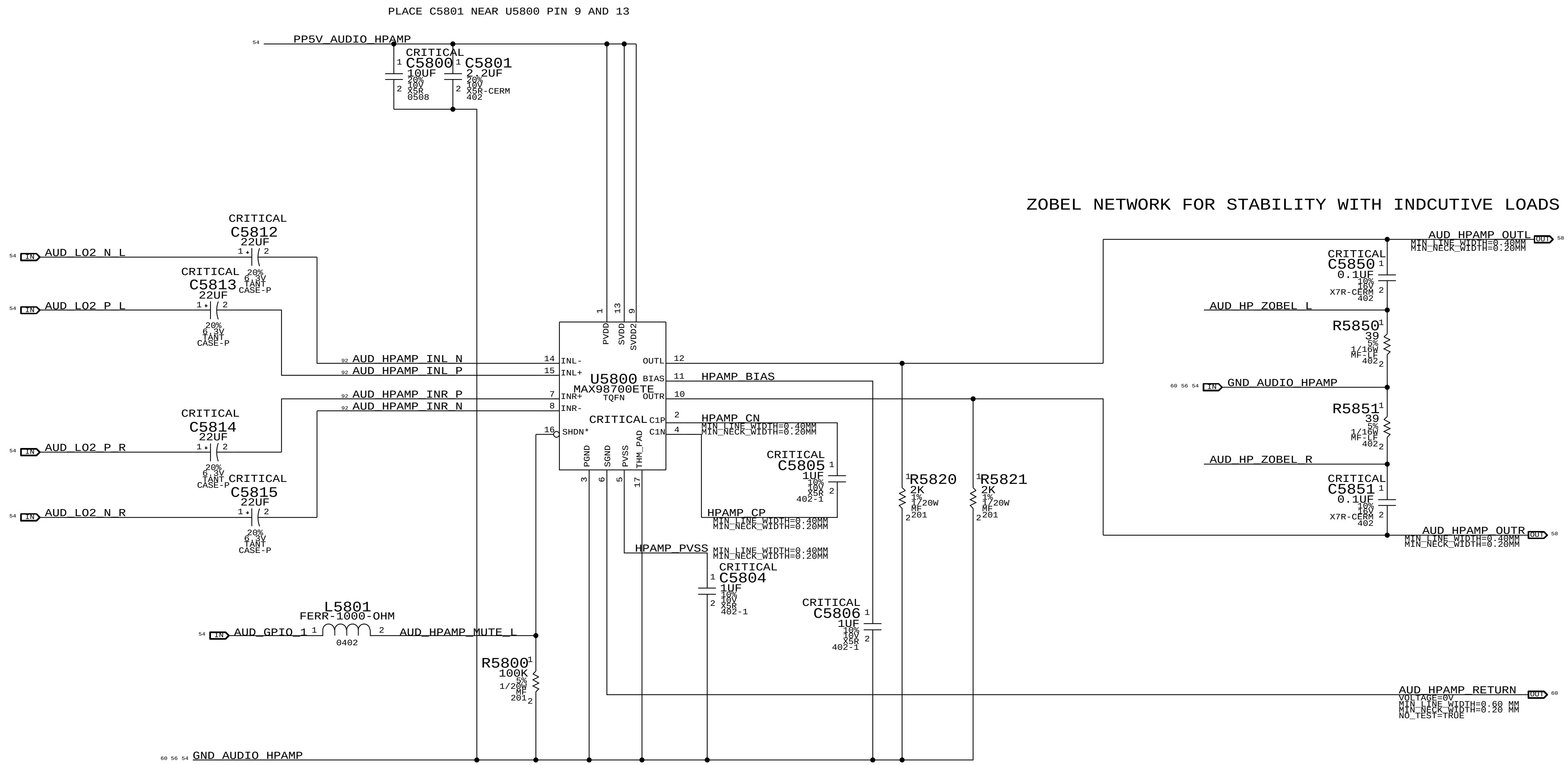
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

HEADPHONE AMPLIFIER (MAX98700)
APN:353S3244



DAC FILTER: @ 168KHZ

CODEC FILTER STOPBAND:26.256KHZ, ATTENUATION FROM STOPBAND TO 100KHZ IS 102DB

SYNC_MASTER=J40I SYNC_DATE=05/26/2011

PAGE TITLE
AUDIO:HEADPHONE AMP

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

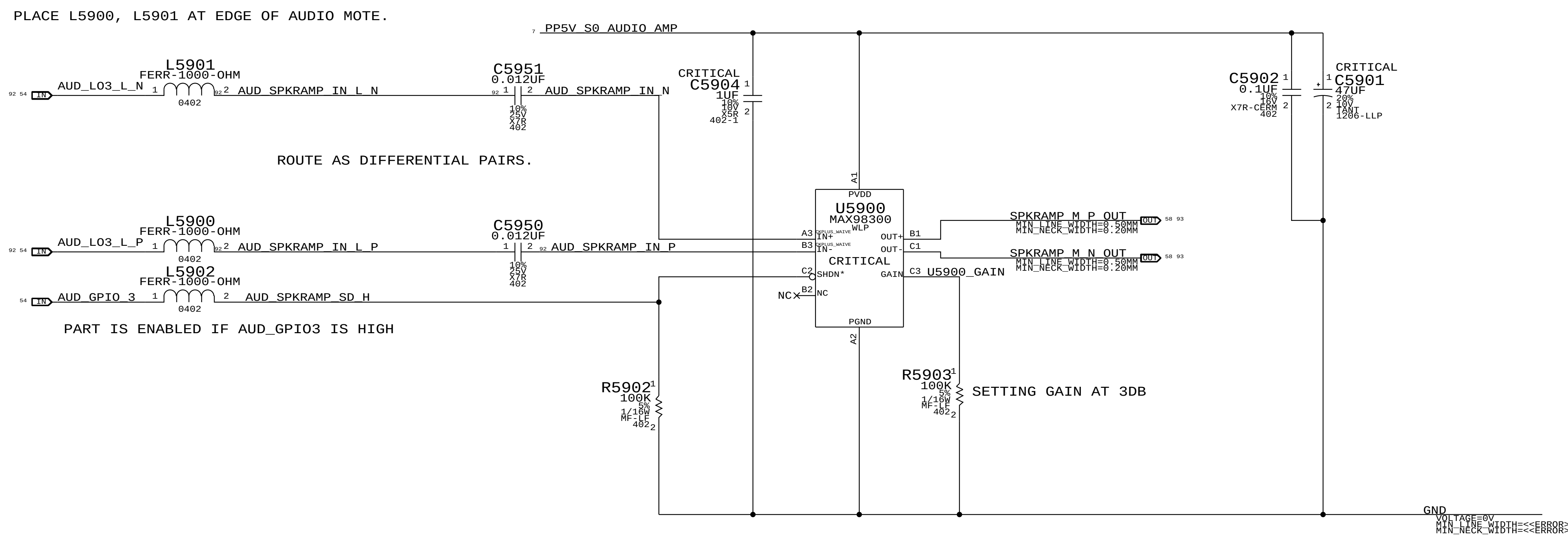
BRANCH <BRANCH>

MONO SPEAKER AMPLIFIER

APN 353S2888

GAIN = 3 DB NOM TO ALLOW A LITTLE MORE THAN 1% THD POWER = 2.1W AT 4OHMS.

FC = 470 HZ. RIN AT 3DB IS 28K.



INTERNAL SWITCHING FREQUENCY: 300KHZ.

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

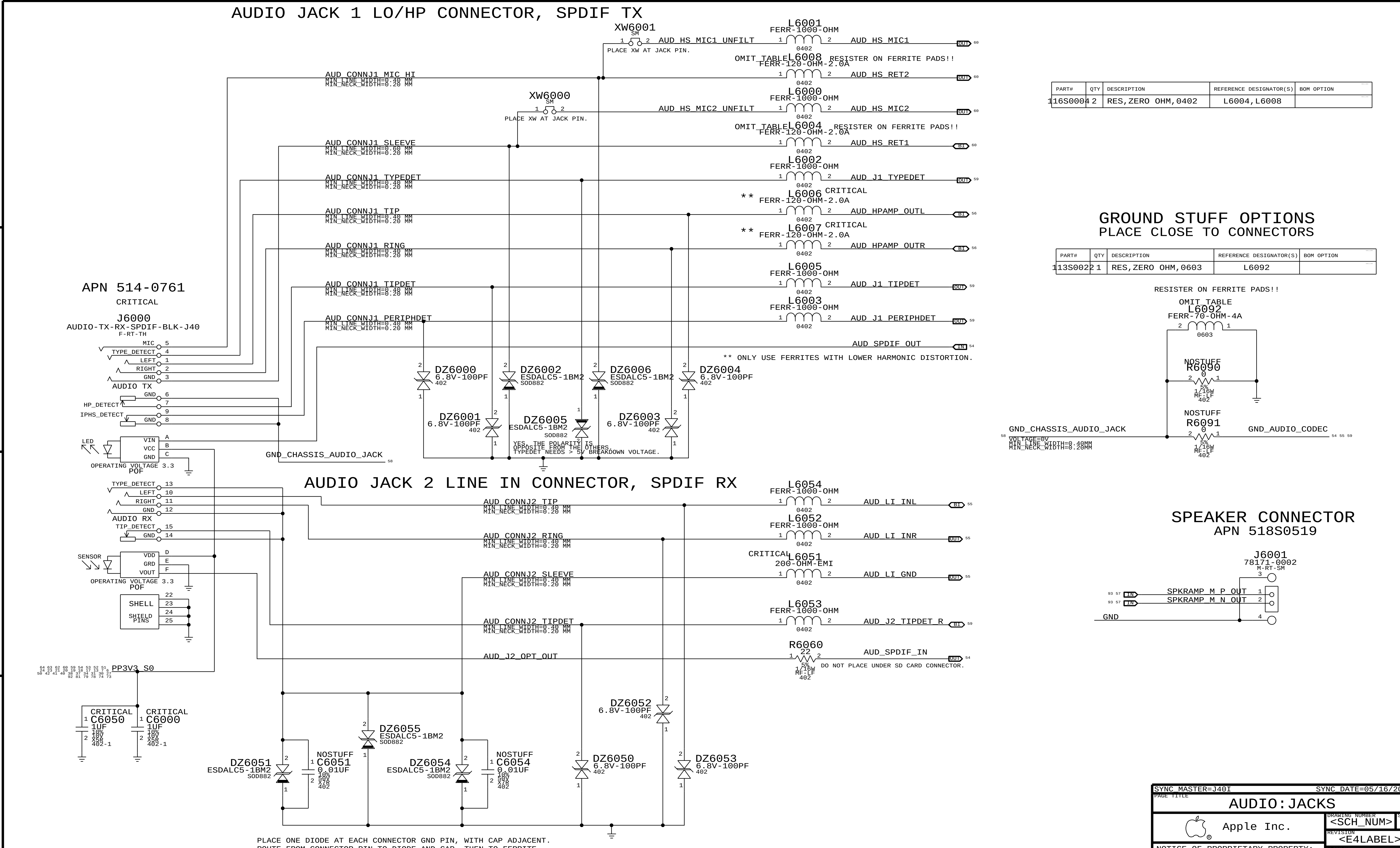
PAGE TITLE: AUDIO:SPEAKER AMP

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0004	2	RES,ZERO OHM,0402	L6004,L6008	

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
113S0022	1	RES,ZERO OHM,0603	L6092	

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: AUDIO:JACKS

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2256

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	~SHDN CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (03)	0X0A (10)	GPIO_1	0X0A(D)
SPEAKER 1	0X04 (4)	0X04 (04)	0X0B (11)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (08)	0X10 (16)	N/A	0X0D(B)

PANTHER-POINT OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	~SHDN CONTROL	DET ASSIGNMENT
HDMI OUT	N/A	-	-	N/A	N/A

CODEC INPUT SIGNAL PATHS

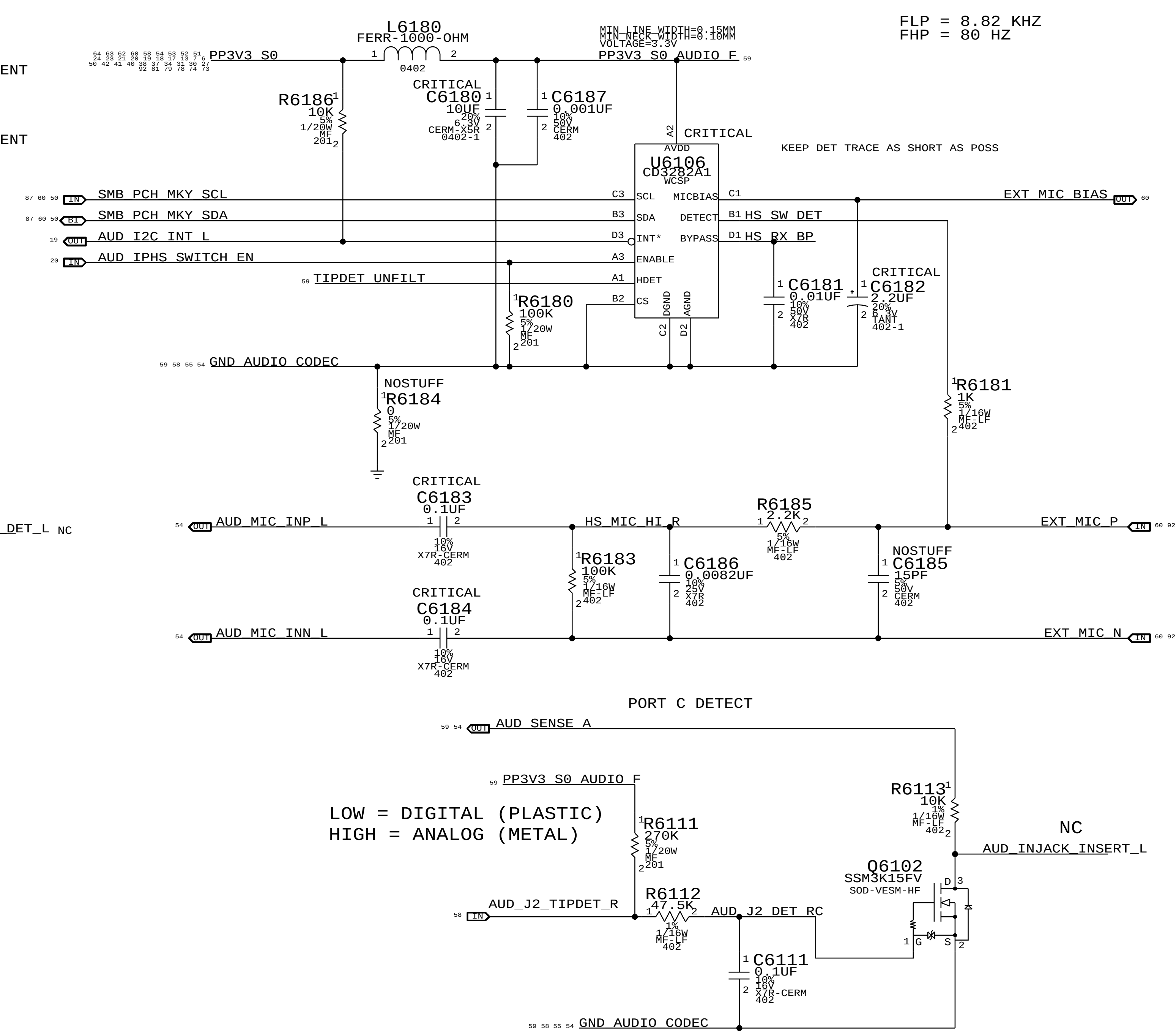
FUNCTION	GAIN/MUTE	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
LINE IN	0X05 (5)	0X05 (05)	0X0C (12)	N/A	0X0C(C)
EXT MIC IN	0X06 (6)	0X06 (06)	0X0D (13,V22,B,LEFT)	MIKEY	MIKEY
SPDIF IN	N/A	0X07 (07)	0X0F (15)	N/A	N/A

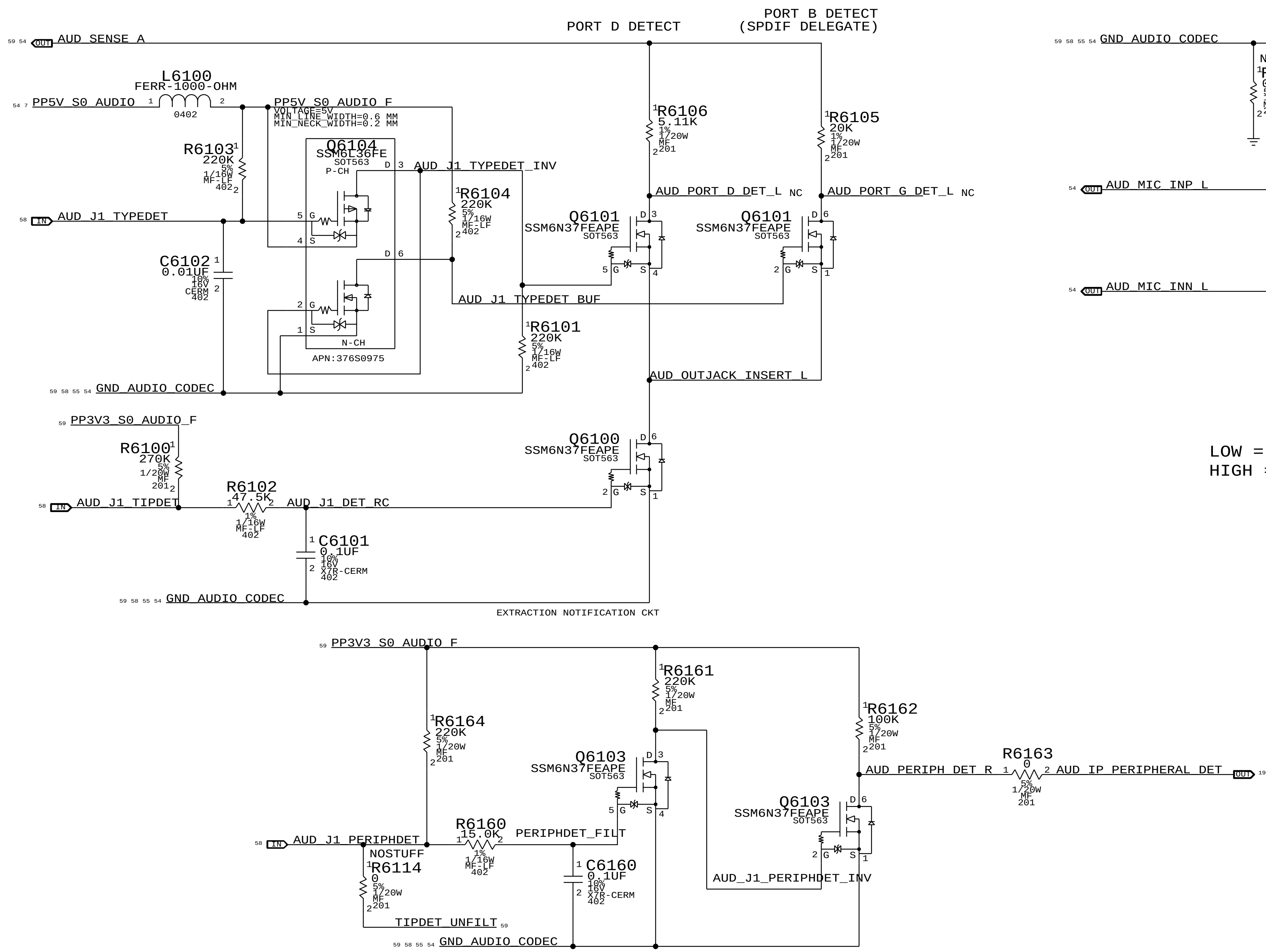
SYSTEM INT AND GPIO LINES

FUNCTION	
MIKEY ENABLE	SATA4GP/GPIO16
MIKEY INTERRUPT	PIRQG*/GPIO5
PERIPHERAL DETECT	PIRQF*/GPIO3

WRITE: 0X72 READ: 0X73

FLP = 8.82 KHZ
FHP = 80 HZ





SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE
AUDIO:JACK TRANSLATORS

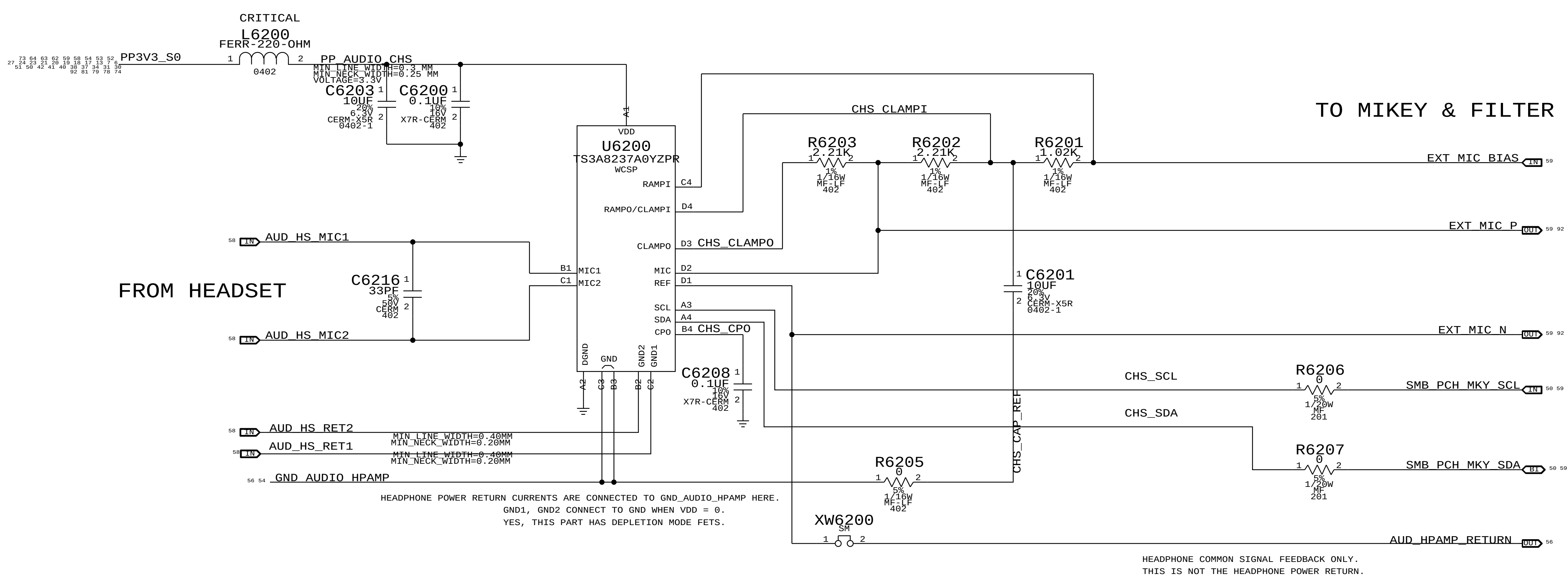
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

EXTERNAL (HEADSET) MIC INPUT CIRCUITRY APN:353S3522



I2C ADDRESSES: CHS uses SMBus 0 connections

CHS	U6200	READ	0111	0111	0X77
CHS	U6200	WRITE	0111	0110	0X76

SYNC_MASTER=J50I SYNC_DATE=05/16/2011

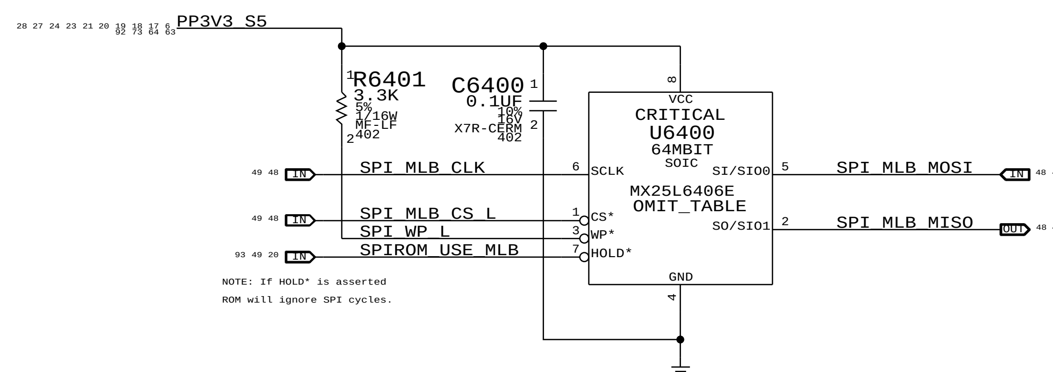
PAGE TITLE
AUDIO:DETECT/MIC BIAS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

SPI BOOT ROM

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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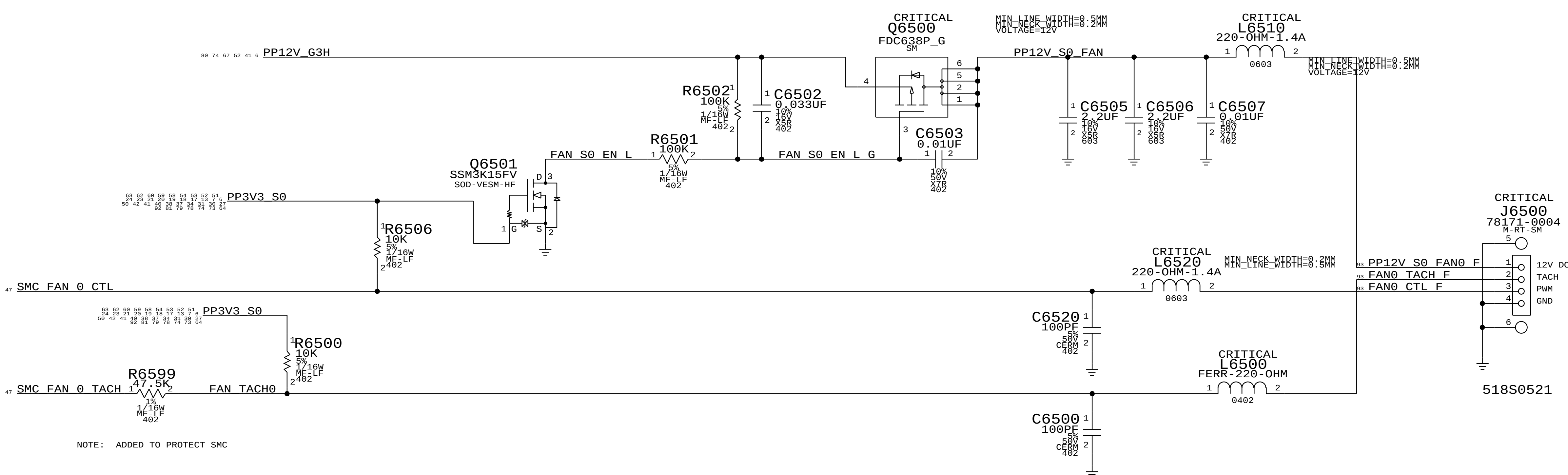
FAN CONTROL CIRCUIT



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: Fan Control Circuit

Apple Inc.

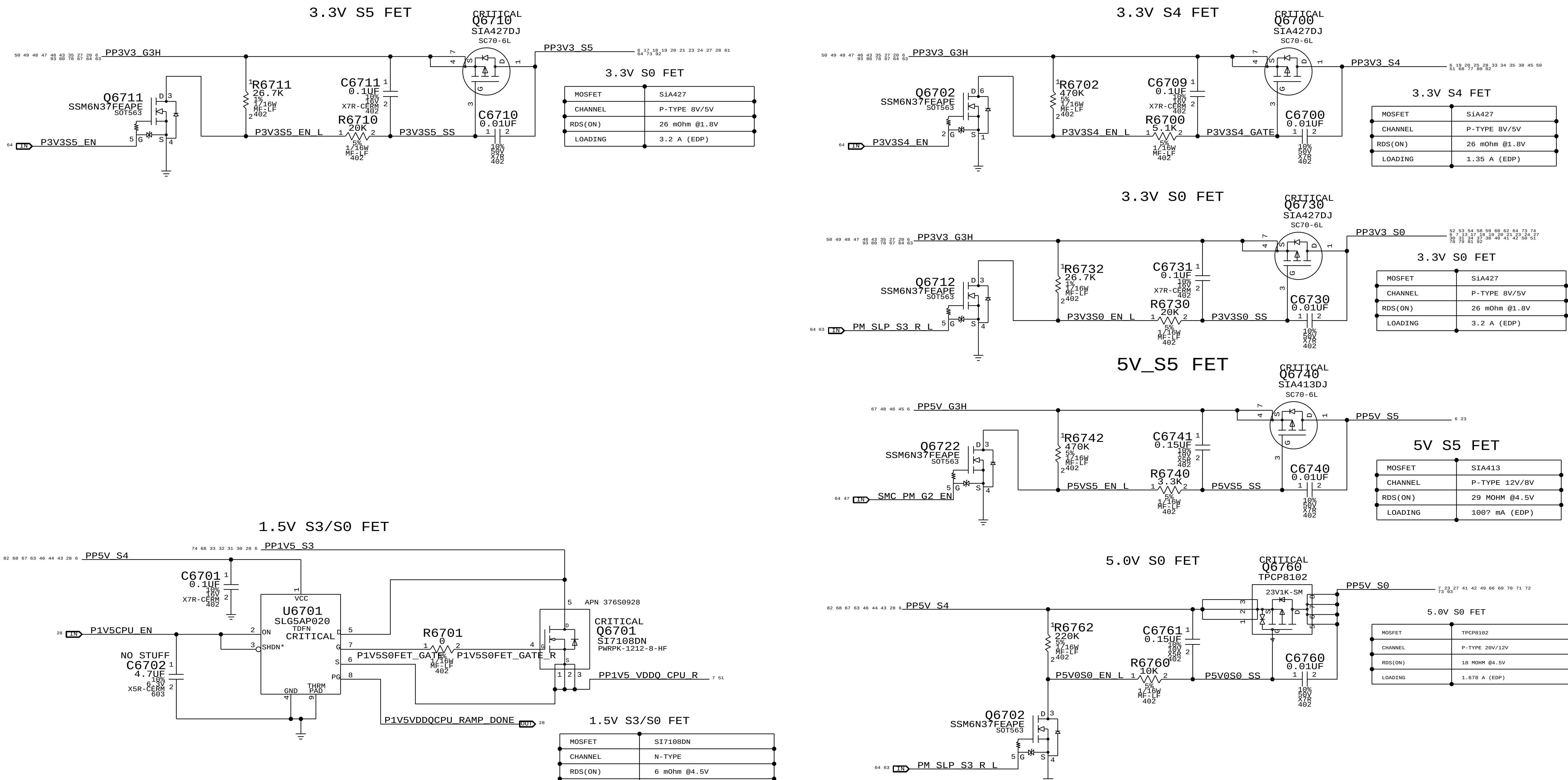
DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

PAGE: 65 OF 500

SHEET: 62 OF 94



3.3V S0 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	3.2 A (EDP)

3.3V S4 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	1.35 A (EDP)

3.3V S0 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	3.2 A (EDP)

5V S5 FET

MOSFET	SIA413
CHANNEL	P-TYPE 12V/8V
RDS(ON)	29 MOHM @4.5V
LOADING	100? mA (EDP)

5.0V S0 FET

MOSFET	TPCP8102
CHANNEL	P-TYPE 20V/12V
RDS(ON)	18 MOHM @4.5V
LOADING	1.678 A (EDP)

1.5V S3/S0 FET

MOSFET	SI7108DN
CHANNEL	N-TYPE
RDS(ON)	6 mOhm @4.5V
LOADING	5 A (EDP)

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

Power FETs

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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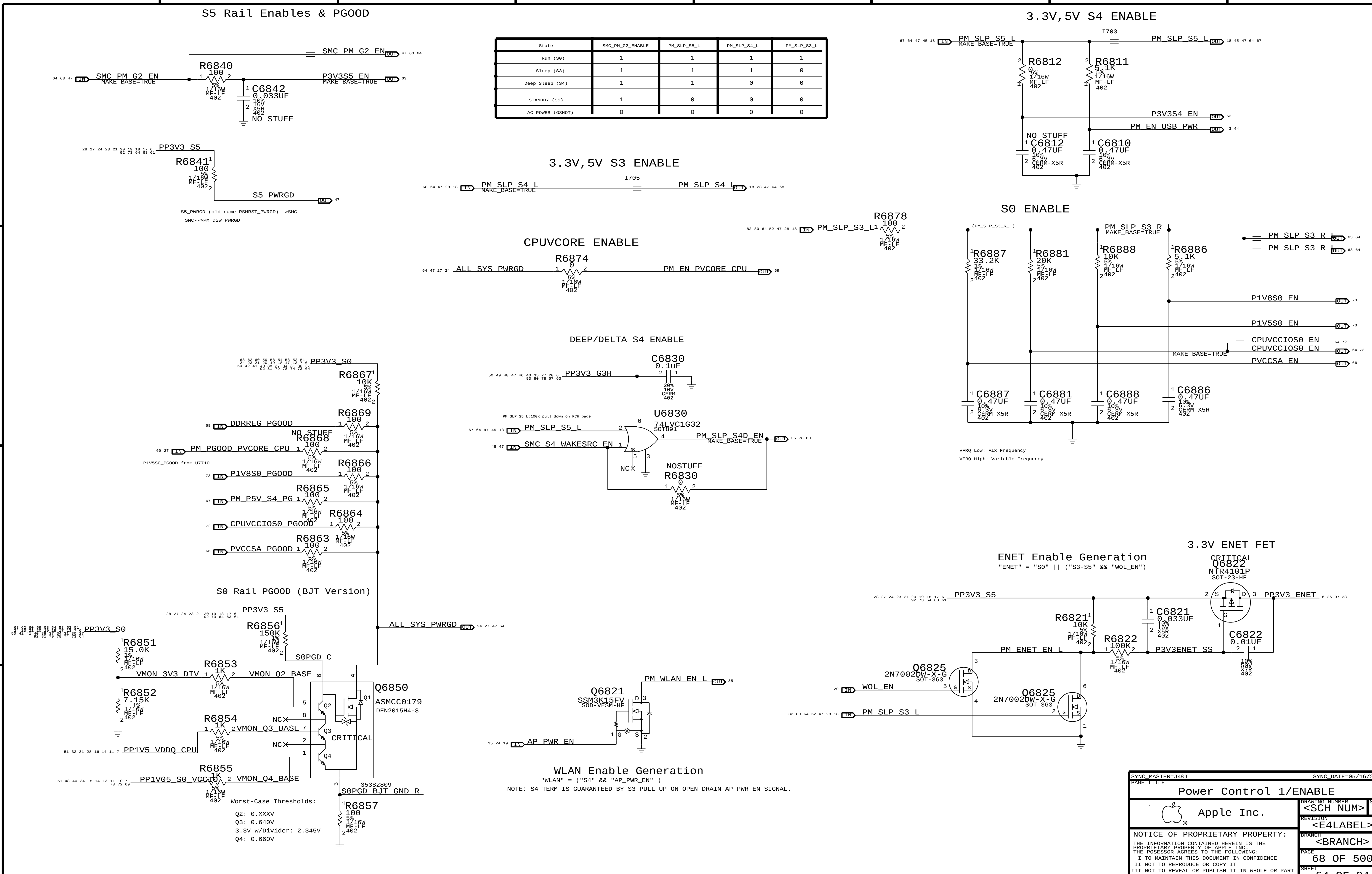

State	SMC_PM_G2_ENABLE	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1	1
Sleep (S3)	1	1	1	0
Deep Sleep (S4)	1	1	0	0
STANDBY (S5)	1	0	0	0
AC POWER (G3HOT)	0	0	0	0

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

Power Control 1/ENABLE

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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ROOM FOR MORE?

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

POWER CONTROL CONT.

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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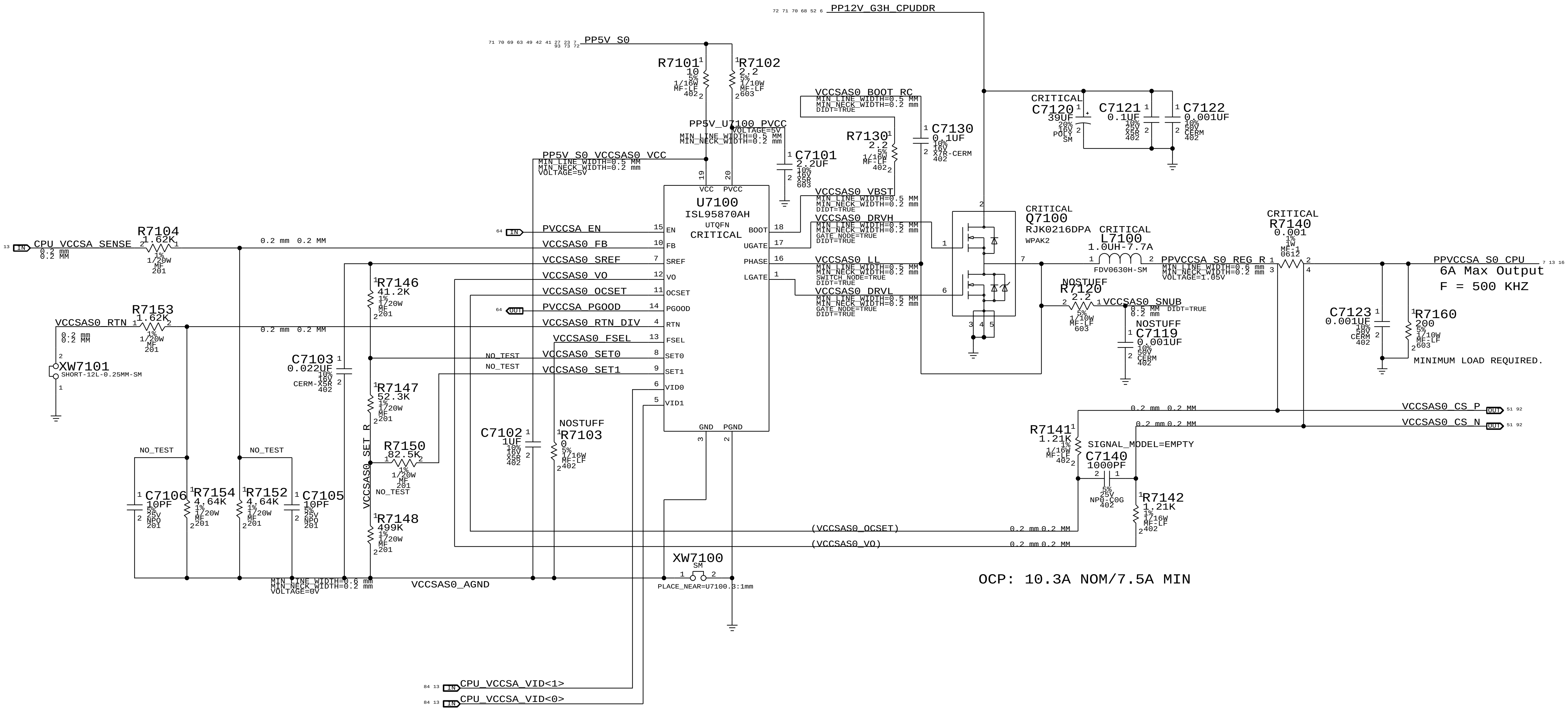
System Agent Power Supply



INTEL TABLE:

VID1	VID0	Voltage
0	0	0.9V
1	0	0.8V
0	1	0.725V
1	1	0.675V

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE
VR - SYSTEM AGENT SUPPLY

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

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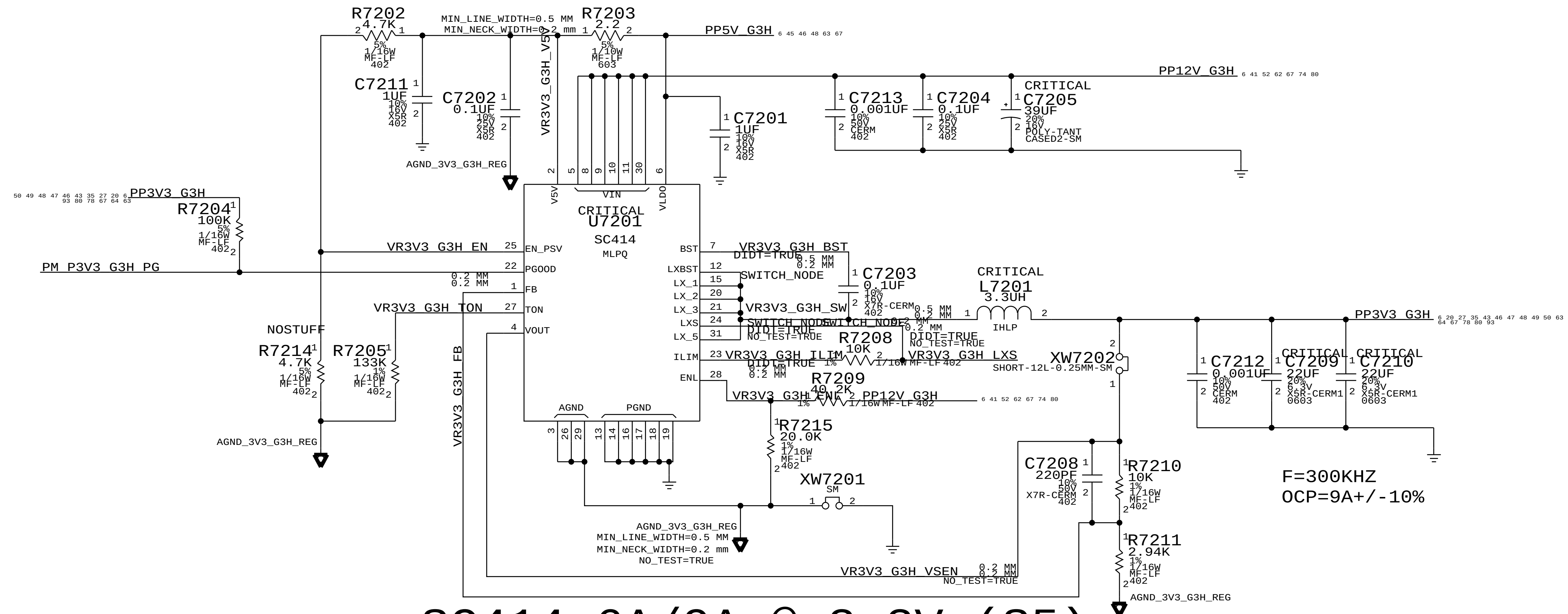
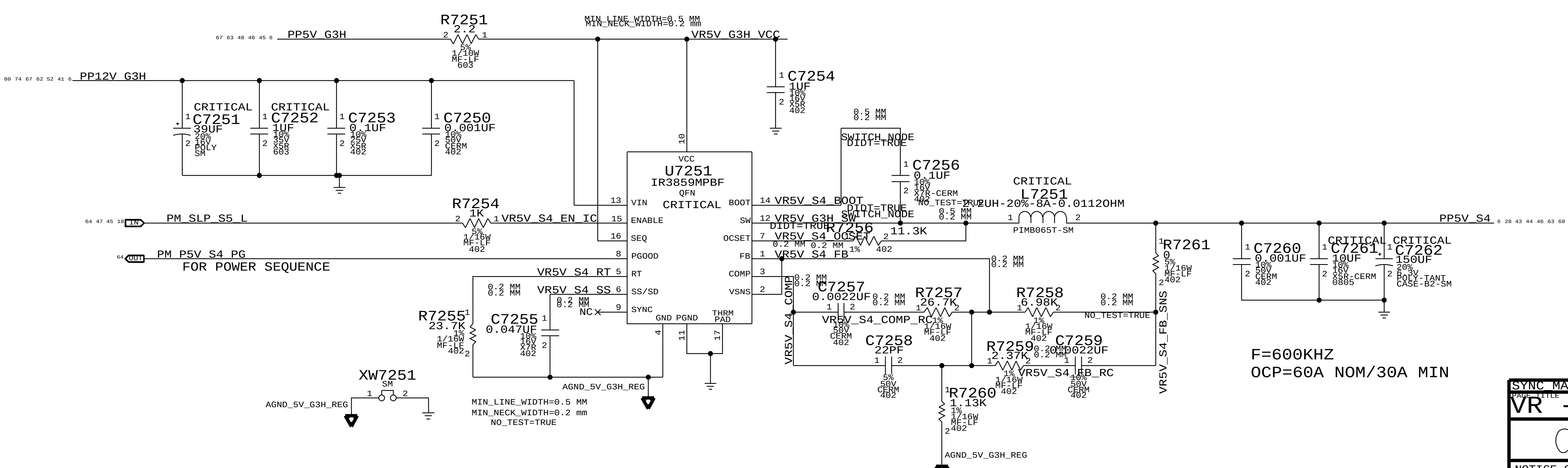


SC414 6A/9A @ 3.3V (S5)



IR3859 5A/9A @ 5V (S4)

SYNC_MASTER=J401 SYNC_DATE=05/16/2011

VR - 3.3V S5 AND 5V S4

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

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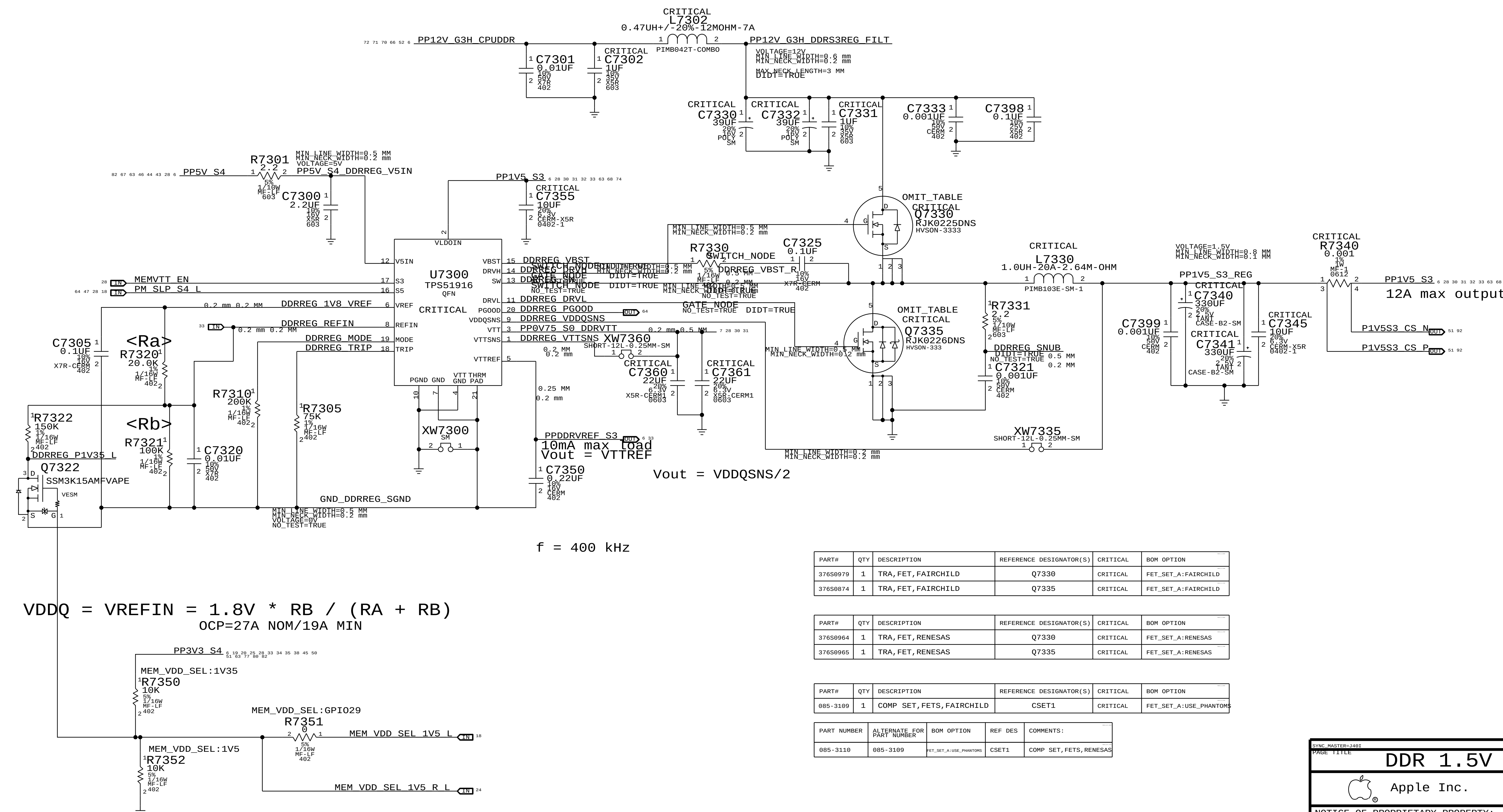

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
376S0979	1	TRA,FET,FAIRCHILD	Q7330	CRITICAL	FET_SET_A:FAIRCHILD
376S0874	1	TRA,FET,FAIRCHILD	Q7335	CRITICAL	FET_SET_A:FAIRCHILD

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
376S0964	1	TRA,FET,RENESAS	Q7330	CRITICAL	FET_SET_A:RENESAS
376S0965	1	TRA,FET,RENESAS	Q7335	CRITICAL	FET_SET_A:RENESAS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
085-3109	1	COMP SET,FETS,FAIRCHILD	CSET1	CRITICAL	FET_SET_A:USE_PHANTOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
085-3110	085-3109	FET_SET_A:USE_PHANTOMS	CSET1	COMP SET,FETS,RENESAS

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

DDR 1.5V VR

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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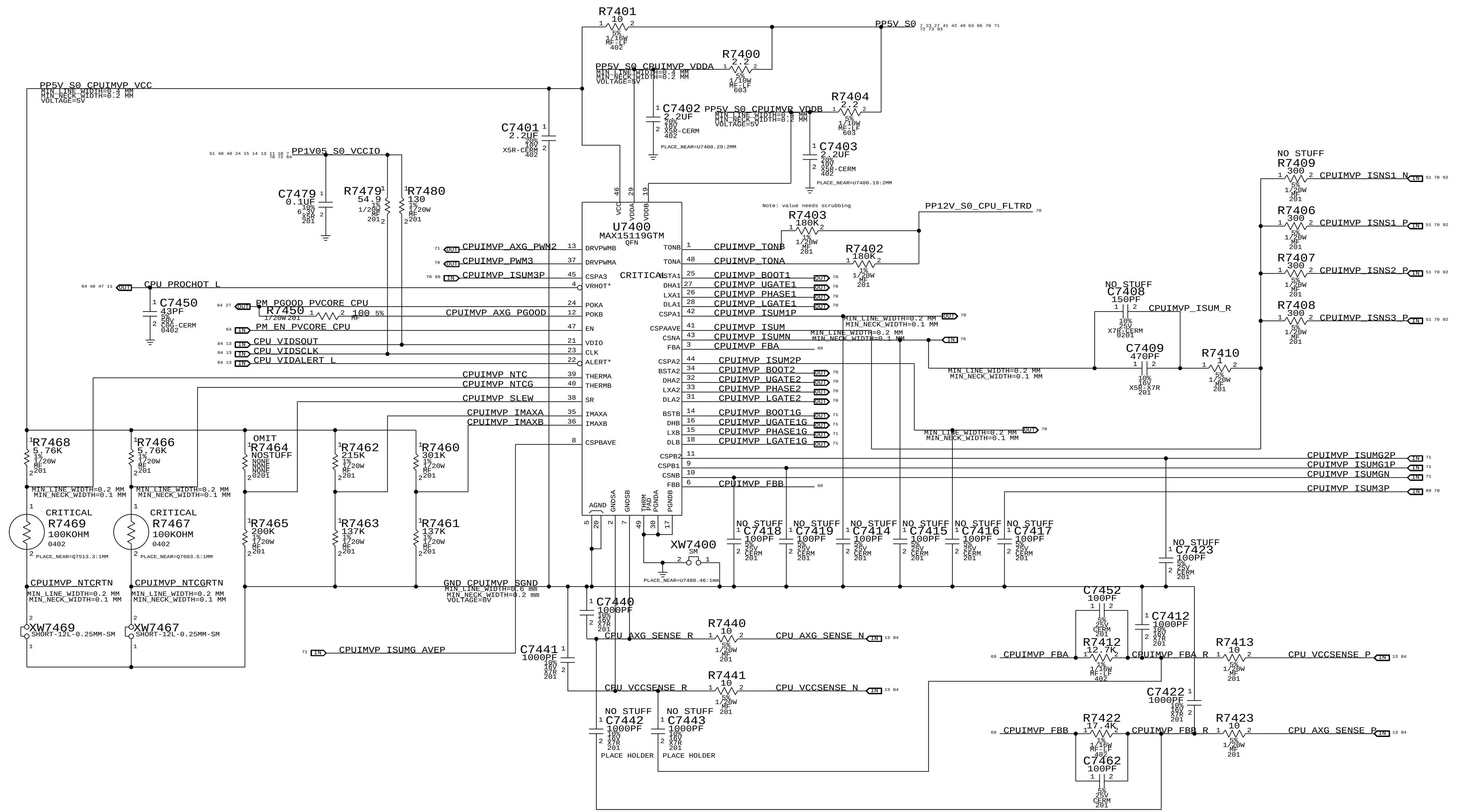
SHEET 68 OF 94



SYNC_MASTER=J31_JACK SYNC_DATE=09/02/2011

PAGE TITLE

CPU IMVP7 & AXG VCore Regulator

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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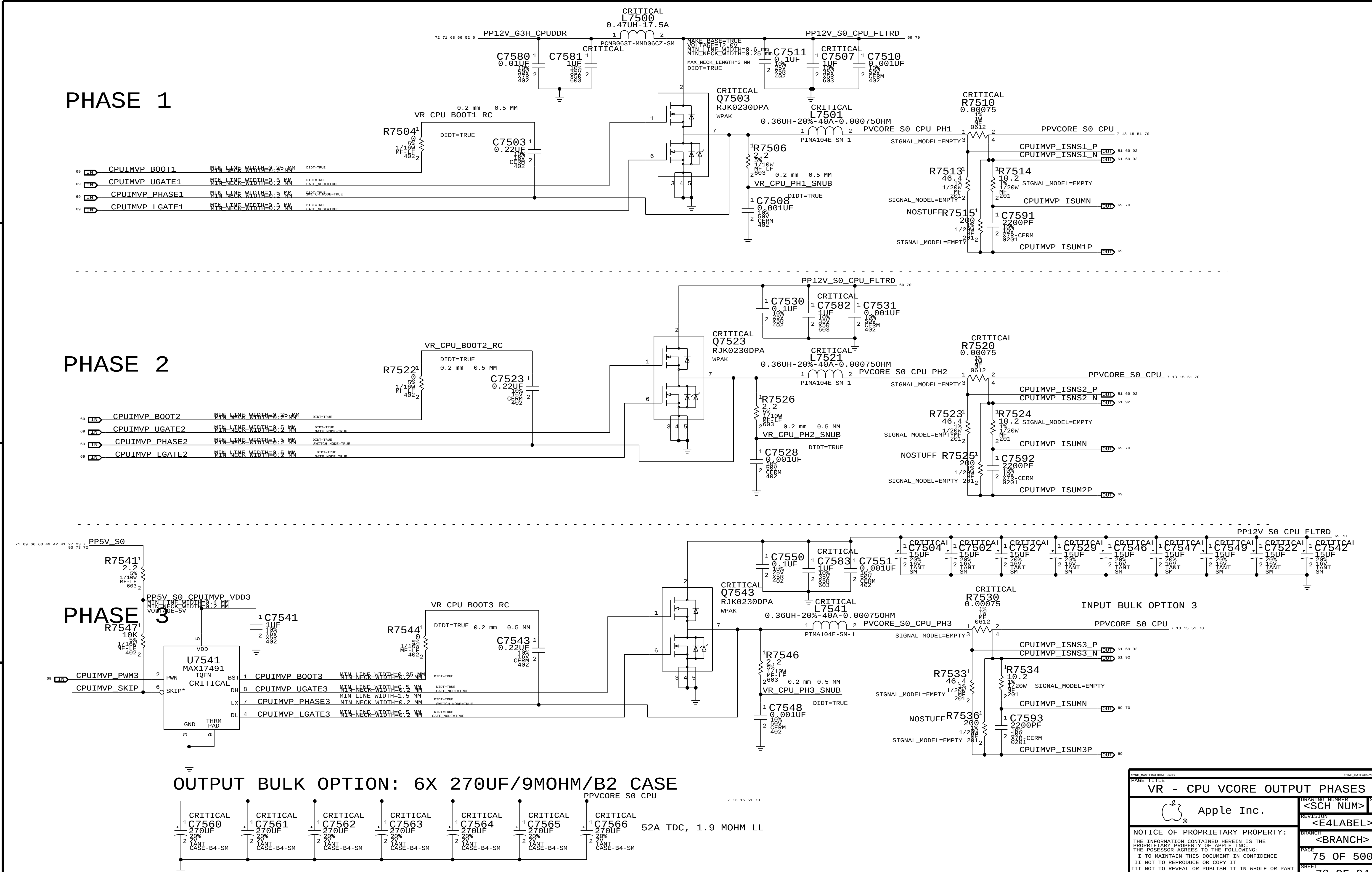
SHEET 69 OF 94



PAGE TITLE
VR - CPU VCORE OUTPUT PHASES

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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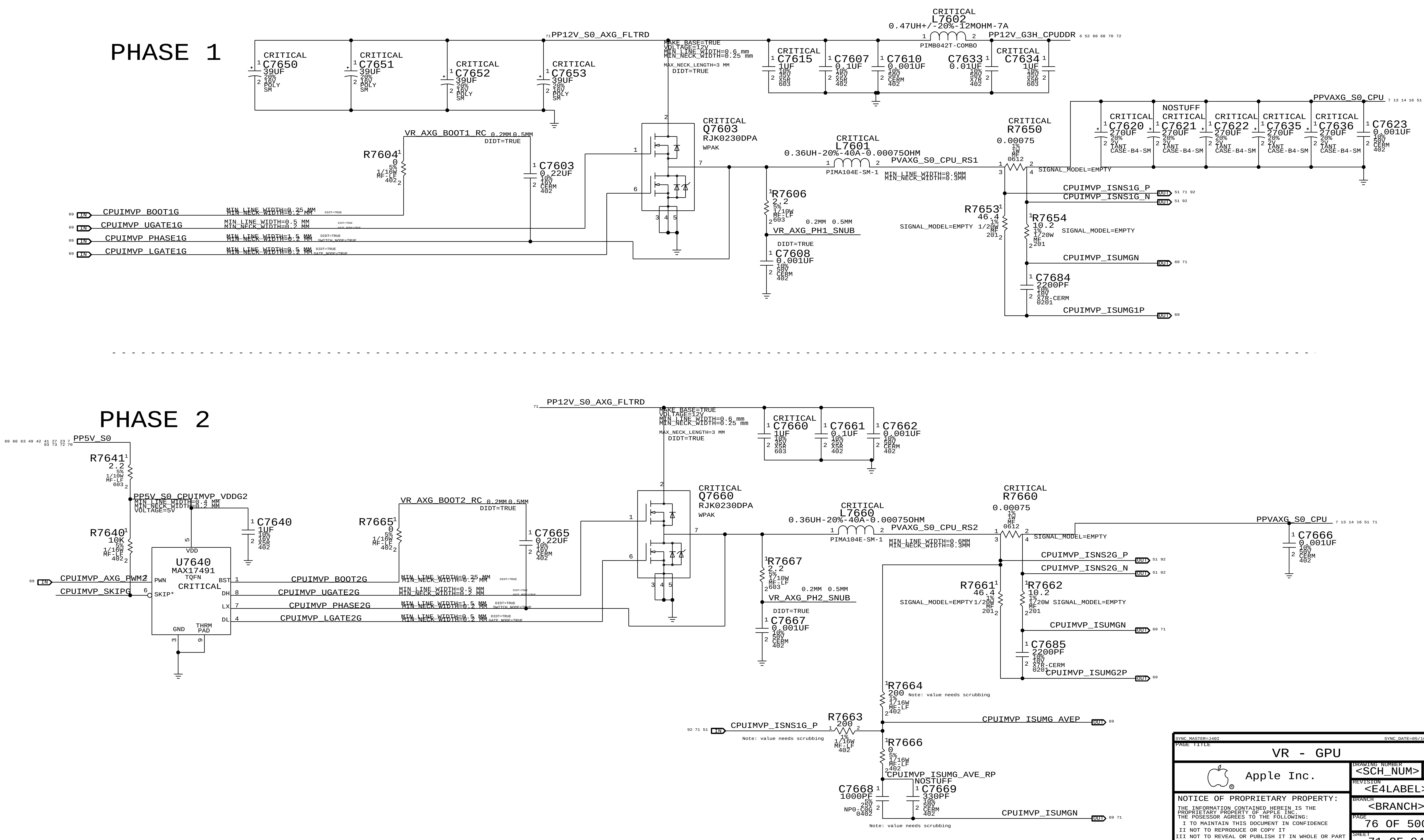
AXG PHASE (35A AVG/50A PEAK, 3.9MOHM LL)



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: VR - GPU

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

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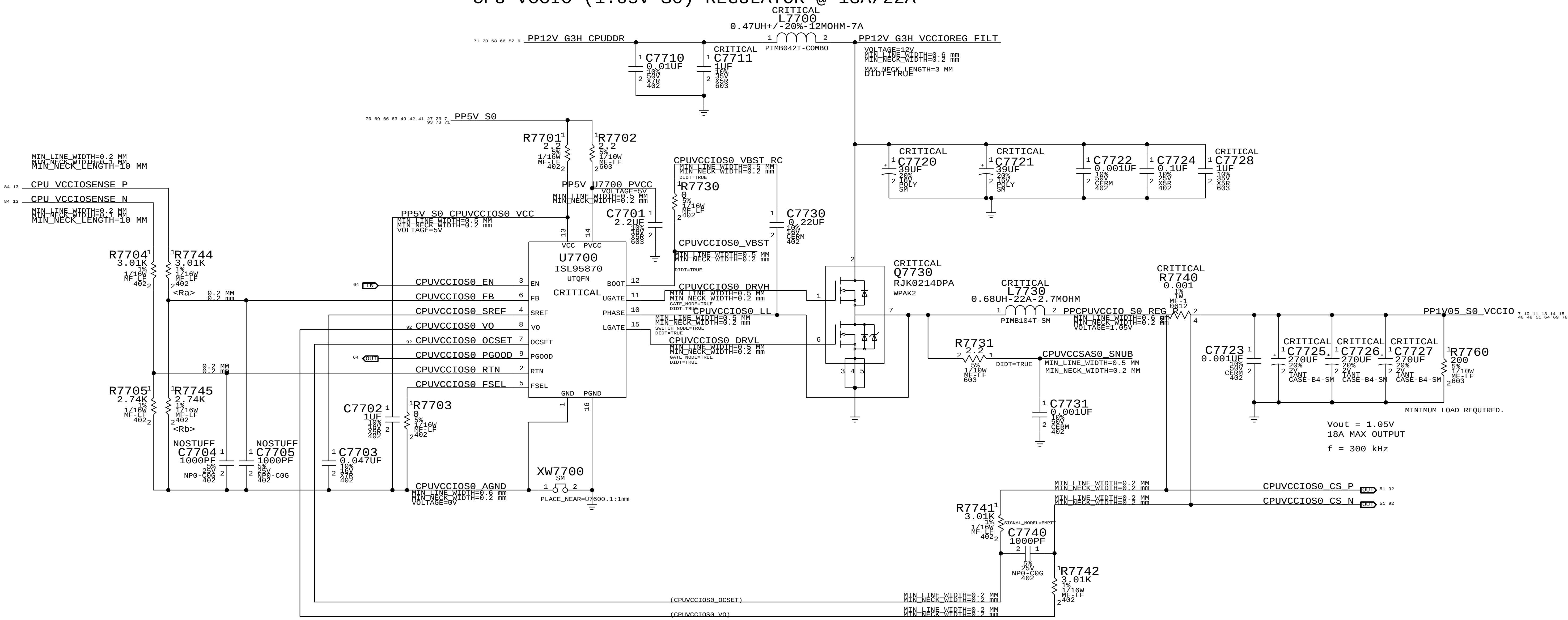
SHEET: 71 OF 94

CPU VCCIO (1.05V S0) REGULATOR @ 18A/22A



OCP = R7641 x 8.5uA / R7640
OCP = 25.6A NOM/21.3A MIN
Vout = 0.5V * (1 + Ra / Rb)

SYNC_MASTER=LOCAL-J40S SYNC_DATE=05/16/2011

PAGE TITLE

VR - 1.05V S0

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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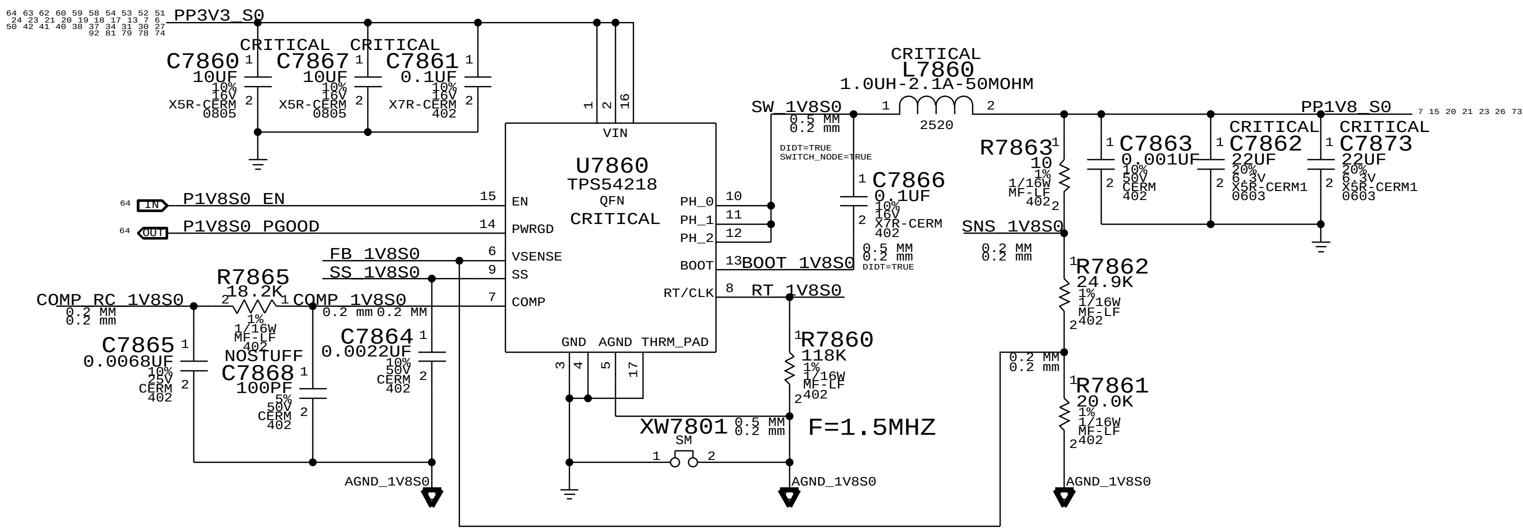
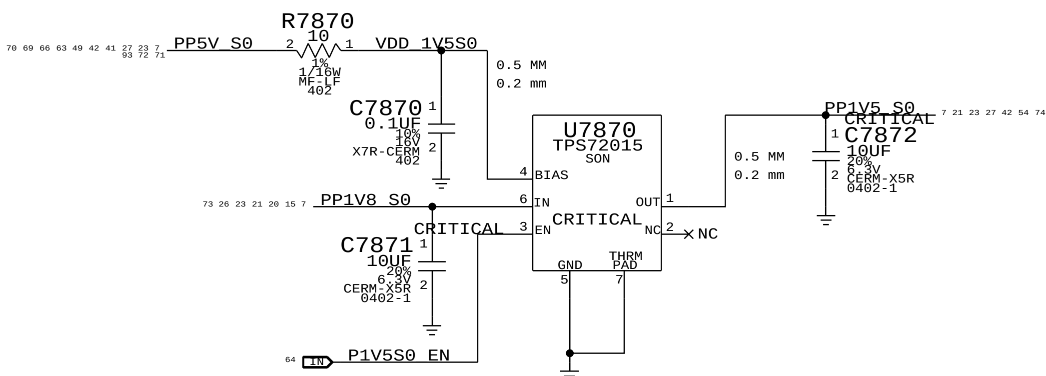
1.8V S0 SWITCHER @ 1.7A/2A

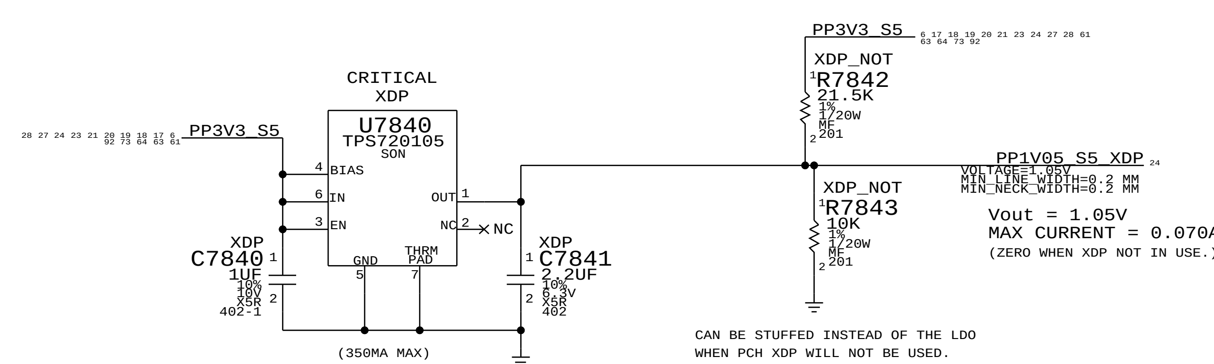


1.5V S0 @ 0.3A/0.5A



1.05V S5 XDP LDO

PANTHER-POINT REQUIRES JTAG PULL-UPS TO BE POWERED AT 1.05V WHEN SUS SUSPEND WELL IS ACTIVE.
Pull-ups (3) must be 51 ohms to support XDP (not required in production).
70mA is required to support pull-ups. Alternative is strong voltage
dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



Vout = 1.05V
MAX CURRENT = 0.070A
(ZERO WHEN XDP NOT IN USE.)

CAN BE STUFFED INSTEAD OF THE LDO
WHEN PCH XDP WILL NOT BE USED.

SYNC_MASTER=J401 SYNC_DATE=05/16/2011

PAGE TITLE
VR - MISC

Apple Inc.

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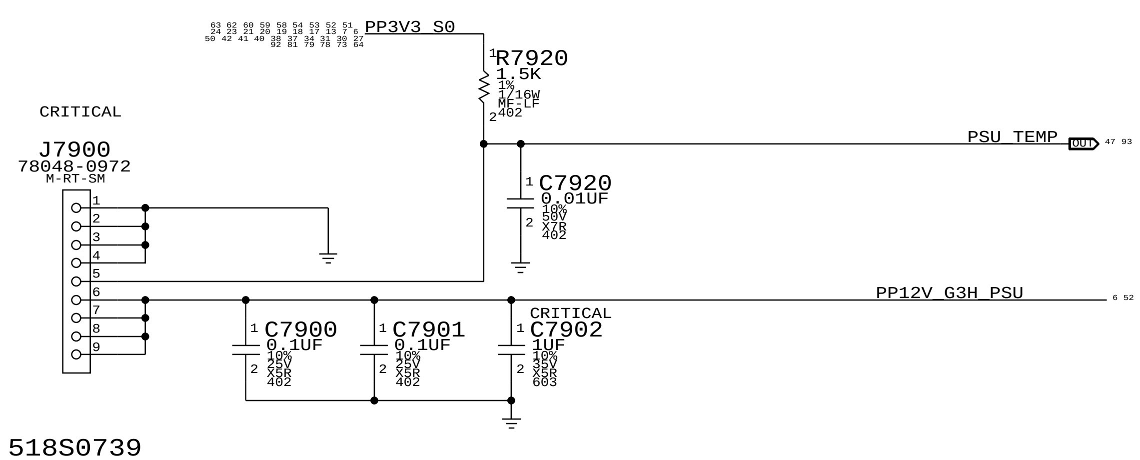
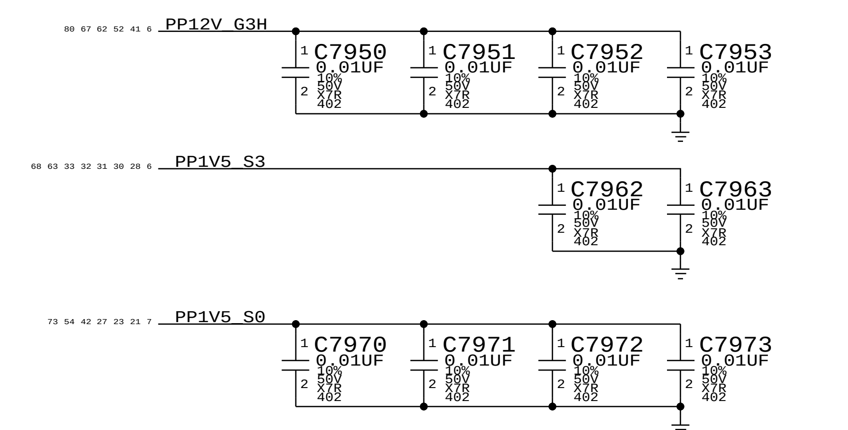


PLACE CAPS AS CLOSE TO J7900 AS POSSIBLE.
BEST IF VIAS ARE AFTER THE CAPS.

SPRINKLE THESE CAPS SOMEWHAT EVENLY AROUND THE PLANES AND ROUTES FOR THESE RAILS.



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

PSU CONNECTOR, MISC CAPS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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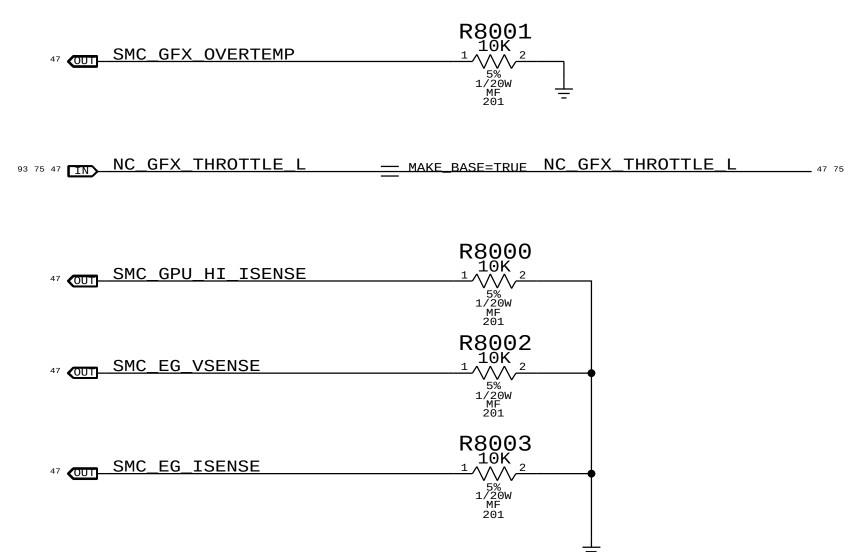

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

NO GPU GLUE

Apple Inc.

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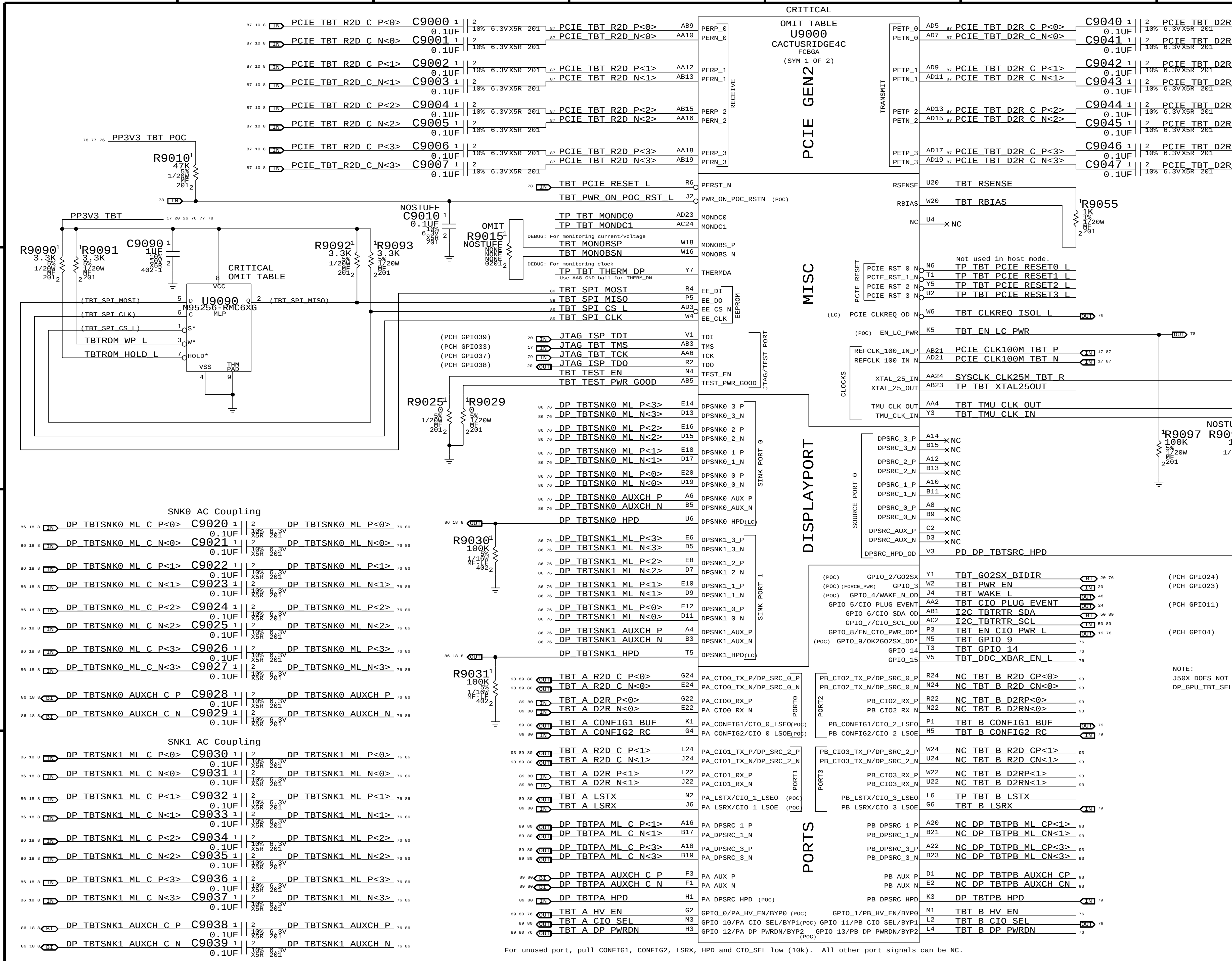

SYNC_MASTER=J13_MLB SYNC_DATE=08/12/2011

PAGE TITLE

Thunderbolt Host (1 of 2)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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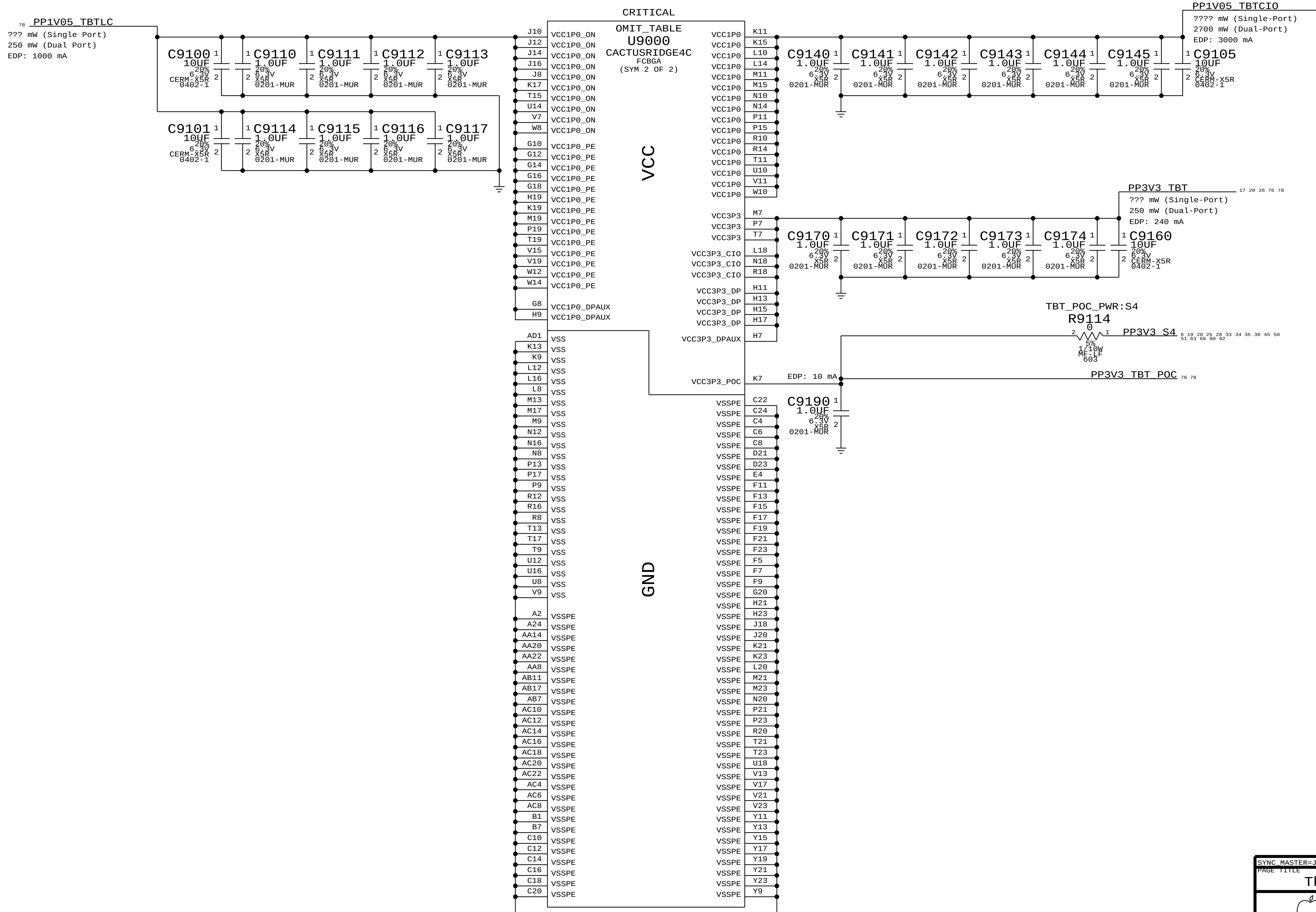

SYNC_MASTER=J13_MLB SYNC_DATE=08/12/2011

PAGE TITLE

Thunderbolt Host (2 of 2)

Apple Inc.

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REVISION <E4LABEL>

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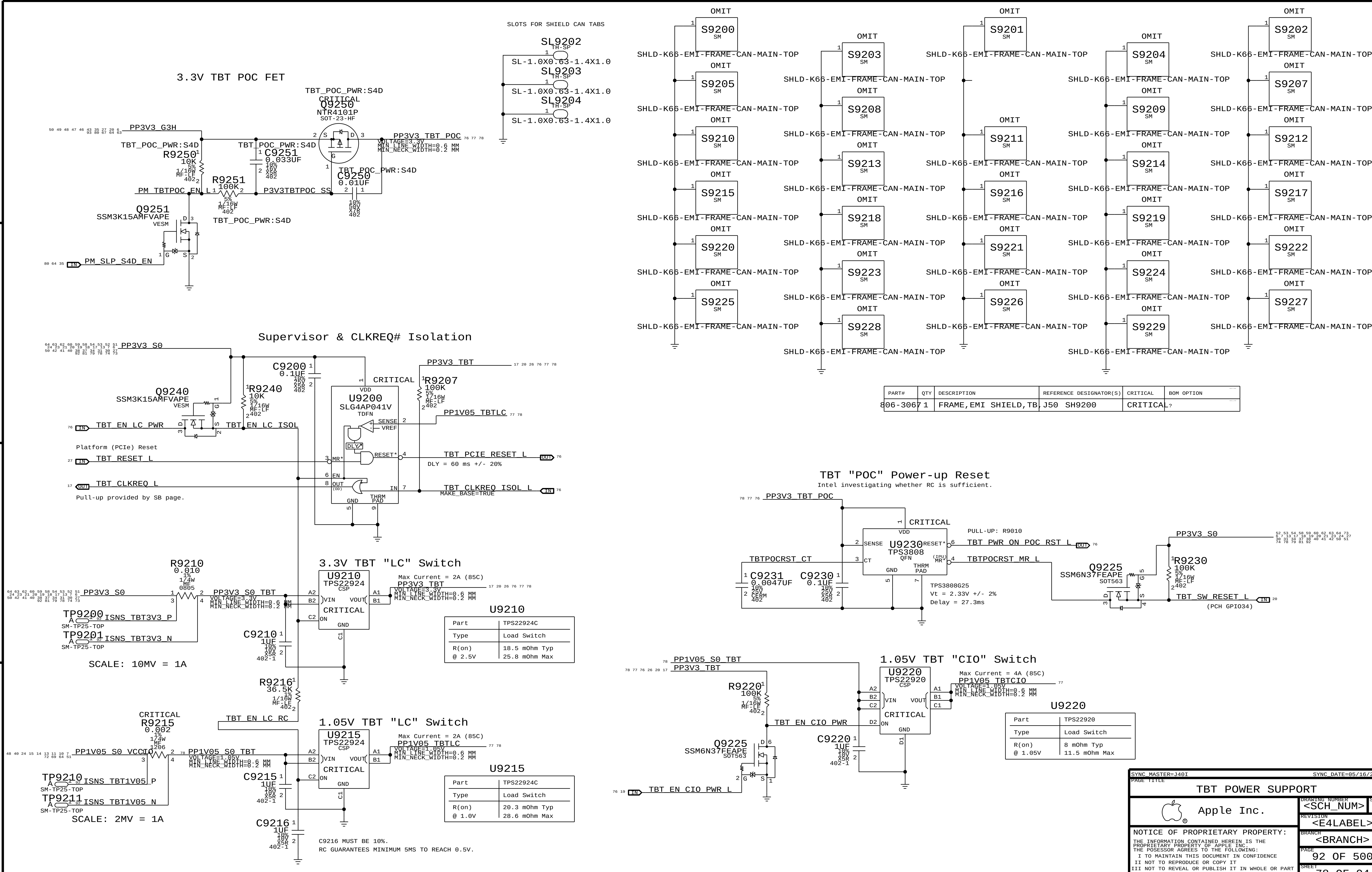

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
806-3067	1	FRAME,EMI SHIELD,TB,	J50 SH9200	CRITICAL	?

U9210

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

U9215

Part	TPS22924C
Type	Load Switch
R(on) @ 1.0V	20.3 mOhm Typ 28.6 mOhm Max

U9220

Part	TPS22920
Type	Load Switch
R(on) @ 1.05V	8 mOhm Typ 11.5 mOhm Max

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

TBT POWER SUPPORT

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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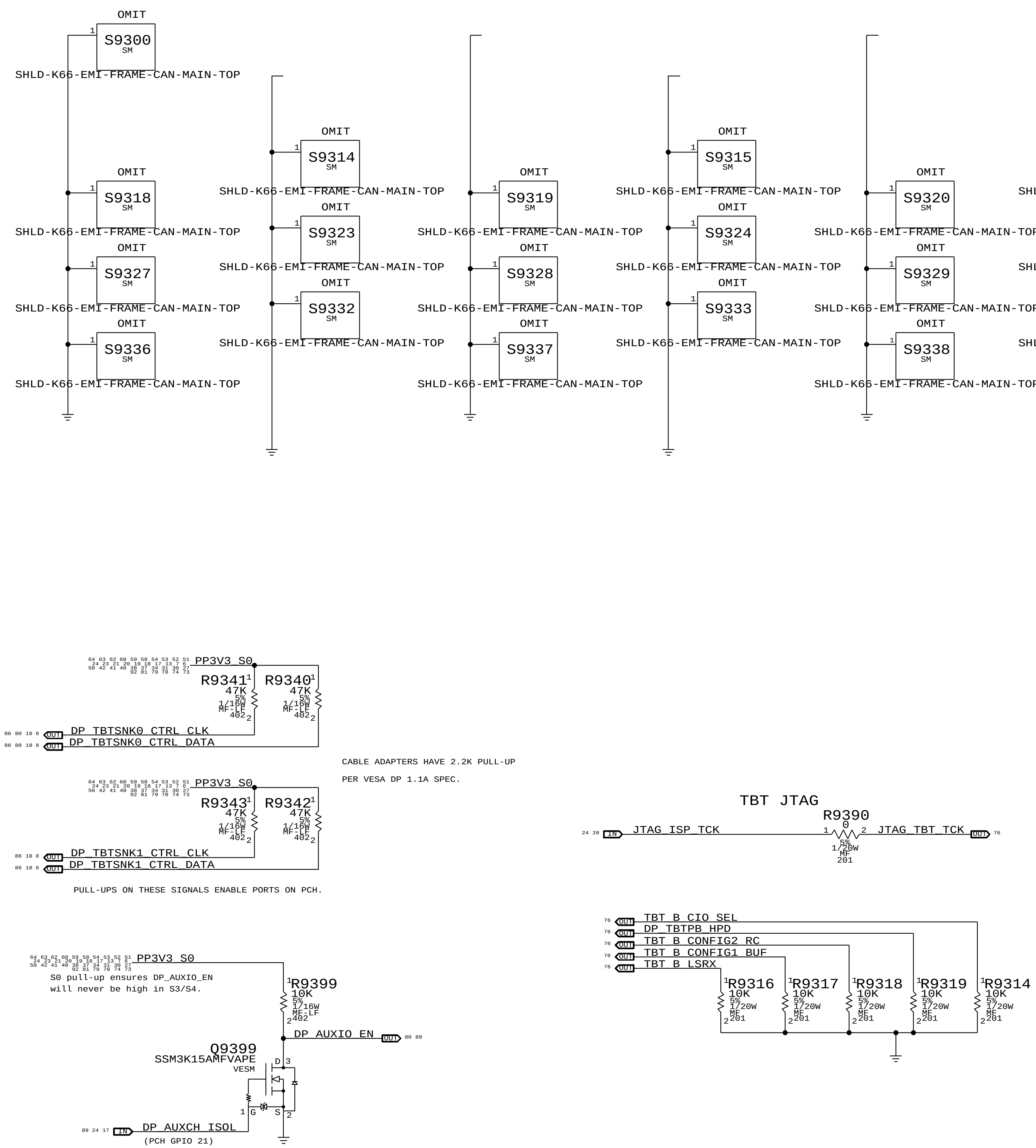

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

TBT MISC

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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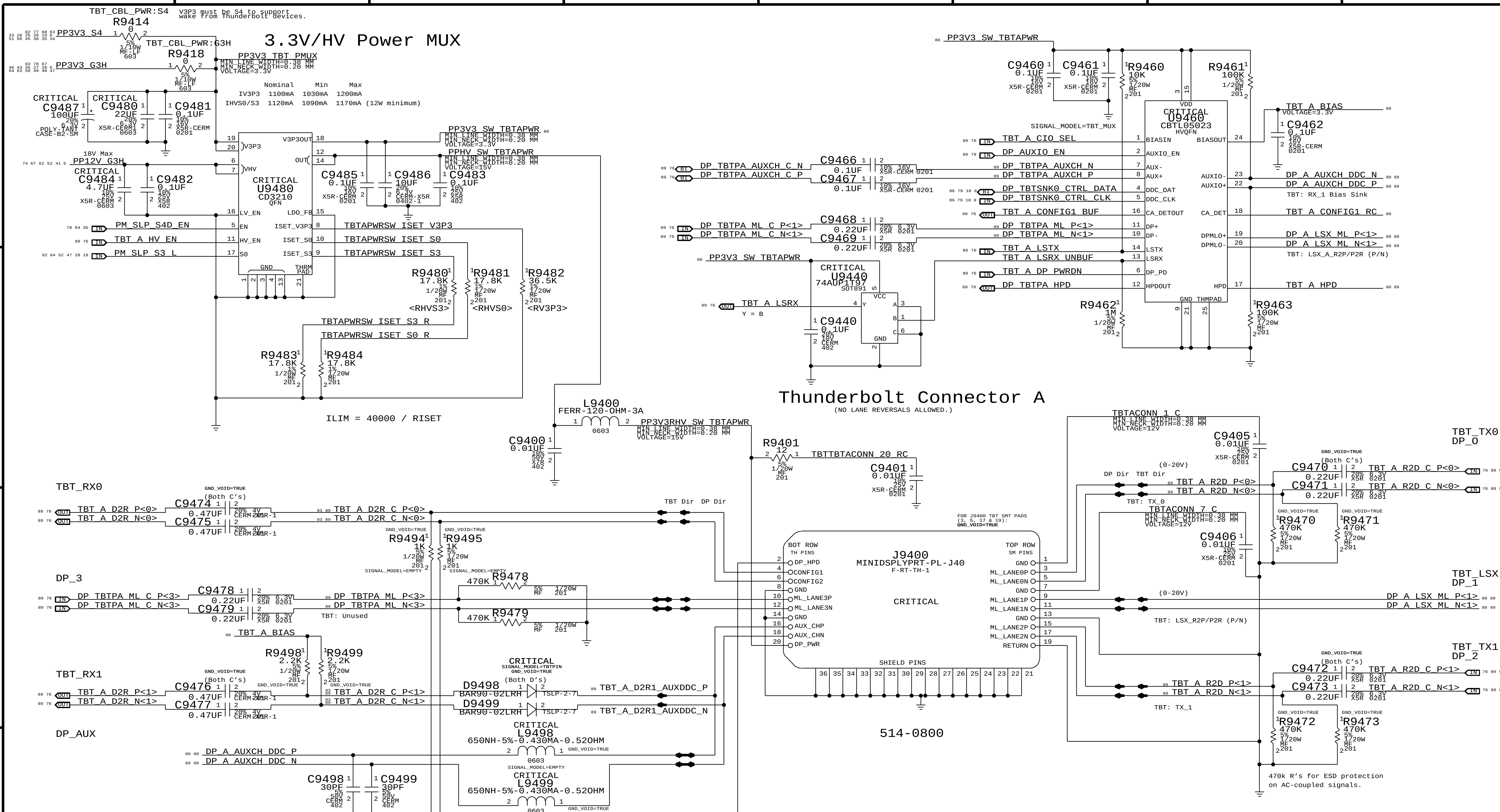

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
870-2198	10	SOLDER PREFORM 0201		CRITICAL	?

SF9400,SF9401,SF9402,SF9403,SF9404,SF9405,SF9406,SF9407,SF9408,SF9409

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
900-0064	870-2198	?	?	ALTERNATE APN

SYNC_MASTER=J5_KEPLER SYNC_DATE=07/20/2011

PAGE TITLE

Thunderbolt Connector A

Apple Inc.

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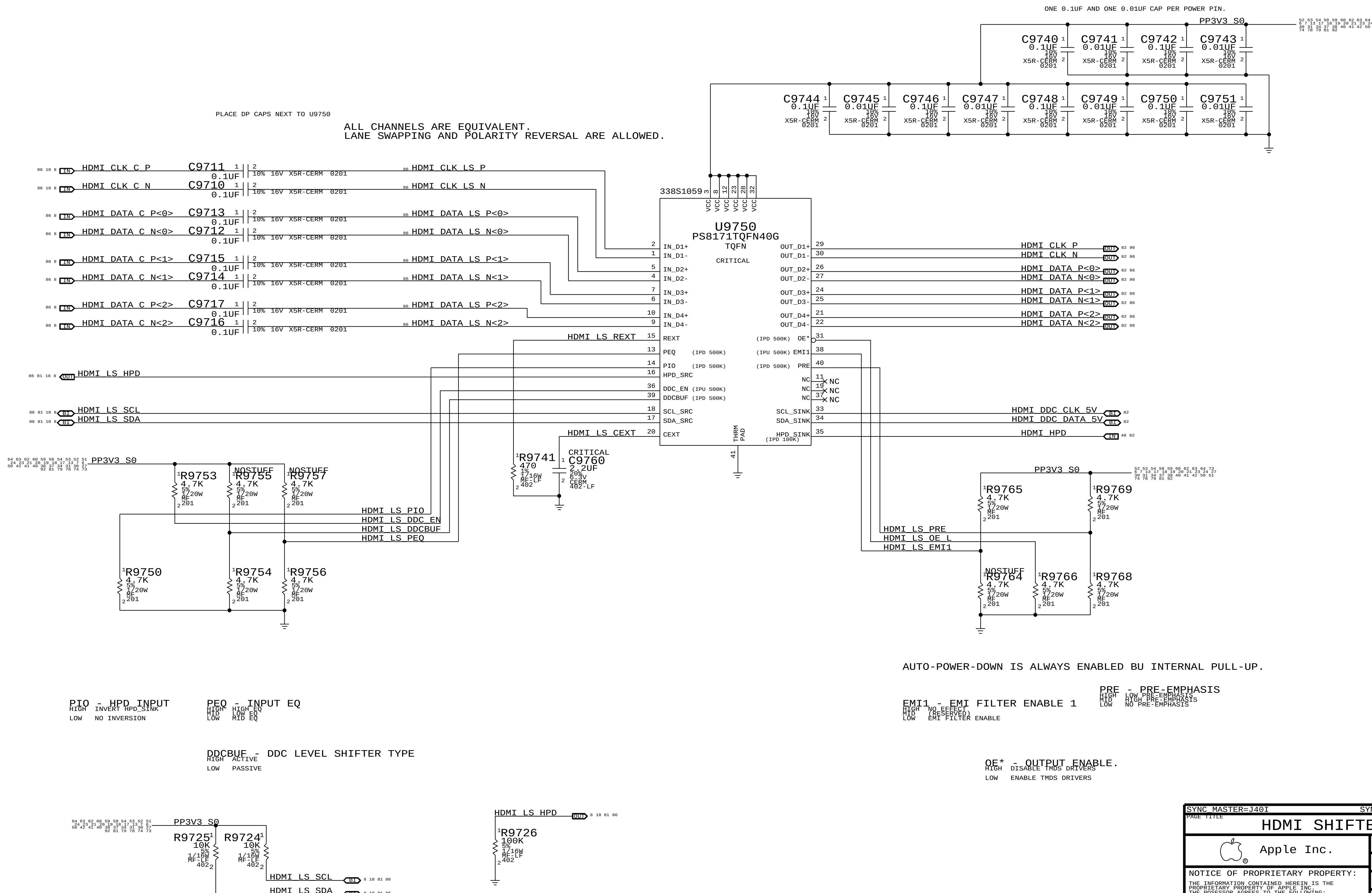

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

HDMI SHIFTER

Apple Inc.

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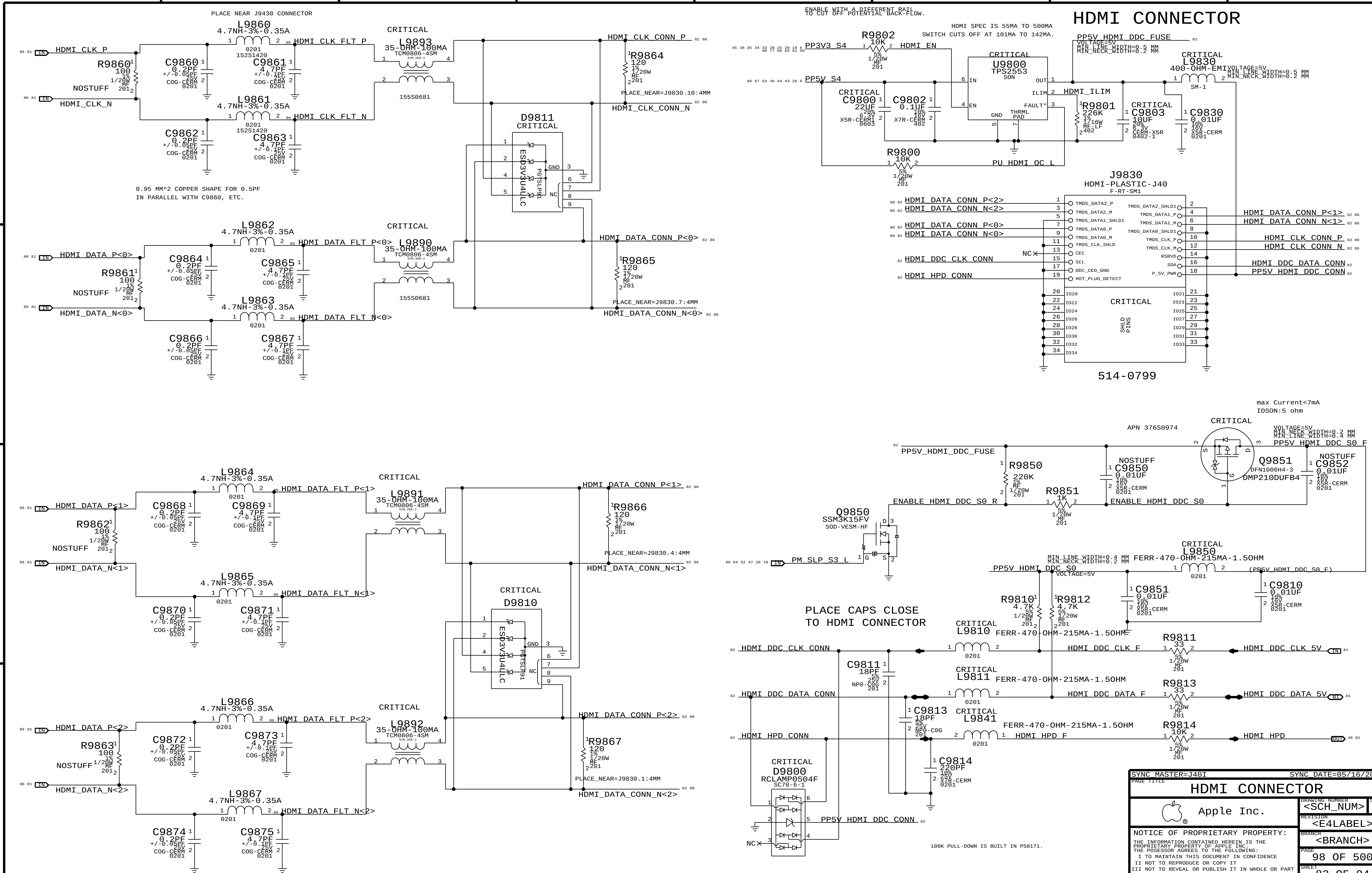
HDMI CONNECTOR



SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: HDMI CONNECTOR

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

8 7 6 5 4 3 2 1

D C B A

J50* 12 LAYER BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA	MM	16.2

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.310 MM	0.090 MM			
27P4_OHM_SE	*	Y	0.250 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP,BOTTOM	Y	0.185 MM	0.090 MM			
37_OHM_SE	*	Y	0.155 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.090 MM			
40_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.13 MM	0.13 MM			
45_OHM_SE	*	Y	0.099 MM	0.099 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.090 MM			
50_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	Y	0.154 MM	0.154 MM	=STANDARD	0.200 MM	0.200 MM
72_OHM_DIFF	TOP,BOTTOM	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.105 MM	0.105 MM	=STANDARD	0.120 MM	0.120 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.135 MM	0.135 MM		0.160 MM	0.160 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.110 MM	0.090 MM	=STANDARD	0.180 MM	0.180 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.090 MM		0.190 MM	0.190 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.102 MM	0.090 MM	=STANDARD	0.220 MM	0.220 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.115 MM	0.090 MM		0.230 MM	0.230 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.080 MM	0.080 MM	=STANDARD	0.200 MM	0.200 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.089 MM	0.089 MM		0.220 MM	0.220 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	Y	0.065 MM	0.065 MM	=STANDARD	0.200 MM	0.200 MM
110_OHM_DIFF	TOP,BOTTOM	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3,ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9,ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
BGA_P075	TOP	Y	0.110 MM	0.080 MM	10 MM		
BGA_P075	BOTTOM	Y	0.110 MM	0.073 MM	10 MM		

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.140 MM	?
3X_DIELECTRIC	*	0.210 MM	?
4X_DIELECTRIC	*	0.280 MM	?
5X_DIELECTRIC	*	0.350 MM	?
7X_DIELECTRIC	*	0.490 MM	?

NOTE: Based on K92 mlb stackup.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?
BGA_P075MM	*	0.075 MM	?

Breakout Constraint Overrides

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P075MM

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE
PCB Rule Definitions

Apple Inc.

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8 7 6 5 4 3 2 1

8	7	6	5	4	3	2	1

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_8MIL	*	8 MIL	?
CPU_COMP	*	20 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?
CPU_VID	*	0.457 MM	?

Most CPU signals with impedance requirements are 50-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: HR PDG section 4.1.9, PDDG section 2.5.1

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	=4X_DIELECTRIC	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
VRSENSE	*	Y	=27P4_OHM_SE	=27P4_OHM_SE	=1:1_DIFFPAIR	0.2 MM	0.2 MM

CPU Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
DMI_S2N	85_OHM_DIFF	PCIE	DMI_S2N_P<3:0>	10 18
DMI_S2N	85_OHM_DIFF	PCIE	DMI_S2N_N<3:0>	10 18
DMI_N2S	85_OHM_DIFF	PCIE	DMI_N2S_P<3:0>	10 18
DMI_N2S	85_OHM_DIFF	PCIE	DMI_N2S_N<3:0>	10 18
DMI_CLK100M	90_OHM_DIFF	CLK_PCIE	DMI_CLK100M_CPU_P	11 17
	90_OHM_DIFF	CLK_PCIE	DMI_CLK100M_CPU_N	11 17
FDI_DATA	85_OHM_DIFF	PCIE	FDI_DATA_P<7:0>	8 18
FDI_DATA	85_OHM_DIFF	PCIE	FDI_DATA_N<7:0>	8 18
	50_OHM_SE	CPU_AGTL	FDI_FSYNC<1..0>	8 18
	50_OHM_SE	CPU_AGTL	FDI_LSYNC<1..0>	8 18
	50_OHM_SE	CPU_AGTL	FDI_INT	8 10 18
	50_OHM_SE	CPU_VID	CPU_VCCSA_VID<1..0>	13 66
	50_OHM_SE	CPU_VID	CPU_VIDALERT_L	13 69
	50_OHM_SE	CPU_VID	CPU_VIDALERT_L_R	13
	50_OHM_SE	CPU_VID	CPU_VIDSCLK	13 69
	50_OHM_SE	CPU_VID	CPU_VIDSCLK_R	13
	50_OHM_SE	CPU_VID	CPU_VIDSOUT	13 69
	50_OHM_SE	CPU_VID	CPU_VIDSOUT_R	13
CPU_PECI	50_OHM_SE	PCIE	CPU_PECI	11 20 47 48
	50_OHM_SE	PCIE	CPU_PECI_R	48
PM_SYNC	50_OHM_SE	CPU_AGTL	PM_SYNC	11 18
PM_MEM_PWRGD	50_OHM_SE	CPU_AGTL	PM_MEM_PWRGD	11 18 28
CPU_SM_RCOMP	27P4_OHM_SE	CPU_COMP	CPU_SM_RCOMP<2..0>	11
CPU_CFG	50_OHM_SE	CPU_ITP	CPU_CFG<11..0>	10 24
CPU_CFG	50_OHM_SE	CPU_ITP	CPU_CFG<17..16>	10 24
CPU_CATERR_L	50_OHM_SE	CPU_AGTL	CPU_CATERR_L	6 11 47
	50_OHM_SE	CPU_AGTL	CPU_PROC_SEL_L	11 20
	50_OHM_SE	CPU_AGTL	CPU_VTTSELECT	
CPU_PROCHOT_L	50_OHM_SE	CPU_AGTL	CPU_PROCHOT_L	11 47 48 69
CPU_PWRGD	50_OHM_SE	CPU_AGTL	CPU_PWRGD	11 20 24
PM_THRMTRIP_L	50_OHM_SE	CPU_8MIL	PM_THRMTRIP_L	11 20 48
	55_OHM_SE	CPU_8MIL	CPU_PSI_L	
PM_DPRSLPVR	50_OHM_SE	CPU_AGTL	PM_DPRSLPVR	
	27P4_OHM_SE	CPU_COMP	CPU_PEG_COMP	10
	27P4_OHM_SE	CPU_COMP	CPU_PEG_RBIAS	
CPU_COMP	27P4_OHM_SE	CPU_COMP	CPU_COMP3	
CPU_COMP	27P4_OHM_SE	CPU_COMP	CPU_COMP2	
CPU_COMP	27P4_OHM_SE	CPU_COMP	CPU_COMP1	
CPU_COMP	27P4_OHM_SE	CPU_COMP	CPU_COMP0	
CPU_VCCSENSE	VRSENSE	CPU_VCCSENSE	CPU_VCCSENSE_P	13 69
CPU_VCCSENSE	VRSENSE	CPU_VCCSENSE	CPU_VCCSENSE_N	13 69
CPU_VCCSENSE	VRSENSE	CPU_VCCSENSE	CPU_VCCIOSENSE_P	13 72
CPU_VCCSENSE	VRSENSE	CPU_VCCSENSE	CPU_VCCIOSENSE_N	13 72
CPU_VCCSENSE	VRSENSE	CPU_VCCSENSE	CPU_AXG_SENSE_P	13 69
CPU_VCCSENSE	VRSENSE	CPU_VCCSENSE	CPU_AXG_SENSE_N	13 69
	VRSENSE	CPU_VCCSENSE	CPU_VCC_VALSENSE_P	13
	VRSENSE	CPU_VCCSENSE	CPU_VCC_VALSENSE_N	13
	VRSENSE	CPU_VCCSENSE	CPU_AXG_VALSENSE_P	13
	VRSENSE	CPU_VCCSENSE	CPU_AXG_VALSENSE_N	13
	VRSENSE		VR_AXG_ISNS_N	
	VRSENSE		VR_AXG_ISNS_P	
	VRSENSE		VR_AXG_VSNS_R_N	
	VRSENSE		VR_AXG_VSNS_R_P	
	VRSENSE		VR_CPU_ISNS1_N	
	VRSENSE		VR_CPU_ISNS1_P	
	VRSENSE		VR_CPU_ISNS2_N	
	VRSENSE		VR_CPU_ISNS2_P	
	VRSENSE		VR_CPU_ISNS3_N	
	VRSENSE		VR_CPU_ISNS3_P	
	VRSENSE		VR_CPU_VSNS_R_N	84
	VRSENSE		VR_CPU_VSNS_R_P	84
	VRSENSE		VR_CPU_VSNS_R_N	84
	VRSENSE		VR_CPU_VSNS_R_P	84
	55_OHM_SE	CPU_8MIL	GFX_VID<6..0>	
PM_DPRSLPVR	50_OHM_SE	CPU_AGTL	GFX_DPRSLPVR	
	50_OHM_SE	CPU_AGTL	GFX_VR_EN	
	50_OHM_SE	CPU_AGTL	GFXIMVP_IMON	
	85_OHM_DIFF	PCIE	PEG_R2D_P<7..0>	
	85_OHM_DIFF	PCIE	PEG_R2D_N<7..0>	
PEG_R2D	85_OHM_DIFF	PCIE	PEG_R2D_C_P<7..0>	
	85_OHM_DIFF	PCIE	PEG_R2D_C_N<7..0>	
PEG_D2R	85_OHM_DIFF	PCIE	PEG_D2R_P<7..0>	
	85_OHM_DIFF	PCIE	PEG_D2R_N<7..0>	
	85_OHM_DIFF	PCIE	PEG_D2R_C_P<7..0>	
	85_OHM_DIFF	PCIE	PEG_D2R_C_N<7..0>	

PEG CLOCK ON PP104 WITH PCIE CONSTRAINTS.

XDP Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
XDP_BPM	50_OHM_SE	CPU_ITP	XDP_BPM_L<3..0>	11 24
XDP_BPM_L	50_OHM_SE	CPU_ITP	XDP_BPM_L<7..4>	11 24
	50_OHM_SE	CPU_ITP	XDP_CPURST_L	24
CPU_CFG	50_OHM_SE	CPU_ITP	XDP_CPU_CFG<0>	24
XDP_PRDY_L	50_OHM_SE	CPU_ITP	XDP_CPU_PRDY_L	11 24
XDP_PREQ_l	50_OHM_SE	CPU_ITP	XDP_CPU_PREQ_L	11 24
XDP_TDI	50_OHM_SE	CPU_ITP	XDP_CPU_TDI	11 24 26
XDP_TDO	50_OHM_SE	CPU_ITP	XDP_CPU_TDO	11 24 26
XDP_TMS	50_OHM_SE	CPU_ITP	XDP_CPU_TMS	11 24 26
XDP_TCK	50_OHM_SE	CPU_ITP	XDP_CPU_TCK	11 24 26
XDP_TRST_L	50_OHM_SE	CPU_ITP	XDP_CPU_TRST_L	11 24 26
	50_OHM_SE	CPU_ITP	XDP_PCH_TCK	17 24 26 87
	50_OHM_SE	CPU_ITP	XDP_PCH_TDI	17 24 26 87
	50_OHM_SE	CPU_ITP	XDP_PCH_TDO	17 24 26 87
	50_OHM_SE	CPU_ITP	XDP_PCH_TMS	17 24 26 87
	50_OHM_SE	CPU_ITP	XDP_TDI	
	50_OHM_SE	CPU_ITP	XDP_TDO	
	50_OHM_SE	CPU_ITP	XDP_TMS	
XDP_CPU_PWRGOOD	50_OHM_SE	CPU_ITP	XDP_CPU_PWRGD	24
XDP_BDRESET_L	50_OHM_SE	CPU_ITP	XDP_DBRESET_L	11 24 27 87
	50_OHM_SE	CPU_ITP	XDP_PRESENT_L	
	50_OHM_SE	CPU_ITP	XDP_RES_RL	
	50_OHM_SE	CPU_ITP	XDP_CPU_PWRBTN_L	24
	50_OHM_SE	CPU_ITP	XDP_VR_READY	24
XDP_CLK_CPU	90_OHM_DIFF	CLK_PCIE	ITPCPU_CLK100M_P	11 17
XDP_CLK_CPU	90_OHM_DIFF	CLK_PCIE	ITPCPU_CLK100M_N	11 17
XDP_CLK_PCH	90_OHM_DIFF	CLK_PCIE	ITPXDP_CLK100M_P	17 24
XDP_CLK_PCH	90_OHM_DIFF	CLK_PCIE	ITPXDP_CLK100M_N	17 24
XDP_CLK_ITP	90_OHM_DIFF	CLK_PCIE	XDP_CPU_CLK100M_P	24
XDP_CLK_ITP	90_OHM_DIFF	CLK_PCIE	XDP_CPU_CLK100M_N	24
	50_OHM_SE	CPU_ITP	XDP_OBSDATA_B<3..0>	24

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CPU Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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A	B	C	D

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=3:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_2OTHER	*	25 MILS	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_CLK	*	MEM_DATA2MEM
MEM_DATA	MEM_CTRL	*	MEM_DATA2MEM
MEM_DATA	MEM_CMD	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA
MEM_DATA	MEM_DQS	*	MEM_DATA2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_2OTHER
MEM_CTRL	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_DATA	*	*	MEM_2OTHER
MEM_DQS	*	*	MEM_2OTHER

Need to support MEM_*-style wildcards!

DDR3:
DQ/DM signals should be matched within 0.508mm of associated DQS pair.
DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.
DQS to clock matching should be within [CLK-12.7mm] and [CLK+25.4mm].
CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.508mm.
CONTROL signals should be matched within [CLK-12.7mm] to [CLK+0.0mm] of CLK pairs.
A/BA/CMD signals should be matched within [CLK-12.7mm] to [CLK+12.7mm] of CLK pairs.
DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.
Maximum length of any signal from die pad to SODIMM pad is 139.7mm, from procesor ball to SODIMM pad is 114.3mm.

SOURCE: Calpella SFF Platform DG, Rev 1.5 (#407364), Section 2.2
SOURCE: HR PDG, Section 4.1.9

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
MEM_A_CLK	85_OHM_DIFF	MEM_CLK	MEM_A_CLK_P<5..0>	12 30
MEM_A_CLK	85_OHM_DIFF	MEM_CLK	MEM_A_CLK_N<5..0>	12 30
MEM_A_CNTL	40_OHM_SE	MEM_CTRL	MEM_A_CKE<3..0>	12 30
MEM_A_CNTL	40_OHM_SE	MEM_CTRL	MEM_A_CS_L<3..0>	12 30
MEM_A_CNTL	40_OHM_SE	MEM_CTRL	MEM_A_ODT<3..0>	12 30
MEM_A_CMD	40_OHM_SE	MEM_CMD	MEM_A_A<15..0>	12 30
MEM_A_CMD	40_OHM_SE	MEM_CMD	MEM_A_BA<2..0>	12 30
MEM_A_CMD	40_OHM_SE	MEM_CMD	MEM_A_RAS_L	12 30
MEM_A_CMD	40_OHM_SE	MEM_CMD	MEM_A_CAS_L	12 30
MEM_A_CMD	40_OHM_SE	MEM_CMD	MEM_A_WE_L	12 30
MEM_A_DQ_BYTE0	50_OHM_SE	MEM_DATA	MEM_A_DQ<7..0>	12 29
MEM_A_DQ_BYTE1	50_OHM_SE	MEM_DATA	MEM_A_DQ<15..8>	12 29
MEM_A_DQ_BYTE2	50_OHM_SE	MEM_DATA	MEM_A_DQ<23..16>	12 29
MEM_A_DQ_BYTE3	50_OHM_SE	MEM_DATA	MEM_A_DQ<31..24>	12 29
MEM_A_DQ_BYTE4	50_OHM_SE	MEM_DATA	MEM_A_DQ<39..32>	12 29 30
MEM_A_DQ_BYTE5	50_OHM_SE	MEM_DATA	MEM_A_DQ<47..40>	12 29
MEM_A_DQ_BYTE6	50_OHM_SE	MEM_DATA	MEM_A_DQ<55..48>	12 29
MEM_A_DQ_BYTE7	50_OHM_SE	MEM_DATA	MEM_A_DQ<63..56>	12 29
MEM_A_DQS0	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_P<0>	12 29
MEM_A_DQS0	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_N<0>	12 29
MEM_A_DQS1	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_P<1>	12 29
MEM_A_DQS1	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_N<1>	12 29
MEM_A_DQS2	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_P<2>	12 29
MEM_A_DQS2	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_N<2>	12 29
MEM_A_DQS3	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_P<3>	12 29
MEM_A_DQS3	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_N<3>	12 29
MEM_A_DQS4	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_P<4>	12 29
MEM_A_DQS4	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_N<4>	12 29
MEM_A_DQS5	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_P<5>	12 29
MEM_A_DQS5	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_N<5>	12 29
MEM_A_DQS6	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_P<6>	12 29 30
MEM_A_DQS6	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_N<6>	12 29 30
MEM_A_DQS7	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_P<7>	12 29
MEM_A_DQS7	85_OHM_DIFF	MEM_DQS	MEM_A_DQS_N<7>	12 29
MEM_B_CLK	85_OHM_DIFF	MEM_CLK	MEM_B_CLK_P<5..0>	12 31
MEM_B_CLK	85_OHM_DIFF	MEM_CLK	MEM_B_CLK_N<5..0>	12 31
MEM_B_CNTL	40_OHM_SE	MEM_CTRL	MEM_B_CKE<3..0>	12 31
MEM_B_CNTL	40_OHM_SE	MEM_CTRL	MEM_B_CS_L<3..0>	12 31
MEM_B_CNTL	40_OHM_SE	MEM_CTRL	MEM_B_ODT<3..0>	12 31
MEM_B_CMD	40_OHM_SE	MEM_CMD	MEM_B_A<15..0>	12 31
MEM_B_CMD	40_OHM_SE	MEM_CMD	MEM_B_BA<2..0>	12 31
MEM_B_CMD	40_OHM_SE	MEM_CMD	MEM_B_RAS_L	12 31
MEM_B_CMD	40_OHM_SE	MEM_CMD	MEM_B_CAS_L	12 31
MEM_B_CMD	40_OHM_SE	MEM_CMD	MEM_B_WE_L	12 31
MEM_B_DQ_BYTE0	50_OHM_SE	MEM_DATA	MEM_B_DQ<7..0>	12 29
MEM_B_DQ_BYTE1	50_OHM_SE	MEM_DATA	MEM_B_DQ<15..8>	12 29
MEM_B_DQ_BYTE2	50_OHM_SE	MEM_DATA	MEM_B_DQ<23..16>	12 29
MEM_B_DQ_BYTE3	50_OHM_SE	MEM_DATA	MEM_B_DQ<31..24>	12 29
MEM_B_DQ_BYTE4	50_OHM_SE	MEM_DATA	MEM_B_DQ<39..32>	12 29 31
MEM_B_DQ_BYTE5	50_OHM_SE	MEM_DATA	MEM_B_DQ<47..40>	12 29
MEM_B_DQ_BYTE6	50_OHM_SE	MEM_DATA	MEM_B_DQ<55..48>	12 29
MEM_B_DQ_BYTE7	50_OHM_SE	MEM_DATA	MEM_B_DQ<63..56>	12 29
MEM_B_DQS0	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_P<0>	12 29
MEM_B_DQS0	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_N<0>	12 29
MEM_B_DQS1	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_P<1>	12 29
MEM_B_DQS1	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_N<1>	12 29
MEM_B_DQS2	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_P<2>	12 29
MEM_B_DQS2	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_N<2>	12 29
MEM_B_DQS3	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_P<3>	12 29
MEM_B_DQS3	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_N<3>	12 29
MEM_B_DQS4	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_P<4>	12 29
MEM_B_DQS4	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_N<4>	12 29
MEM_B_DQS5	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_P<5>	12 29
MEM_B_DQS5	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_N<5>	12 29
MEM_B_DQS6	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_P<6>	12 29 31
MEM_B_DQS6	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_N<6>	12 29 31
MEM_B_DQS7	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_P<7>	12 29
MEM_B_DQS7	85_OHM_DIFF	MEM_DQS	MEM_B_DQS_N<7>	12 29

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PAGE TITLE
Memory Constraints
Apple Inc.
DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	-	=4:1_SPACING	?
TMDS	-	=4:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	TOP,BOTTOM	=4:1_SPACING	?
TMDS	TOP,BOTTOM	=4:1_SPACING	?

SOURCE: HR PDG, TABLES 191,193, Radeon TRM section 7.7.1
PER ANIL: MODIFIED DP CONNECTOR NEEDS 90 OHMS. MEASURE PROTO BOARD AND ADJUST IF NEEDED.

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=5:1_SPACING	?
SATA_ICOMP	*	15 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	TOP,BOTTOM	=5:1_SPACING	?

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

USB 2.0 AND 3.0 INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=4:1_SPACING	?
USB3	*	=5:1_SPACING	?
USB_RBIAS	*	15 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=4:1_SPACING	?

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

PCH USB NET PROPERTIES

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	USB_HUB_UP	85_OHM_DIFF	USB	USB_HUB_UP_N	19 25
		85_OHM_DIFF	USB	USB_HUB_UP_P	19 25
	USB_EXTA	85_OHM_DIFF	USB	USB_EXTA_N	19 43
		85_OHM_DIFF	USB	USB_EXTA_P	19 43
	USB_EXTB	85_OHM_DIFF	USB	USB_EXTB_N	25 43
		85_OHM_DIFF	USB	USB_EXTB_P	25 43
	USB_EXTC	85_OHM_DIFF	USB	USB_EXTC_N	19 44
		85_OHM_DIFF	USB	USB_EXTC_P	19 44
	USB_EXTD	85_OHM_DIFF	USB	USB_EXTD_N	25 44
		85_OHM_DIFF	USB	USB_EXTD_P	25 44
I325		85_OHM_DIFF	USB	USB2_EXTA_F_N	43
I326		85_OHM_DIFF	USB	USB2_EXTA_F_P	43
I327		85_OHM_DIFF	USB	USB2_EXTB_F_N	43
I328		85_OHM_DIFF	USB	USB2_EXTB_F_P	43
I329		85_OHM_DIFF	USB	USB2_EXTC_F_N	44
I330		85_OHM_DIFF	USB	USB2_EXTC_F_P	44
I331		85_OHM_DIFF	USB	USB2_EXTD_F_N	44
I332		85_OHM_DIFF	USB	USB2_EXTD_F_P	44
I365		85_OHM_DIFF	USB	USB2_EXTA_MUXED_N	43
I366		85_OHM_DIFF	USB	USB2_EXTA_MUXED_P	43
		85_OHM_DIFF	USB	USB_EXTB_XHCI_P	19 25
		85_OHM_DIFF	USB	USB_EXTB_XHCI_N	19 25
		85_OHM_DIFF	USB	USB_EXTD_XHCI_P	19 25
		85_OHM_DIFF	USB	USB_EXTD_XHCI_N	19 25
		85_OHM_DIFF	USB	USB_EXTB_EHCI_P	19 25
		85_OHM_DIFF	USB	USB_EXTB_EHCI_N	19 25
		85_OHM_DIFF	USB	USB_EXTD_EHCI_P	19 25
		85_OHM_DIFF	USB	USB_EXTD_EHCI_N	19 25
	USB_SMC	85_OHM_DIFF	USB	USB_SMC_P	25 47
		85_OHM_DIFF	USB	USB_SMC_N	25 47
		85_OHM_DIFF	USB	USB_SMC_SR_P	25
		85_OHM_DIFF	USB	USB_SMC_SR_N	25
	USB_IR	85_OHM_DIFF	USB	USB_IR_N	25 45
		85_OHM_DIFF	USB	USB_IR_P	25 45
		85_OHM_DIFF	USB	USB_IR_R_N	45
		85_OHM_DIFF	USB	USB_IR_R_P	45
	USB_BT	85_OHM_DIFF	USB	USB_BT_N	25 35
		85_OHM_DIFF	USB	USB_BT_P	25 35
		85_OHM_DIFF	USB	USB_BT_CONN_N	35 93
		85_OHM_DIFF	USB	USB_BT_CONN_P	35 93
I386		85_OHM_DIFF	USB	USB_BT_SW_N	35
I387		85_OHM_DIFF	USB	USB_BT_SW_P	35
I340	USB3_EXTA_RX	85_OHM_DIFF	USB3	USB3_EXTA_RX_N	19 43
I339		85_OHM_DIFF	USB3	USB3_EXTA_RX_P	19 43
I338	USB3_EXTB_RX	85_OHM_DIFF	USB3	USB3_EXTB_RX_N	19 43
I337		85_OHM_DIFF	USB3	USB3_EXTB_RX_P	19 43
I336	USB3_EXTC_RX	85_OHM_DIFF	USB3	USB3_EXTC_RX_N	19 44
I335		85_OHM_DIFF	USB3	USB3_EXTC_RX_P	19 44
I334	USB3_EXTD_RX	85_OHM_DIFF	USB3	USB3_EXTD_RX_N	19 44
I333		85_OHM_DIFF	USB3	USB3_EXTD_RX_P	19 44
I348	USB3_EXTA_TX	85_OHM_DIFF	USB3	USB3_EXTA_TX_N	19 43
I347		85_OHM_DIFF	USB3	USB3_EXTA_TX_P	19 43
I345	USB3_EXTB_TX	85_OHM_DIFF	USB3	USB3_EXTB_TX_N	19 43
I346		85_OHM_DIFF	USB3	USB3_EXTB_TX_P	19 43
I344	USB3_EXTC_TX	85_OHM_DIFF	USB3	USB3_EXTC_TX_N	19 44
I343		85_OHM_DIFF	USB3	USB3_EXTC_TX_P	19 44
I341	USB3_EXTD_TX	85_OHM_DIFF	USB3	USB3_EXTD_TX_N	19 44
I342		85_OHM_DIFF	USB3	USB3_EXTD_TX_P	19 44
I349		85_OHM_DIFF	USB3	USB3_EXTA_RX_F_N	43 93
I350		85_OHM_DIFF	USB3	USB3_EXTA_RX_F_P	43 93
I351		85_OHM_DIFF	USB3	USB3_EXTB_RX_F_N	43
I352		85_OHM_DIFF	USB3	USB3_EXTB_RX_F_P	43
I353		85_OHM_DIFF	USB3	USB3_EXTC_RX_F_N	44 93
I354		85_OHM_DIFF	USB3	USB3_EXTC_RX_F_P	44 93
I355		85_OHM_DIFF	USB3	USB3_EXTD_RX_F_N	44
I356		85_OHM_DIFF	USB3	USB3_EXTD_RX_F_P	44
I357		85_OHM_DIFF	USB3	USB3_EXTA_TX_F_N	43 93
I358		85_OHM_DIFF	USB3	USB3_EXTA_TX_F_P	43 93
I359		85_OHM_DIFF	USB3	USB3_EXTB_TX_F_N	43
I360		85_OHM_DIFF	USB3	USB3_EXTB_TX_F_P	43
I361		85_OHM_DIFF	USB3	USB3_EXTC_TX_F_N	44 93
I362		85_OHM_DIFF	USB3	USB3_EXTC_TX_F_P	44 93
I363		85_OHM_DIFF	USB3	USB3_EXTD_TX_F_N	44
I364		85_OHM_DIFF	USB3	USB3_EXTD_TX_F_P	44
I379		85_OHM_DIFF	USB3	USB3_EXTA_RX_C_P	
I380		85_OHM_DIFF	USB3	USB3_EXTB_RX_C_N	
I381		85_OHM_DIFF	USB3	USB3_EXTB_RX_C_P	
I383		85_OHM_DIFF	USB3	USB3_EXTC_RX_C_N	
I382		85_OHM_DIFF	USB3	USB3_EXTC_RX_C_P	
I384		85_OHM_DIFF	USB3	USB3_EXTD_RX_C_N	
I385		85_OHM_DIFF	USB3	USB3_EXTD_RX_C_P	
I371		85_OHM_DIFF	USB3	USB3_EXTA_TX_C_N	43
I373		85_OHM_DIFF	USB3	USB3_EXTA_TX_C_P	43
I372		85_OHM_DIFF	USB3	USB3_EXTB_TX_C_N	43
I374		85_OHM_DIFF	USB3	USB3_EXTB_TX_C_P	43
I375		85_OHM_DIFF	USB3	USB3_EXTC_TX_C_N	44
I377		85_OHM_DIFF	USB3	USB3_EXTC_TX_C_P	44
I376		85_OHM_DIFF	USB3	USB3_EXTD_TX_C_N	44
I378		85_OHM_DIFF	USB3	USB3_EXTD_TX_C_P	44

DP Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	8 18 76
		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	8 18 76
		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	76
		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	76
		55_OHM_SE	DISPLAYPORT	DP_TBTSNK0_CTRL_CLK	8 18 79 80
		55_OHM_SE	DISPLAYPORT	DP_TBTSNK0_CTRL_DATA	8 18 79 80
		55_OHM_SE	DISPLAYPORT	DP_TBTSNK0_HPD	8 18 76
		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_N	8 18 76
		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_P	8 18 76
		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK0_AUXCH_N	76
		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK0_AUXCH_P	76
I318		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	8 18 76
I319		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	8 18 76
I320		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	76
I321		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	76
I322		55_OHM_SE	DISPLAYPORT	DP_TBTSNK1_CTRL_CLK	8 18 79
I323		55_OHM_SE	DISPLAYPORT	DP_TBTSNK1_CTRL_DATA	8 18 79
I324		55_OHM_SE	DISPLAYPORT	DP_TBTSNK1_HPD	8 18 76
I313		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_N	8 18 76
I314		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_P	8 18 76
I315		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK1_AUXCH_N	76
I316		85_OHM_DIFF	DISPLAYPORT	DP_TBTSNK1_AUXCH_P	76

HDMI Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
		100_OHM_DIFF	TMDS	HDMI_CLK_CONN_N	82
		100_OHM_DIFF	TMDS	HDMI_CLK_CONN_P	82
I367		100_OHM_DIFF	TMDS	HDMI_CLK_FLT_N	82
I368		100_OHM_DIFF	TMDS	HDMI_CLK_FLT_P	82
		85_OHM_DIFF	TMDS	HDMI_CLK_C_N	8 18 81
		85_OHM_DIFF	TMDS	HDMI_CLK_C_P	8 18 81
		85_OHM_DIFF	TMDS	HDMI_CLK_LS_N	81
		85_OHM_DIFF	TMDS	HDMI_CLK_LS_P	81
		100_OHM_DIFF	TMDS	HDMI_CLK_N	81 82
		100_OHM_DIFF	TMDS	HDMI_CLK_P	81 82
		100_OHM_DIFF	TMDS	HDMI_DATA_CONN_N<2..0>	82
		100_OHM_DIFF	TMDS	HDMI_DATA_CONN_P<2..0>	82
I369		100_OHM_DIFF	TMDS	HDMI_DATA_FLT_N<2..0>	82
I370		100_OHM_DIFF	TMDS	HDMI_DATA_FLT_P<2..0>	82
		85_OHM_DIFF	TMDS	HDMI_DATA_C_N<2..0>	8 81
		85_OHM_DIFF	TMDS	HDMI_DATA_C_P<2..0>	8 81
		85_OHM_DIFF	TMDS	HDMI_DATA_LS_N<2..0>	81
		85_OHM_DIFF	TMDS	HDMI_DATA_LS_P<2..0>	81
		100_OHM_DIFF	TMDS	HDMI_DATA_N<2..0>	81 82
		100_OHM_DIFF	TMDS	HDMI_DATA_P<2..0>	81 82
		55_OHM_SE	TMDS	HDMI_LS_SCL	8 18 81
		55_OHM_SE	TMDS	HDMI_LS_SDA	8 18 81
		55_OHM_SE	TMDS	HDMI_LS_HPD	8 18 81

SATA PROPERTIES

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	SATA_HDD_D2R	90_OHM_DIFF	SATA	SATA_HDD1_D2R_CONN_N	42 93
		90_OHM_DIFF	SATA	SATA_HDD1_D2R_CONN_P	42 93
	SATA_HDD_D2R	90_OHM_DIFF	SATA	SATA_HDD1_D2R_N	17 42
		90_OHM_DIFF	SATA	SATA_HDD1_D2R_P	17 42
	SATA_HDD_R2D	90_OHM_DIFF	SATA	SATA_HDD1_R2D_CONN_N	42 93
		90_OHM_DIFF	SATA	SATA_HDD1_R2D_CONN_P	42 93
	SATA_HDD_R2D	90_OHM_DIFF	SATA	SATA_HDD1_R2D_C_N	17 42
		90_OHM_DIFF	SATA	SATA_HDD1_R2D_C_P	17 42
	SATA_HDD_D2R	90_OHM_DIFF	SATA	SATA_HDD2_D2R_CONN_N	42 93
		90_OHM_DIFF	SATA	SATA_HDD2_D2R_CONN_P	42 93
	SATA_HDD_D2R	90_OHM_DIFF	SATA	SATA_HDD2_D2R_N	17 42
		90_OHM_DIFF	SATA	SATA_HDD2_D2R_P	17 42
	SATA_HDD_R2D	90_OHM_DIFF	SATA	SATA_HDD2_R2D_CONN_N	42 93
		90_OHM_DIFF	SATA	SATA_HDD2_R2D_CONN_P	42 93
	SATA_HDD_R2D	90_OHM_DIFF	SATA	SATA_HDD2_R2D_C_N	17 42
		90_OHM_DIFF	SATA	SATA_HDD2_R2D_C_P	17 42
I312	SATA_HDD_D2R	90_OHM_DIFF	SATA	SATA_HDD1_D2R_DF_N	42
I311		90_OHM_DIFF	SATA	SATA_HDD1_D2R_DF_P	42
I309	SATA_HDD_R2D	90_OHM_DIFF	SATA	SATA_HDD1_R2D_DF_N	42
I310		90_OHM_DIFF	SATA	SATA_HDD1_R2D_DF_P	42
I307	SATA_HDD_D2R	90_OHM_DIFF	SATA	SATA_HDD2_D2R_DF_N	42
I308		90_OHM_DIFF	SATA	SATA_HDD2_D2R_DF_P	42
I305	SATA_HDD_R2D	90_OHM_DIFF	SATA	SATA_HDD2_R2D_DF_N	42
I306		90_OHM_DIFF	SATA	SATA_HDD2_R2D_DF_P	42
	PCH_SATA3_ICOMP	50_OHM_SE	SATA_ICOMP	PCH_SATA3COMP	17
	PCH_SATA_ICOMP	37_OHM_SE	SATA_ICOMP	PCH_SATAICOMP	17
	PCH_USB_RBIAS		USB_RBIAS	PCH_USB_RBIAS	19

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PAGE TITLE
PCH Constraints 1

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	0.2 MM	0.2 MM

PSEUDO DIFF PAIR WITH 0.2MM SPACING

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	LPC_AD	50_OHM_SE	LPC	LPC_AD<3..0>	17 47 49
I273		50_OHM_SE	LPC	LPC_AD_R<3..0>	17
	LPC_FRAME_L	50_OHM_SE	LPC	LPC_FRAME_L	17 47 49
		50_OHM_SE	LPC	LPC_FRAME_R_L	17
I272	LPC_RESET_L	50_OHM_SE	LPC	LPCPLUS_RESET_L	27 49
	PCH_LPC_CLK0	50_OHM_SE	CLK_LPC	LPC_CLK33M_SMC_R	19 27
		50_OHM_SE	CLK_LPC	LPC_CLK33M_SMC	27 47
		50_OHM_SE	CLK_LPC	LPC_CLK33M_LPCPLUS	27 49
I271		50_OHM_SE	CLK_LPC	LPC_CLK33M_LPCPLUS_R	19 27
I282		SMB_50S	SMB	SMB_PCH_STR_SCL	50
I283		SMB_50S	SMB	SMB_PCH_STR_SDA	50
I280		SMB_50S	SMB	SMB_PCH_DIMM_SCL	30 31 50
I281		SMB_50S	SMB	SMB_PCH_DIMM_SDA	30 31 50
I278		SMB_50S	SMB	SMB_PCH_MKY_SCL	50 59 60
I279		SMB_50S	SMB	SMB_PCH_MKY_SDA	50 59 60
I276		SMB_50S	SMB	SMB_PCH_VRF_SCL	33 50
I277		SMB_50S	SMB	SMB_PCH_VRF_SDA	33 50
	SMBUS_PCH_CLK	SMB_50S	SMB	SMBUS_PCH_CLK	
		SMB_50S	SMB	SMBUS_PCH_DATA	
	SMBUS_PCH_0_CLK	SMB_50S	SMB	SMB_PCH_XDP_SCL	
		SMB_50S	SMB	SMB_PCH_XDP_SDA	
		SMB_50S	SMB	SML_PCH_1_CLK	17 50
		SMB_50S	SMB	SML_PCH_1_DATA	17 50
	SPI_CLK	55_OHM_SE	SPI	SPI_CLK_R	17 49
		55_OHM_SE	SPI	SPI_CLK	49 93
	SPI_MOSI	55_OHM_SE	SPI	SPI_MOSI_R	17 49
		55_OHM_SE	SPI	SPI_MOSI	49 93
	SPI_MISO	55_OHM_SE	SPI	SPI_MISO	17 49 93
	SPI_CS0	55_OHM_SE	SPI	SPI_CS0_R_L	17 49
		55_OHM_SE	SPI	SPI_CS0_L	49 93
	HDA_BIT_CLK	50_OHM_SE	HDA	HDA_BIT_CLK	17 54
		50_OHM_SE	HDA	HDA_BIT_CLK_R	17
	HDA_SYNC	50_OHM_SE	HDA	HDA_SYNC	17 54
		50_OHM_SE	HDA	HDA_SYNC_R	17
	HDA_RST_L	50_OHM_SE	HDA	HDA_RST_R_L	17
		50_OHM_SE	HDA	HDA_RST_L	17 54
	HDA_SDIN0	50_OHM_SE	HDA	HDA_SDIN0	17 54
		50_OHM_SE	HDA	AUD_SDI_R	54
	HDA_SDOUT	50_OHM_SE	HDA	HDA_SDOUT	17 54
		50_OHM_SE	HDA	HDA_SDOUT_R	17 27
I255		90_OHM_DIFF	CLK_PCIE	PCH_CLK96M_DOT_P	17
I257		90_OHM_DIFF	CLK_PCIE	PCH_CLK96M_DOT_N	17
I256		90_OHM_DIFF	CLK_PCIE	PCH_CLK100M_SATA_P	17
I259		90_OHM_DIFF	CLK_PCIE	PCH_CLK100M_SATA_N	17
I258		50_OHM_SE	CLK_PCIE	PCH_CLK14P3M_REFCLK	17
I260		50_OHM_SE	CLK_PCIE	PCH_CLK33M_PCIIN	17 27
I285		50_OHM_SE	PCIE	PCH_PECI	20
I284		50_OHM_SE	CPU_ITP	XDP_DA0_USB_EXTA_OC_L	24
I295		50_OHM_SE	CPU_ITP	XDP_DA1_USB_EXTB_OC_L	24
I294		50_OHM_SE	CPU_ITP	XDP_DA2_USB_EXTC_OC_L	24
I293		50_OHM_SE	CPU_ITP	XDP_DA3_USB_EXTD_OC_L	24
I292		50_OHM_SE	CPU_ITP	XDP_DB0_USB_EXTB_OC_EHCI_L	24
I291		50_OHM_SE	CPU_ITP	XDP_DB1_USB_EXTD_OC_EHCI_L	24
I290		50_OHM_SE	CPU_ITP	XDP_DB2_AP_PWR_EN	24
I289		50_OHM_SE	CPU_ITP	XDP_DB3_SDCONN_STATE_CHANGE	24
I288		50_OHM_SE	CPU_ITP	XDP_PCH_S5_PWRGD	24
I287		50_OHM_SE	CPU_ITP	XDP_PCH_PWRBTN_L	24
I286		SMB_50S	CPU_ITP	SMB_CPU_XDP_SDA	24 50
I305		SMB_50S	CPU_ITP	SMB_CPU_XDP_SCL	24 50
I304		50_OHM_SE	CPU_ITP	XDP_PCH_TCK	17 24 26 84
I303		50_OHM_SE	CPU_ITP	XDP_FC0	24
I302		50_OHM_SE	CPU_ITP	XDP_FC1	24
I301		50_OHM_SE	CPU_ITP	XDP_DC0_ISOLATE_CPU_MEM_L	24
I300		50_OHM_SE	CPU_ITP	XDP_DC1_MXM_GOOD	24
I299		50_OHM_SE	CPU_ITP	XDP_DC2_DP_AUXCH_ISOL	24
I298		50_OHM_SE	CPU_ITP	XDP_DC3_SATARDRVR_EN	24
I297		50_OHM_SE	CPU_ITP	XDP_DD0_DP_GPU_TBT_SEL	24
I296		50_OHM_SE	CPU_ITP	XDP_DD1_JTAG_ISP_TCK	24
I315		50_OHM_SE	CPU_ITP	XDP_DD2_AUD_IPHS_SWITCH_EN	24
I314		50_OHM_SE	CPU_ITP	XDP_DD3_ENET_LOW_PWR	24
I313		50_OHM_SE	CPU_ITP	XDPPCH_PLTRST_L	24 27
I312		50_OHM_SE	CPU_ITP	XDP_DBRESET_L	11 24 27 84
I311		50_OHM_SE	CPU_ITP	XDP_PCH_TDO	17 24 26 84
I309		50_OHM_SE	CPU_ITP	XDP_PCH_TDI	17 24 26 84 87
I308		50_OHM_SE	CPU_ITP	XDP_PCH_TDI	17 24 26 84 87
I307		50_OHM_SE	CPU_ITP	XDP_PCH_TMS	17 24 26 84

NET	DIFFERENTIAL_PAIR	PAGES
SMBUS_PCH_CLK	SMBUS_PCH	17 50
SMBUS_PCH_DATA	SMBUS_PCH	17 50
SMB_PCH_XDP_SCL	SMBUS_PCH_0	24 50
SMB_PCH_XDP_SDA	SMBUS_PCH_0	24 50

	ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
		85_OHM_DIFF	PCIE	PCIE_ENET_R2D_P	37
		85_OHM_DIFF	PCIE	PCIE_ENET_R2D_N	37
	PCIE_ENET_R2D	85_OHM_DIFF	PCIE	PCIE_ENET_R2D_C_P	17 37
		85_OHM_DIFF	PCIE	PCIE_ENET_R2D_C_N	17 37
	PCIE_ENET_D2R	85_OHM_DIFF	PCIE	PCIE_ENET_D2R_P	17 37
		85_OHM_DIFF	PCIE	PCIE_ENET_D2R_N	17 37
		85_OHM_DIFF	PCIE	PCIE_ENET_D2R_C_P	37
		85_OHM_DIFF	PCIE	PCIE_ENET_D2R_C_N	37
		85_OHM_DIFF	PCIE	PCIE_AP_R2D_P	35 93
		85_OHM_DIFF	PCIE	PCIE_AP_R2D_N	35 93
	PCIE_AP_R2D	85_OHM_DIFF	PCIE	PCIE_AP_R2D_C_P	17 35
		85_OHM_DIFF	PCIE	PCIE_AP_R2D_C_N	17 35
	PCIE_AP_D2R	85_OHM_DIFF	PCIE	PCIE_AP_D2R_P	17 35 93
		85_OHM_DIFF	PCIE	PCIE_AP_D2R_N	17 35 93
		85_OHM_DIFF	PCIE	PCIE_FW_R2D_P	39
		85_OHM_DIFF	PCIE	PCIE_FW_R2D_N	39
	PCIE_FW_R2D	85_OHM_DIFF	PCIE	PCIE_FW_R2D_C_P	17 39
		85_OHM_DIFF	PCIE	PCIE_FW_R2D_C_N	17 39
	PCIE_FW_D2R	85_OHM_DIFF	PCIE	PCIE_FW_D2R_P	17 39
		85_OHM_DIFF	PCIE	PCIE_FW_D2R_N	17 39
		85_OHM_DIFF	PCIE	PCIE_FW_D2R_C_P	39
		85_OHM_DIFF	PCIE	PCIE_FW_D2R_C_N	39
I253		90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_PCH_P	17
I254		90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_PCH_N	17
I262	PCIE_CLK100M_T29_	90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_TBT_P	17 76
I261		90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_TBT_N	17 76
	PCIE_CLK100M	90_OHM_DIFF	CLK_PCIE	NC_PEG_CLK100MP	8 17 93
		90_OHM_DIFF	CLK_PCIE	NC_PEG_CLK100MN	8 17 93
	PCIE_CLK100M_ENET	90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_ENET_P	17 37
		90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_ENET_N	17 37
I275	PCIE_CLK100M_AP	90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_AP_CONN_P	35 93
I274		90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_AP_CONN_N	35 93
	PCIE_CLK100M_AP	90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_AP_P	17 35
		90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_AP_N	17 35
	PCIE_CLK100M_FW	90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_FW_P	17 39
		90_OHM_DIFF	CLK_PCIE	PCIE_CLK100M_FW_N	17 39
I263	PCIE_T29_R2D	85_OHM_DIFF	PCIE	PCIE_TBT_R2D_C_P<3..0>	8 10 76
I264	PCIE_T29_R2D	85_OHM_DIFF	PCIE	PCIE_TBT_R2D_C_N<3..0>	8 10 76
I265	PCIE_T29_R2D	85_OHM_DIFF	PCIE	PCIE_TBT_R2D_P<3..0>	76
I267	PCIE_T29_R2D	85_OHM_DIFF	PCIE	PCIE_TBT_R2D_N<3..0>	76
I266	PCIE_T29_D2R	85_OHM_DIFF	PCIE	PCIE_TBT_D2R_P<3..0>	8 10 76
I268	PCIE_T29_D2R	85_OHM_DIFF	PCIE	PCIE_TBT_D2R_N<3..0>	8 10 76
I270	PCIE_T29_D2R	85_OHM_DIFF	PCIE	PCIE_TBT_D2R_C_P<3..0>	76
I269	PCIE_T29_D2R	85_OHM_DIFF	PCIE	PCIE_TBT_D2R_C_N<3..0>	76

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

PCH Constraints 2

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

ETHERNET CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_3X	*	=3:1_SPACING	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	0.6 MM	?
2KV_SPACING	*	0.50 MM	100
2KV_SPACING	TOP,BOTTOM	1.27 MM	100

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_HV	*	*	2KV_SPACING
ENET_HV	ENET_HV	*	ENET_MDI

NOTE: 2.2kV isolation needed on all primary-side ethernet signals. Min 1.27mm separation from all other copper on the board!

SD CARD INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_INTERFACE	*	=3X_DIELECTRIC	?

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

Ethernet Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
		50_OHM_SE	ENET_3X	SYSCLK_CLK25M_ENET	26 37
		50_OHM_SE	ENET_3X	TP_ENET_CLK25M_XTALO	26 37
		50_OHM_SE	ENET_3X	ENET_RESET_L	27 34
	ENET_MDI	100_OHM_DIFF	ENET_MDI	ENET_MDI_P<3..0>	36 37
		100_OHM_DIFF	ENET_MDI	ENET_MDI_N<3..0>	36 37
I174		100_OHM_DIFF	ENET_HV	ENET_MDI_TRAN_N<3..0>	36
I175		100_OHM_DIFF	ENET_HV	ENET_MDI_TRAN_P<3..0>	36
I176		50_OHM_SE	ENET_HV	ENET_CENTER_TAP<3..0>	36
I177			ENET_HV	ENET_CMODE_REF	36

High-voltage isolated ethernet signals.

SD Interface Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
I178		55_OHM_SE	SD_INTERFACE	SDCONN_CLK	34 37
I187		55_OHM_SE	SD_INTERFACE	SDCONN_CLK_FF	34
I180		55_OHM_SE	SD_INTERFACE	SDCONN_CMD	34 37
I181		55_OHM_SE	SD_INTERFACE	SDCONN_DATA<7..0>	34 37
I188		55_OHM_SE	SD_INTERFACE	SDCONN_CLK_R	34
I189		55_OHM_SE	SD_INTERFACE	SDCONN_CMD_R	34
I190		55_OHM_SE	SD_INTERFACE	SDCONN_DATA_R<7..0>	34
I182		55_OHM_SE	SD_INTERFACE	ENET_CR_CLK	37
I183		55_OHM_SE	SD_INTERFACE	ENET_CR_CMD	37
I184		55_OHM_SE	SD_INTERFACE	ENET_CR_DATA<7..0>	37
I191		55_OHM_SE	SD_INTERFACE	SDCONNF_CLK	34
I192		55_OHM_SE	SD_INTERFACE	SDCONNF_CLK_FF	34

FireWire Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
I166		110_OHM_DIFF	FW_TP	FW_PORT0_TPAL_N	
I167		110_OHM_DIFF	FW_TP	FW_PORT0_TPAL_P	
I168		110_OHM_DIFF	FW_TP	FW_PORT1_TPA_N	39 41
I169	FW_P1_TPA	110_OHM_DIFF	FW_TP	FW_PORT1_TPA_P	39 41
I170		110_OHM_DIFF	FW_TP	FW_PORT0_TPBL_N	
I171		110_OHM_DIFF	FW_TP	FW_PORT0_TPBL_P	
I172		110_OHM_DIFF	FW_TP	FW_PORT1_TPB_N	39 41
I173	FW_P1_TPB	110_OHM_DIFF	FW_TP	FW_PORT1_TPB_P	39 41
I185		50_OHM_SE	CLK_PCIE	FW_CLK24P576M_XI	39
I186		50_OHM_SE	CLK_PCIE	FW_CLK24P576M_XO	39
I193		50_OHM_SE	CLK_PCIE	FW_CLK24P576M_XO_R	39

Port 0 and 2 Not Used

DISPLAYPORT SIGNAL CONSTRAINTS

NOTE: DISPLAYPORT PHYSICAL/SPACING CONSTRAINTS PROVIDED BY CHIPSET OR GPU PAGE.

TBT I2C SIGNAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_I2C	*	=2X_DIELECTRIC	?

TBT SPI SIGNAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2X_DIELECTRIC	?

TBT/DP CONNECTOR SIGNAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP	*	=5X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP	TOP,BOTTOM	=7X_DIELECTRIC	?

SOURCE: BILL CORNELIUS'S TBT ROUTING NOTES

TBT IC NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	SMB_50S	TBT_I2C	I2C_TBTRTR_SCL	50 76
	SMB_50S	TBT_I2C	I2C_TBTRTR_SDA	50 76
TBT_SPI_CLK	55_OHM_SE	TBT_SPI	TBT_SPI_CLK	76
TBT_SPI_MOSI	55_OHM_SE	TBT_SPI	TBT_SPI_MOSI	76
TBT_SPI_MISO	55_OHM_SE	TBT_SPI	TBT_SPI_MISO	76
TBT_SPI_CS_L	55_OHM_SE	TBT_SPI	TBT_SPI_CS_L	76
I276		TBT_I2C	TBT_A_LSTX	76 80
I286		TBT_I2C	TBT_A_LSRX	76 80
I285		TBT_I2C	TBT_A_CONFIG1_BUF	76 80
I284		TBT_I2C	TBT_A_CONFIG2_RC	76 80
I283		TBT_I2C	TBT_A_HV_EN	76 80
I282		TBT_I2C	TBT_A_CIO_SEL	76 80
I281		TBT_I2C	TBT_A_DP_PWRDN	76 80
I280		TBT_I2C	DP_TBTPA_HPD	76 80
I279		TBT_I2C	TBT_A_HPD	80
I278		TBT_I2C	DP_AUXIO_EN	79 80
I277		TBT_I2C	DP_AUXCH_ISOL	17 24 79

TBT/DP NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
TBT_R2D0	85_OHM_DIFF	TBTDP	TBT_A_R2D_P<0>	80
TBT_R2D0	85_OHM_DIFF	TBTDP	TBT_A_R2D_N<0>	80
TBT_R2D1	85_OHM_DIFF	TBTDP	TBT_A_R2D_P<1>	80
TBT_R2D1	85_OHM_DIFF	TBTDP	TBT_A_R2D_N<1>	80
TBT_D2R0	85_OHM_DIFF	TBTDP	TBT_A_D2R_P<0>	76 80
TBT_D2R0	85_OHM_DIFF	TBTDP	TBT_A_D2R_N<0>	76 80
TBT_D2R1	85_OHM_DIFF	TBTDP	TBT_A_D2R_P<1>	76 80
TBT_D2R1	85_OHM_DIFF	TBTDP	TBT_A_D2R_N<1>	76 80
	85_OHM_DIFF	TBTDP	TBT_A_R2D_C_P<1..0>	76 80 93
	85_OHM_DIFF	TBTDP	TBT_A_R2D_C_N<1..0>	76 80 93
	85_OHM_DIFF	TBTDP	TBT_A_D2R_C_P<1..0>	80 93
	85_OHM_DIFF	TBTDP	TBT_A_D2R_C_N<1..0>	80 93
I287	85_OHM_DIFF	TBTDP	TBT_A_D2R1_AUXDDC_P	80
I288	85_OHM_DIFF	TBTDP	TBT_A_D2R1_AUXDDC_N	80
	85_OHM_DIFF	TBTDP	DP_A_AUXCH_DDC_P	80
	85_OHM_DIFF	TBTDP	DP_A_AUXCH_DDC_N	80
	85_OHM_DIFF	DISPLAYPORT	DP_A_LSX_ML_P<1>	80
	85_OHM_DIFF	DISPLAYPORT	DP_A_LSX_ML_N<1>	80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_AUXCH_P	80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_AUXCH_N	80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_AUXCH_C_P	76 80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_AUXCH_C_N	76 80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_ML_P<3>	80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_ML_N<3>	80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	76 80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	76 80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_ML_P<1>	80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_ML_N<1>	80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	76 80
	85_OHM_DIFF	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	76 80

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: TBT CONSTRAINTS

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1TO1_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	DIFFERENTIAL_PAIR	NET	PAGES
SMBUS_SMC_A_S3_SCL	SMB_50S	SMB	SMBUS_SMC_2	SMBUS_SMC_2_S3_SCL	47 50
	SMB_50S	SMB	SMBUS_SMC_2	SMBUS_SMC_2_S3_SDA	47 50
SMBUS_SMC_B_S0_SCL	SMB_50S	SMB	SMBUS_SMC_1	SMBUS_SMC_1_S0_SCL	47 50
	SMB_50S	SMB	SMBUS_SMC_1	SMBUS_SMC_1_S0_SDA	47 50
SMBUS_SMC_0_S0_SCL	SMB_50S	SMB	SMBUS_SMC_0	SMBUS_SMC_0_S0_SCL	47 50 53
	SMB_50S	SMB	SMBUS_SMC_0	SMBUS_SMC_0_S0_SDA	47 50 53
SMBUS_SMC_BSA_SCL	SMB_50S	SMB	SMBUS_SMC_ASF	ENET_ASF_SMB_CLK	37 47 50
	SMB_50S	SMB	SMBUS_SMC_ASF	ENET_ASF_SMB_DATA	37 47 50
SMBUS_SMC_MGMT_SCL	SMB_50S	SMB	SMBUS_SMC_MGMT	SMBUS_SMC_MGMT_SCL	
	SMB_50S	SMB	SMBUS_SMC_MGMT	SMBUS_SMC_MGMT_SDA	
I83	SMB_50S	SMB		SMB_SMC_1_STR_SCL	50
I84	SMB_50S	SMB		SMB_SMC_1_STR_SDA	50
I85	SMB_50S	SMB		SMB_SMC_B_TSN_SCL	50 53
I86	SMB_50S	SMB		SMB_SMC_B_TSN_SDA	50 53

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

SMC Constraints

DRAWING NUMBER <SCH_NUM> SIZE D

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GDDR5 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR5_45R50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12.7 MM	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	*	=5x_DIELECTRIC	?
GDDR5_CMD	*	=2x_DIELECTRIC	?
GDDR5_DATA	*	=3x_DIELECTRIC	?
GDDR5_EDC	*	=7x_DIELECTRIC	?

GDDR5 FB B Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	
FB_B0_CLK	80_OHM_DIFF	GDDR5_CLK	FB_B0_CLK_P
FB_B0_CLK	80_OHM_DIFF	GDDR5_CLK	FB_B0_CLK_N
FB_B1_CLK	80_OHM_DIFF	GDDR5_CLK	FB_B1_CLK_P
FB_B1_CLK	80_OHM_DIFF	GDDR5_CLK	FB_B1_CLK_N
FB_B0_CMD	45_OHM_SE	GDDR5_CMD	FB_B0_A<8..0>
FB_B1_CMD	45_OHM_SE	GDDR5_CMD	FB_B1_A<8..0>
FB_B0_CMD	45_OHM_SE	GDDR5_CMD	FB_B0_ABI_L
FB_B1_CMD	45_OHM_SE	GDDR5_CMD	FB_B1_ABI_L
FB_B0_CMD	45_OHM_SE	GDDR5_CMD	FB_B0_RAS_L
FB_B1_CMD	45_OHM_SE	GDDR5_CMD	FB_B1_RAS_L
FB_B0_CMD	45_OHM_SE	GDDR5_CMD	FB_B0_CAS_L
FB_B1_CMD	45_OHM_SE	GDDR5_CMD	FB_B1_CAS_L
FB_B0_CMD	45_OHM_SE	GDDR5_CMD	FB_B0_WE_L
FB_B1_CMD	45_OHM_SE	GDDR5_CMD	FB_B1_WE_L
FB_B0_CMD_R	45_OHM_SE	GDDR5_CMD	FB_B0_CKE_L
FB_B1_CMD_R	45_OHM_SE	GDDR5_CMD	FB_B1_CKE_L
FB_B0_CMD	45_OHM_SE	GDDR5_CMD	FB_B0_CS_L
FB_B1_CMD	45_OHM_SE	GDDR5_CMD	FB_B1_CS_L
FB_B0_EDC0	45_OHM_SE	GDDR5_EDC	FB_B0_EDC<0>
FB_B0_EDC1	45_OHM_SE	GDDR5_EDC	FB_B0_EDC<1>
FB_B0_EDC2	45_OHM_SE	GDDR5_EDC	FB_B0_EDC<2>
FB_B0_EDC3	45_OHM_SE	GDDR5_EDC	FB_B0_EDC<3>
FB_B1_EDC0	45_OHM_SE	GDDR5_EDC	FB_B1_EDC<0>
FB_B1_EDC1	45_OHM_SE	GDDR5_EDC	FB_B1_EDC<1>
FB_B1_EDC2	45_OHM_SE	GDDR5_EDC	FB_B1_EDC<2>
FB_B1_EDC3	45_OHM_SE	GDDR5_EDC	FB_B1_EDC<3>
FB_B0_DBI_L0	45_OHM_SE	GDDR5_DATA	FB_B0_DBI_L<0>
FB_B0_DBI_L1	45_OHM_SE	GDDR5_DATA	FB_B0_DBI_L<1>
FB_B0_DBI_L2	45_OHM_SE	GDDR5_DATA	FB_B0_DBI_L<2>
FB_B0_DBI_L3	45_OHM_SE	GDDR5_DATA	FB_B0_DBI_L<3>
FB_B1_DBI_L0	45_OHM_SE	GDDR5_DATA	FB_B1_DBI_L<0>
FB_B1_DBI_L1	45_OHM_SE	GDDR5_DATA	FB_B1_DBI_L<1>
FB_B1_DBI_L2	45_OHM_SE	GDDR5_DATA	FB_B1_DBI_L<2>
FB_B1_DBI_L3	45_OHM_SE	GDDR5_DATA	FB_B1_DBI_L<3>
FB_B0_WCLK0	80_OHM_DIFF	GDDR5_CMD	FB_B0_WCLK_P<0>
FB_B0_WCLK0	80_OHM_DIFF	GDDR5_CMD	FB_B0_WCLK_N<0>
FB_B0_WCLK1	80_OHM_DIFF	GDDR5_CMD	FB_B0_WCLK_P<1>
FB_B0_WCLK1	80_OHM_DIFF	GDDR5_CMD	FB_B0_WCLK_N<1>
FB_B1_WCLK0	80_OHM_DIFF	GDDR5_CMD	FB_B1_WCLK_P<0>
FB_B1_WCLK0	80_OHM_DIFF	GDDR5_CMD	FB_B1_WCLK_N<0>
FB_B1_WCLK1	80_OHM_DIFF	GDDR5_CMD	FB_B1_WCLK_P<1>
FB_B1_WCLK1	80_OHM_DIFF	GDDR5_CMD	FB_B1_WCLK_N<1>
FB_B0_DQ_BYTE0	45_OHM_SE	GDDR5_DATA	FB_B0_DQ<7..0>
FB_B0_DQ_BYTE1	45_OHM_SE	GDDR5_DATA	FB_B0_DQ<15..8>
FB_B0_DQ_BYTE2	45_OHM_SE	GDDR5_DATA	FB_B0_DQ<23..16>
FB_B0_DQ_BYTE3	45_OHM_SE	GDDR5_DATA	FB_B0_DQ<31..24>
FB_B1_DQ_BYTE0	45_OHM_SE	GDDR5_DATA	FB_B1_DQ<7..0>
FB_B1_DQ_BYTE1	45_OHM_SE	GDDR5_DATA	FB_B1_DQ<15..8>
FB_B1_DQ_BYTE2	45_OHM_SE	GDDR5_DATA	FB_B1_DQ<23..16>
FB_B1_DQ_BYTE3	45_OHM_SE	GDDR5_DATA	FB_B1_DQ<31..24>
FB_B0_CMD_R	45_OHM_SE	GDDR5_CMD	FB_B0_RESET_L
FB_B1_CMD_R	45_OHM_SE	GDDR5_CMD	FB_B1_RESET_L

Kepler Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	
GPU_CLK27M	55_OHM_SE	CLK_SLOW	GPU_OSC_27M_XTALIN
GPU_CLK27M	55_OHM_SE	CLK_SLOW	GPU_OSC_27M_XTALOUT
GPU_CLK27M	55_OHM_SE	CLK_SLOW	GPU_OSC_27M_XTAL_BUFFOUT
GPU_CLK27M	55_OHM_SE	CLK_SLOW	GPU_OSC_27M_SSIN
	1:1_DIFFPAIR		PEX_TSTCLK_O_P
	1:1_DIFFPAIR		PEX_TSTCLK_O_N

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

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GPU (KEPLAR) CONSTRAINTS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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BRANCH <BRANCH>

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1TO1_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1TO1_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR
AUDIODIFF	*	=1:1_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SATA	SB_POWER	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP

J50* SPECIFIC NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_VDDQ_P	51
	SENSE_1TO1_55S	SENSE	ISNS_VDDQ_N	51
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_P1V05PCH_P	51
	SENSE_1TO1_55S	SENSE	ISNS_P1V05PCH_N	51
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	P1V5S3_CS_P	51 68
	SENSE_1TO1_55S	SENSE	P1V5S3_CS_N	51 68
SENSE_DIFFPAIR	VRSENSE	SENSE	CPUVCCIOS0_CS_P	51 72
	VRSENSE	SENSE	CPUVCCIOS0_CS_N	51 72
	VRSENSE	SENSE	CPUVCCIOS0_VO	72
	VRSENSE	SENSE	CPUVCCIOS0_OCSET	72
SENSE_DIFFPAIR	VRSENSE	SENSE	VCCSAS0_CS_P	51 66
	VRSENSE	SENSE	VCCSAS0_CS_N	51 66
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_P12VG3H_P	52
	SENSE_1TO1_55S	SENSE	ISNS_P12VG3H_N	52
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_HS_CPUDDR_P	52
	SENSE_1TO1_55S	SENSE	ISNS_HS_CPUDDR_N	52
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_WLAN_P	35 52
	SENSE_1TO1_55S	SENSE	ISNS_WLAN_N	35 52
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_HS_GPU_P	
	SENSE_1TO1_55S	SENSE	ISNS_HS_GPU_N	
	50_OHM_SE		AUD_HPAMP_INL_N	56
	50_OHM_SE		AUD_HPAMP_INL_P	56
	50_OHM_SE		AUD_HPAMP_INR_N	56
	50_OHM_SE		AUD_HPAMP_INR_P	56
	50_OHM_SE		EXT_MIC_N	59 60
	50_OHM_SE		EXT_MIC_P	59 60
AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO3_L_P	54 57
	AUDIODIFF	AUDIO	AUD_LO3_L_N	54 57
	AUDIODIFF	AUDIO	AUD_SPKRAMP_IN_L_P	57
	AUDIODIFF	AUDIO	AUD_SPKRAMP_IN_L_N	57
	AUDIODIFF	AUDIO	AUD_SPKRAMP_IN_P	57
	AUDIODIFF	AUDIO	AUD_SPKRAMP_IN_N	57
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	AMBIENT_THRMD_N	53
	THERM_1TO1_55S	THERM	AMBIENT_THRMD_P	53
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	CPU_PROX_THRM_N	53
	THERM_1TO1_55S	THERM	CPU_PROX_THRM_P	53
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	DIMM_PROX_THRM_N	53
	THERM_1TO1_55S	THERM	DIMM_PROX_THRM_P	53
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	WIFI_PROX_THRM_N	53
	THERM_1TO1_55S	THERM	WIFI_PROX_THRM_P	53
SENSE_DIFFPAIR	DIFFPAIR		ISNS_TBT1V05_N	78
	DIFFPAIR		ISNS_TBT1V05_P	78
SENSE_DIFFPAIR	DIFFPAIR		ISNS_TBT3V3_N	78
	DIFFPAIR		ISNS_TBT3V3_P	78
SENSE_DIFFPAIR	DIFFPAIR		CPU_VDDQ_SENSE_N	
	DIFFPAIR		CPU_VDDQ_SENSE_P	
	50_OHM_SE		VR_CPU_VSNS_XW_N	
	50_OHM_SE		VR_CPU_VSNS_XW_P	
	50_OHM_SE		VR_AXG_VSNS_XW_N	
	50_OHM_SE		VR_AXG_VSNS_XW_P	
	50_OHM_SE		POWER_BTN_N	46 93
	50_OHM_SE		POWER_BTN_P	46 93
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GFXCORE_N	
	SENSE_1TO1_55S	SENSE	GFXCORE_P	
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GFXIMVP6_CS_R_P	
	SENSE_1TO1_55S	SENSE	GFXIMVP6_CS_R_N	
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GFXIMVP6_CS_P	
	SENSE_1TO1_55S	SENSE	GFXIMVP6_CS_N	
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GPU_FBVDDQ_SENSE_P	
	SENSE_1TO1_55S	SENSE	GPU_FBVDDQ_SENSE_N	
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GPUFB_CS_P	
	SENSE_1TO1_55S	SENSE	GPUFB_CS_N	
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GPUVCORE_SENSE_P	
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GPUVCORE_SENSE_N	
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	P1V05_GPU_CS_P	
	SENSE_1TO1_55S	SENSE	P1V05_GPU_CS_N	
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	P1V05_GPU_PEX_IOVDD_SNS_P	
	SENSE_1TO1_55S	SENSE	P1V05_GPU_PEX_IOVDD_SNS_N	
SENSE_DIFFPAIR	VRSENSE	SENSE	CPUIMVP_ISNS1_N	51 69 70
	VRSENSE	SENSE	CPUIMVP_ISNS1_P	51 69 70
SENSE_DIFFPAIR	VRSENSE	SENSE	CPUIMVP_ISNS2_N	51 70
	VRSENSE	SENSE	CPUIMVP_ISNS2_P	51 69 70
SENSE_DIFFPAIR	VRSENSE	SENSE	CPUIMVP_ISNS3_N	51 70
	VRSENSE	SENSE	CPUIMVP_ISNS3_P	51 69 70
SENSE_DIFFPAIR	VRSENSE	SENSE	CPUIMVP_ISNS1G_N	51 71
	VRSENSE	SENSE	CPUIMVP_ISNS1G_P	51 71
SENSE_DIFFPAIR	VRSENSE	SENSE	CPUIMVP_ISNS2G_N	51 71
	VRSENSE	SENSE	CPUIMVP_ISNS2G_P	51 71

J50* SPECIFIC NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
		SB_POWER	PP3V3_S5	6 17 18 19 20 21 23 24 27 28 61 63 64 73
		SB_POWER	PP3V3_S0	52 53 54 58 59 60 62 63 64 73 6 7 13 17 18 19 20 21 23 24 27 30 31 34 37 38 40 41 42 50 51 74 78 79 81
		GND	GND	

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	CPU_THERMD_P	10 53
	THERM_1TO1_55S	THERM	CPU_THERMD_N	10 53
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	GPUTHMSNS_D_P	
	THERM_1TO1_55S	THERM	GPUTHMSNS_D_N	
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	GPU_TDIODE_P	
	THERM_1TO1_55S	THERM	GPU_TDIODE_N	
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	CPU_PROX_THERM_P	
	THERM_1TO1_55S	THERM	CPU_PROX_THERM_N	
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	WIFI_PROX_THERM_P	
	THERM_1TO1_55S	THERM	WIFI_PROX_THERM_N	
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	DIMM_PROX_THERM_P	
	THERM_1TO1_55S	THERM	DIMM_PROX_THERM_N	
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	PCH_THRMD_P	53
	THERM_1TO1_55S	THERM	PCH_THRMD_N	53
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	SMT_THRM_SNS3_P	53
	THERM_1TO1_55S	THERM	SMT_THRM_SNS3_N	53
SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	TBT_THRM_SNS3_P	
	THERM_1TO1_55S	THERM	TBT_THRM_SNS3_N	

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PAGE TITLE
Project Specific Constraints

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AIRPORT/BT/CAMERA CONN.

FUNC_TEST

	TRUE	
PCIE_AP_D2R_P	TRUE	17 35 87
PCIE_AP_D2R_N	TRUE	17 35 87
PCIE_AP_R2D_P	TRUE	35 87
PCIE_AP_R2D_N	TRUE	35 87
PCIE_CLK100M_AP_CONN_P	TRUE	35 87
PCIE_CLK100M_AP_CONN_N	TRUE	35 87
AP_CLKREQ_Q_L	TRUE	35
PCIE_WAKE_L	TRUE	18 35 38
AP_RESET_CONN_L	TRUE	35
WLAN_THROTTLE_L	TRUE	35
PP3V3_G3H_BT_F	TRUE	35
PP3V3_WLAN (3 TPS)	TRUE	35 48
USB_BT_CONN_P	TRUE	35 86
USB_BT_CONN_N	TRUE	35 86
GND (3 TPS)	TRUE	

Speaker Connectors

FUNC_TEST

	TRUE	
SPKRAMP_M_P_OUT	TRUE	57 58
SPKRAMP_M_N_OUT	TRUE	57 58

FAN CONNECTORS

FUNC_TEST

	TRUE	
PP12V_S0_FAN0_F	TRUE	62
FAN0_CTL_F	TRUE	62
FAN0_TACH_F	TRUE	62
GND (1 TPS)	TRUE	

PSU CONNECTOR

FUNC_TEST

	TRUE	
PP12V_G3H_PSU (4 TPS)	TRUE	6 52 74
PSU_TEMP	TRUE	47 74
GND (4 TPS)	TRUE	

SATA HDD CONNECTOR

FUNC_TEST

	TRUE	
PP5V_S0 (2 TPS)	TRUE	70 71 72 73 / 7 23 27 41 42 / 49 63 66 69 / 93
SATA_HDD1_D2R_CONN_P	TRUE	42 86
SATA_HDD1_D2R_CONN_N	TRUE	42 86
SATA_HDD1_R2D_CONN_N	TRUE	42 86
SATA_HDD1_R2D_CONN_P	TRUE	42 86
SSD1_OOBD2R_L	TRUE	42
SSD1_OOBR2D_L	TRUE	42
GND (2 TPS)	TRUE	

SATA HDD2 CONNECTOR

FUNC_TEST

	TRUE	
PP5V_S0 (2 TPS)	TRUE	70 71 72 73 / 7 23 27 41 42 / 49 63 66 69 / 93
SATA_HDD2_D2R_CONN_N	TRUE	42 86
SATA_HDD2_D2R_CONN_P	TRUE	42 86
SATA_HDD2_R2D_CONN_N	TRUE	42 86
SATA_HDD2_R2D_CONN_P	TRUE	42 86
SSD2_OOBD2R_L	TRUE	42
SSD2_OOBR2D_L	TRUE	42
GND (2 TPS)	TRUE	

IR/LED,POWER BUTTON CONNECTORS

FUNC_TEST

	TRUE	
PPIRRCVR_F (1 TPS)	TRUE	46
SMC_SYS_LED_DR	TRUE	46
IRRCVR_OUT	TRUE	45 46
POWER_BTN_N	TRUE	46 92
POWER_BTN_P	TRUE	46 92
GND_IR_F	TRUE	46
GND (1 TPS)	TRUE	

SPI BOOTROM SIGNALS

FUNC_TEST

	TRUE	
PP3V3_G3H (1 TPS)	TRUE	63 64 67 78 80 / 6 20 27 35 43 / 46 47 48 49 50
SPI_CS0_L	TRUE	49 87
SPI_CLK	TRUE	49 87
SPI_MOSI	TRUE	49 87
SPI_MISO	TRUE	17 49 87
SPIROM_USE_MLB	TRUE	20 49 61
GND (1 TPS)	TRUE	

NC_ NETS

NOT ALL NETS APPLY TO ALL SCHEMATIC VARIANTS.

NO_TEST

	TRUE	
NC_AUD_GPIO_0	TRUE	54
NC_AUD_HP_PORT_R	TRUE	54
NC_AUD_MIC_INN_R	TRUE	54
NC_CPU_VCCIO_SEL	TRUE	
NC_DP_TBTPB_AUXCH_CN	TRUE	76
NC_DP_TBTPB_ML_CP<1..3>	TRUE	76
NC_FDI_PCH_DATAN<7..0>	TRUE	
NC_FDI_PCH_INT	TRUE	
NC_FW643_FW620_L	TRUE	39
NC_FW643_VBUF	TRUE	39
NC_GPU_GPIO_20_RSVD	TRUE	
NC_HDA_SDIN2	TRUE	
NC_IG_D_AUXN	TRUE	18
NC_IG_DP_TBTSNK0_AUXCH_CP	TRUE	
NC_IG_DP_TBTSNK0_HPD	TRUE	
NC_IG_DP_TBTSNK1_AUXCH_CN	TRUE	
NC_IG_DP_TBTSNK1_CTRL_DATA	TRUE	
NC_IG_DP_TBTSNK1_ML_CP<3..0>	TRUE	
NC_IG_HDMI_DATA_CN<2..0>	TRUE	
NC_IG_HDMI_LS_SCL	TRUE	
NC_PCH_CLKOUT_DPP	TRUE	17
NC_PCH_GPIO66_CLKOUTFLEX2	TRUE	17
NC_PCI_PME_L	TRUE	
NC_PCIE_CLK100M_PE4N	TRUE	17
NC_PCIE_CLK100M_PE5P	TRUE	17
NC_PCIE_CLK100M_PEGAN	TRUE	17
NC_PE_1F_D2RP<3..0>	TRUE	8 10
NC_PEG_CLK100MN	TRUE	8 17 87
NC_PEG_D2RP<7..0>	TRUE	8 10
NC_POWER_LEVEL	TRUE	
NC_SMC_ADC22	TRUE	47
NC_SMC_FAN_1_TACH	TRUE	47
NC_SMC_ONOFF_L	TRUE	48
NC_SMC_PN2	TRUE	47
NC_SMC_PN5	TRUE	47
NC_SMC_TPAD_RST_L	TRUE	48
NC_TBT_B_D2RN<0..1>	TRUE	76
NC_TBT_B_R2D_CP<0..1>	TRUE	76

NO_TEST

	TRUE	
NC_AUD_GPIO_2	TRUE	54
NC_AUD_LO3_N_R	TRUE	54
NC_AUD_MIC_INP_R	TRUE	54
NC_CS4206_FP	TRUE	54
NC_DP_TBTPB_AUXCH_CP	TRUE	76
NC_FDI_CPU_DATAN<7..0>	TRUE	
NC_FDI_PCH_DATAP<7..0>	TRUE	
NC_FDI_PCH_LSYNC<1..0>	TRUE	
NC_FW643_OCR10_CTL	TRUE	39
NC_GFX_THROTTLE_L	TRUE	47 75
NC_GPU_GPIO_21_RSVD	TRUE	
NC_HDA_SDIN3	TRUE	
NC_IG_D_AUXP	TRUE	18
NC_IG_DP_TBTSNK0_CTRL_CLK	TRUE	
NC_IG_DP_TBTSNK0_ML_CN<3..0>	TRUE	
NC_IG_DP_TBTSNK1_AUXCH_CP	TRUE	
NC_IG_DP_TBTSNK1_HPD	TRUE	
NC_IG_HDMI_CLK_CN	TRUE	
NC_IG_HDMI_DATA_CP<2..0>	TRUE	
NC_IG_HDMI_LS_SDA	TRUE	
NC_PCH_GPIO64_CLKOUTFLEX0	TRUE	17
NC_PCH_GPIO67_CLKOUTFLEX3	TRUE	17
NC_PCIE_CLK100M_EXCARDN	TRUE	17
NC_PCIE_CLK100M_PE4P	TRUE	17
NC_PCIE_CLK100M_PEBN	TRUE	17
NC_PCIE_CLK100M_PEGAP	TRUE	17
NC_PE_1F_R2D_CN<3..0>	TRUE	8 10
NC_PEG_CLK100MP	TRUE	8 17 87
NC_PEG_R2D_CN<7..0>	TRUE	8 10
NC_SMC_ADC20	TRUE	47
NC_SMC_ADC23	TRUE	47
NC_SMC_HIB_L	TRUE	47
NC_SMC_PH2	TRUE	47
NC_SMC_PN3	TRUE	47
NC_SMC_PN6	TRUE	47
NC_SMC_TRST_L	TRUE	49
NC_TBT_B_D2RP<0..1>	TRUE	76
NC_SMC_ADC18	TRUE	47

NO_TEST

	TRUE	
NC_AUD_HP_PORT_L	TRUE	54
NC_AUD_LO3_P_R	TRUE	54
NC_BCM57765_CE_L_MS_INS_L	TRUE	37
NC_CS4206_MICBIAS	TRUE	54
NC_DP_TBTPB_ML_CN<1..3>	TRUE	76
NC_FDI_CPU_DATAP<7..0>	TRUE	
NC_FDI_PCH_FSYNC<1..0>	TRUE	
NC_FW643_AVREG	TRUE	39
NC_FW643_SM	TRUE	39
NC_GPU_GPIO_7	TRUE	
NC_HDA_SDIN1	TRUE	
NC_HPD_D	TRUE	
NC_IG_DP_TBTSNK0_AUXCH_CN	TRUE	
NC_IG_DP_TBTSNK0_CTRL_DATA	TRUE	
NC_IG_DP_TBTSNK0_ML_CP<3..0>	TRUE	
NC_IG_DP_TBTSNK1_CTRL_CLK	TRUE	
NC_IG_DP_TBTSNK1_ML_CN<3..0>	TRUE	
NC_IG_HDMI_CLK_CP	TRUE	
NC_IG_HDMI_LS_HPD	TRUE	
NC_PCH_CLKOUT_DPN	TRUE	17
NC_PCH_GPIO65_CLKOUTFLEX1	TRUE	17
NC_PCI_CLK33M_OUT3	TRUE	19
NC_PCIE_CLK100M_EXCARDP	TRUE	17
NC_PCIE_CLK100M_PE5N	TRUE	17
NC_PCIE_CLK100M_PEBP	TRUE	17
NC_PE_1F_D2RN<3..0>	TRUE	8 10
NC_PE_1F_R2D_CP<3..0>	TRUE	8 10
NC_PEG_D2RN<7..0>	TRUE	8 10
NC_PEG_R2D_CP<7..0>	TRUE	8 10
NC_SMC_ADC21	TRUE	47
NC_SMC_FAN_1_CTL	TRUE	47
NC_SMC_MD1	TRUE	49
NC_SMC_PH3	TRUE	47
NC_SMC_PN4	TRUE	47
NC_SMC_PN7	TRUE	47
NC_SMC_XOSC1	TRUE	47
NC_TBT_B_R2D_CN<0..1>	TRUE	76
NC_SMC_ADC19	TRUE	47

BEAD-PROBES

TP			
TP0C00	TP 1 SIGNAL_MODEL=EMPTY	USB3_EXTA_RX_F_P	43 86
TP0C01	TP 1 SIGNAL_MODEL=EMPTY	USB3_EXTA_RX_F_N	43 86
TP0C02	TP 1 SIGNAL_MODEL=EMPTY	USB3_EXTA_TX_F_P	43 86
TP0C03	TP 1 SIGNAL_MODEL=EMPTY	USB3_EXTA_TX_F_N	43 86
TP0C08	TP 1 SIGNAL_MODEL=EMPTY	USB3_EXTC_RX_F_P	44 86
TP0C09	TP 1 SIGNAL_MODEL=EMPTY	USB3_EXTC_RX_F_N	44 86
TP0C10	TP 1 SIGNAL_MODEL=EMPTY	USB3_EXTC_TX_F_P	44 86
TP0C11	TP 1 SIGNAL_MODEL=EMPTY	USB3_EXTC_TX_F_N	44 86
TP0C16	TP 1 SIGNAL_MODEL=EMPTY	TBT_A_R2D_C_P<0>	76 80 89
TP0C17	TP 1 SIGNAL_MODEL=EMPTY	TBT_A_R2D_C_N<0>	76 80 89
TP0C18	TP 1 SIGNAL_MODEL=EMPTY	TBT_A_R2D_C_P<1>	76 80 89
TP0C19	TP 1 SIGNAL_MODEL=EMPTY	TBT_A_R2D_C_N<1>	76 80 89
TP0C20	TP 1 SIGNAL_MODEL=EMPTY	TBT_A_D2R_C_P<0>	80 89
TP0C21	TP 1 SIGNAL_MODEL=EMPTY	TBT_A_D2R_C_N<0>	80 89
TP0C22	TP 1 SIGNAL_MODEL=EMPTY	TBT_A_D2R_C_P<1>	80 89
TP0C23	TP 1 SIGNAL_MODEL=EMPTY	TBT_A_D2R_C_N<1>	80 89

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